

Retrocausal capacity of a quantum channel: Communicating through noisy closed timelike curves

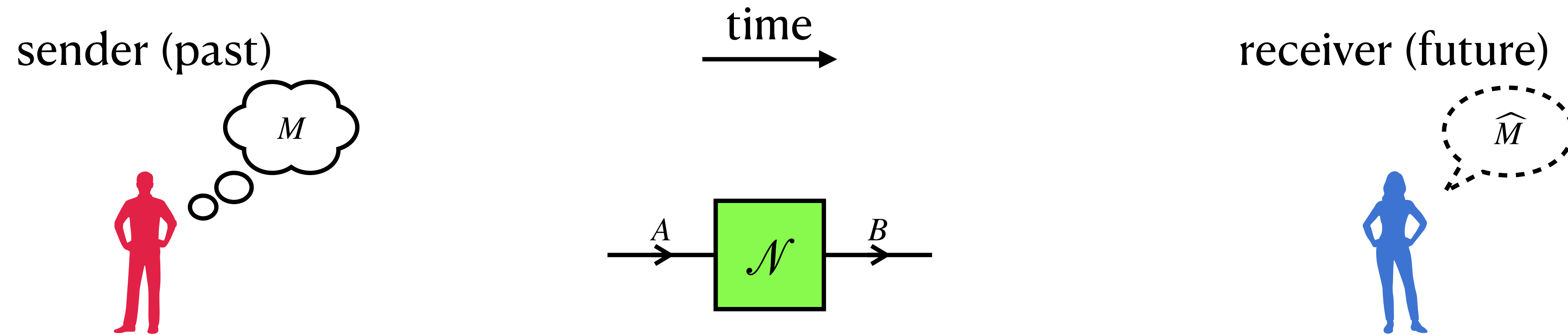
Kaiyuan Ji (Cornell)

Joint work with Seth Lloyd (MIT) and Mark M. Wilde (Cornell)

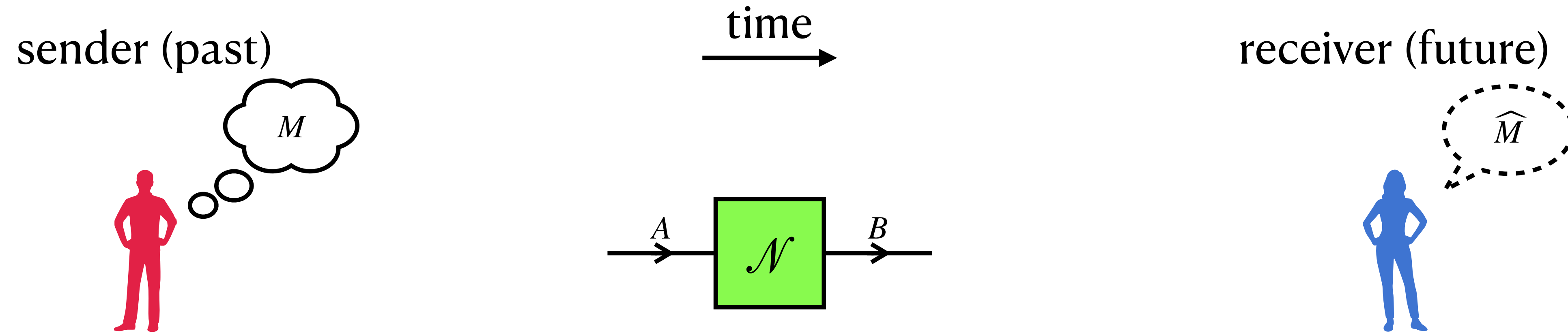


[PRL 136, 230801 \(2026\)](#)

Communication over a noisy channel

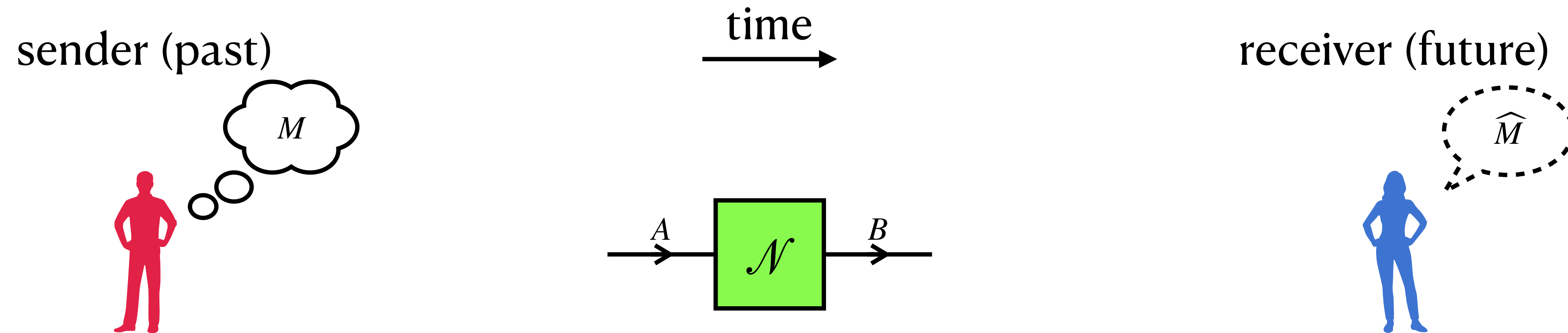


Communication over a noisy channel



- A classical channel is a stochastic map
- A quantum channel is a CPTP linear map

Communication over a noisy channel

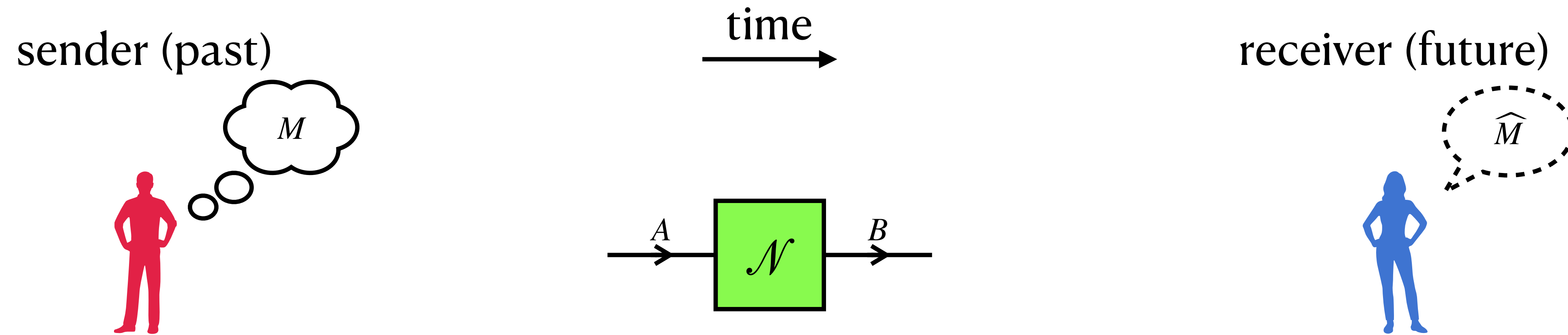


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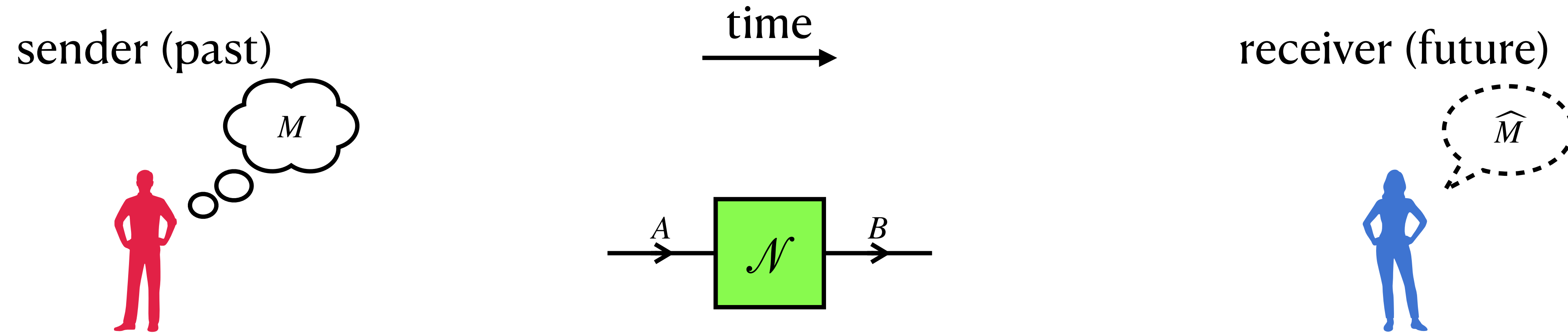
$$\rho_{RA} \geq 0 \quad \Rightarrow \quad \text{id}_R \otimes \mathcal{N}_{A \rightarrow B}[\rho_{RA}] \geq 0$$

$$\text{Tr} \circ \mathcal{N}_{A \rightarrow B} = \text{Tr}$$

Communication over a noisy channel

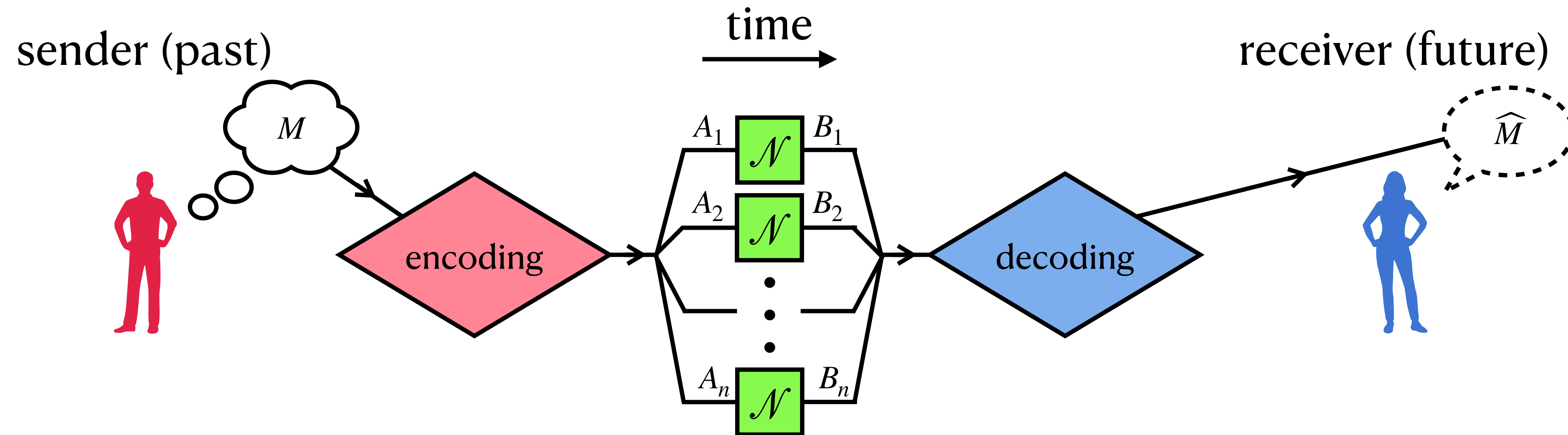


Communication over a noisy channel



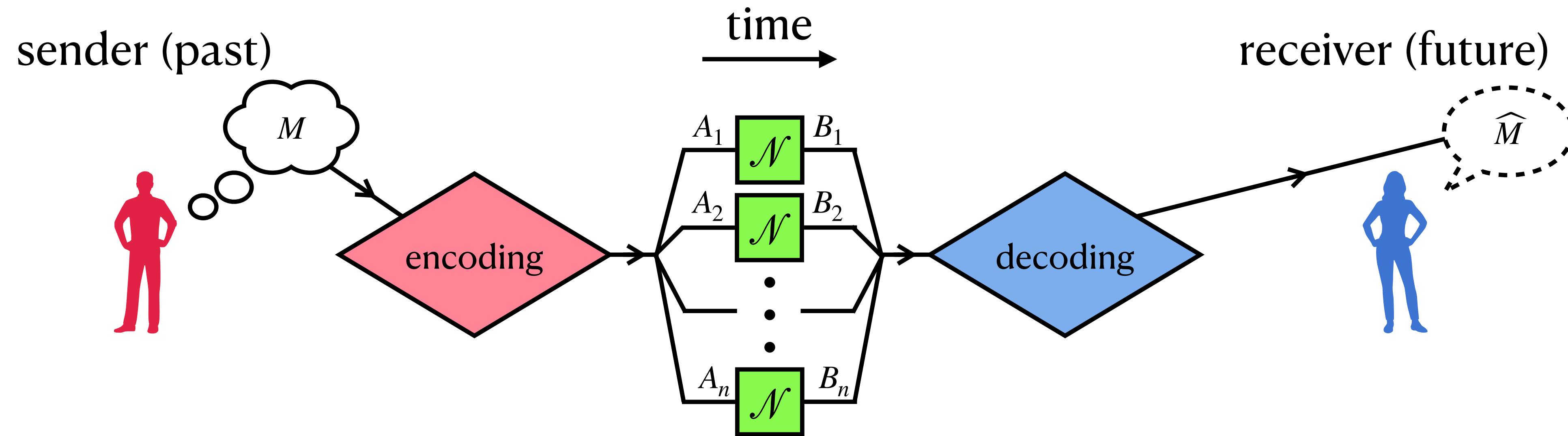
- Shannon theory: fundamental limit on efficiency of reliable communication over given $\mathcal{N}_{A \rightarrow B}$

Communication over a noisy channel



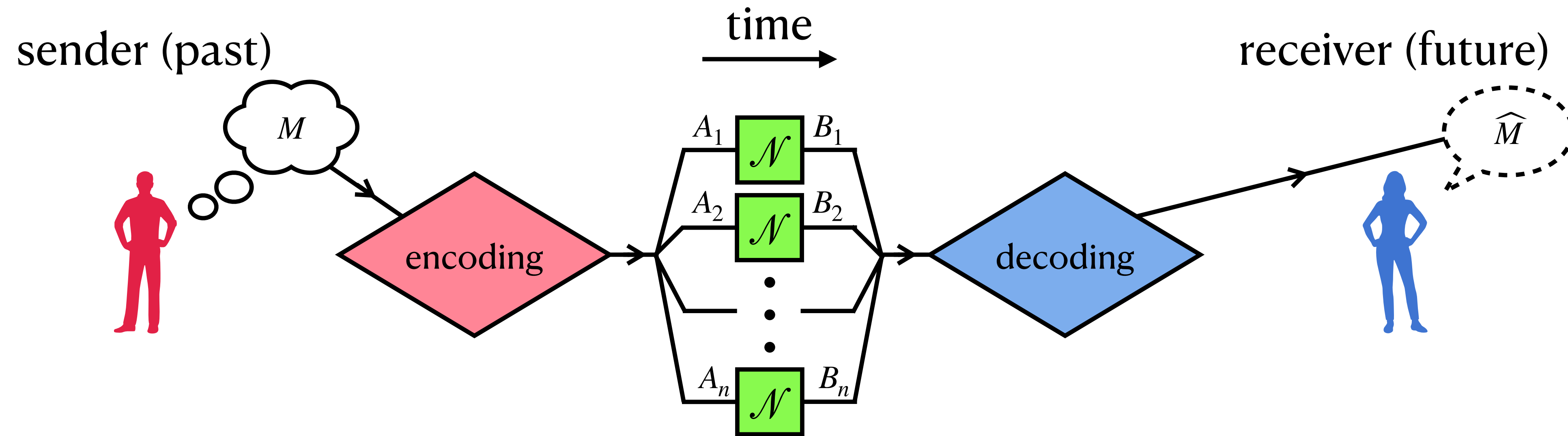
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Communication over a noisy channel



- Shannon theory: fundamental limit on efficiency of reliable communication over given $\mathcal{N}_{A \rightarrow B}$
- Capacity: maximum # bits transmitted per channel use with vanishing error

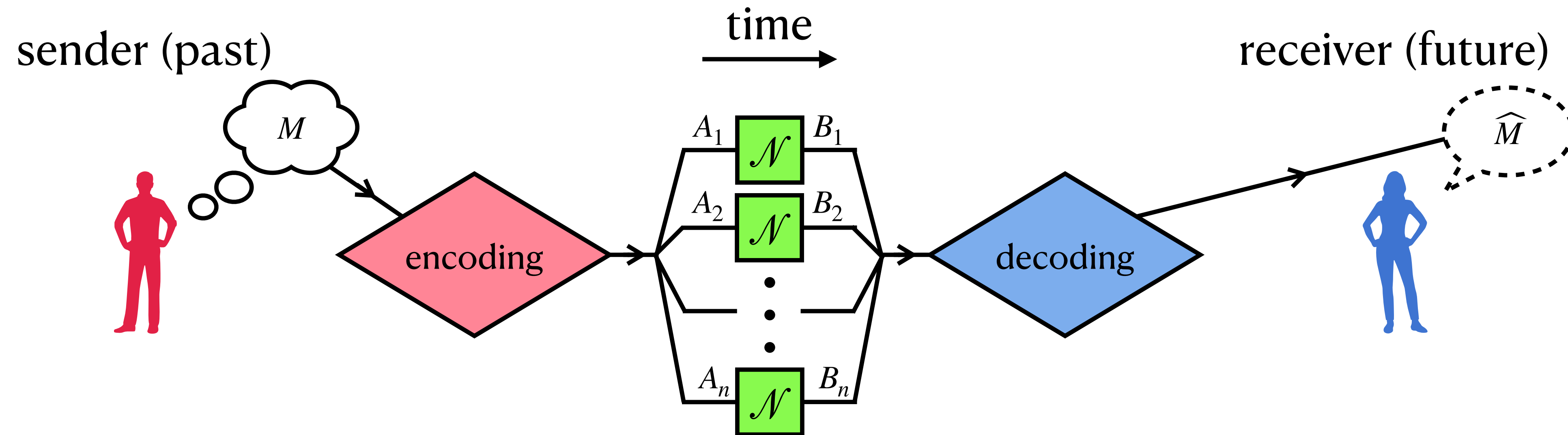
Communication over a noisy channel



- Shannon theory: fundamental limit on efficiency of reliable communication over given $\mathcal{N}_{A \rightarrow B}$
- Capacity: maximum # bits transmitted per channel use with vanishing error
- Noisy-channel coding theorem: $C(\mathcal{N}) = I(\mathcal{N})$ for classical $\mathcal{N}_{A \rightarrow B}$

[Shannon 1948]

Communication over a noisy channel



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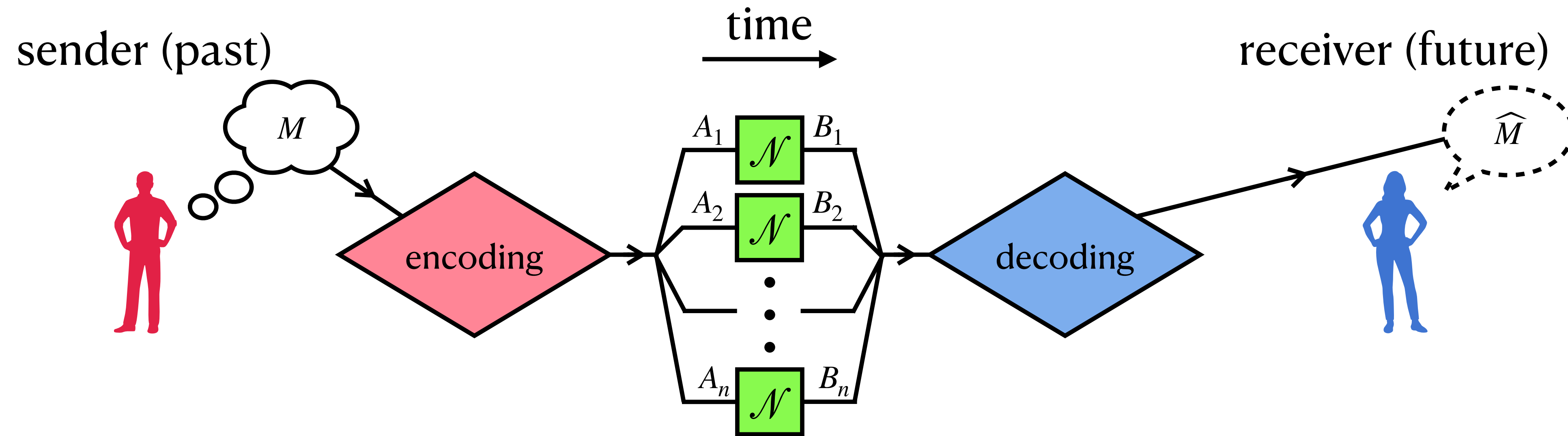
[Shannon 1948]

channel capacity

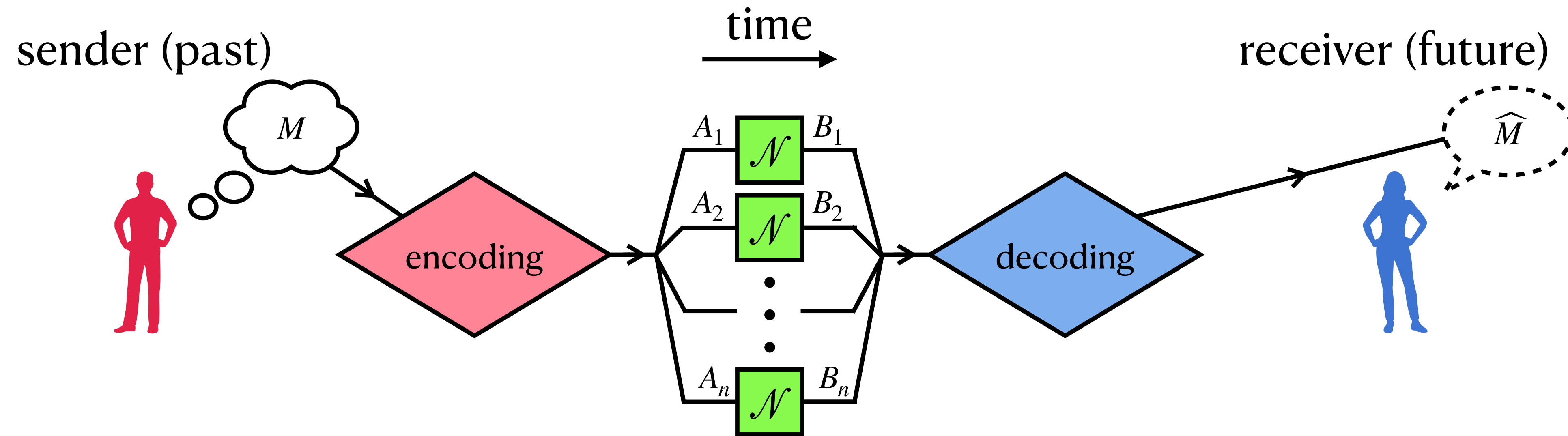
channel mutual information

$$I(\mathcal{N}) := \max_{\rho_{X'X}} I(X'; Y)_{\mathcal{N}_{X \rightarrow Y}[\rho_{X'X}]}$$

Communication over a noisy channel

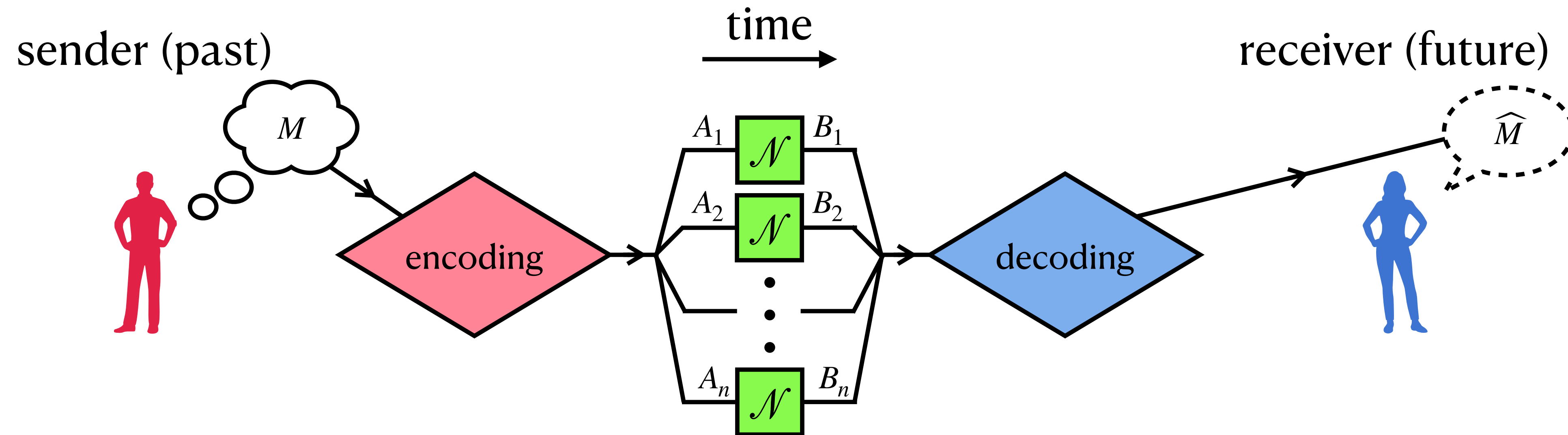


Communication over a noisy channel



For quantum $\mathcal{N}_{A \rightarrow B}$:

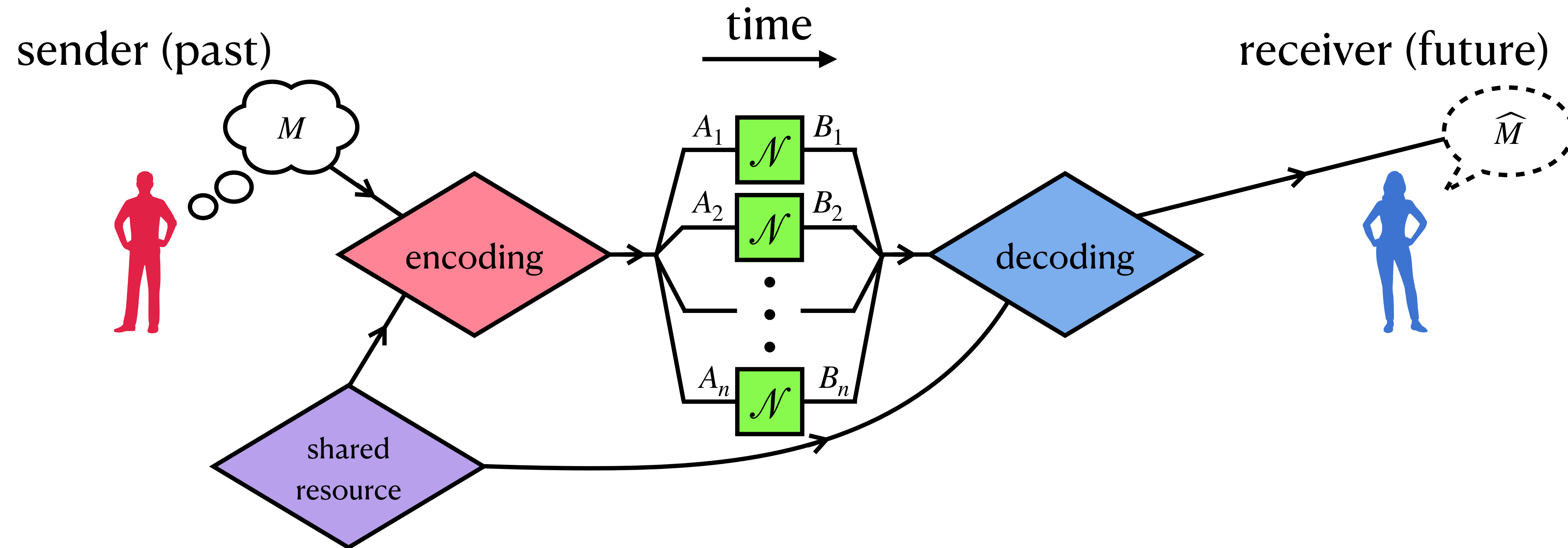
Communication over a noisy channel



For quantum $\mathcal{N}_{A \rightarrow B}$:

- classical message, unassisted: $C(\mathcal{N}) = \chi^\infty(\mathcal{N}) \equiv \lim_{n \rightarrow \infty} \frac{1}{n} \chi(\mathcal{N}^{\otimes n})$ [Holevo 1998]
[Schumacher & Westmoreland 1997]
- quantum message, unassisted: $Q(\mathcal{N}) = I_{\text{coh}}^\infty(\mathcal{N}) \equiv \lim_{n \rightarrow \infty} \frac{1}{n} I_{\text{coh}}(\mathcal{N}^{\otimes n})$ [Lloyd 1997]
[Shor 2002]
[Devetak 2005]

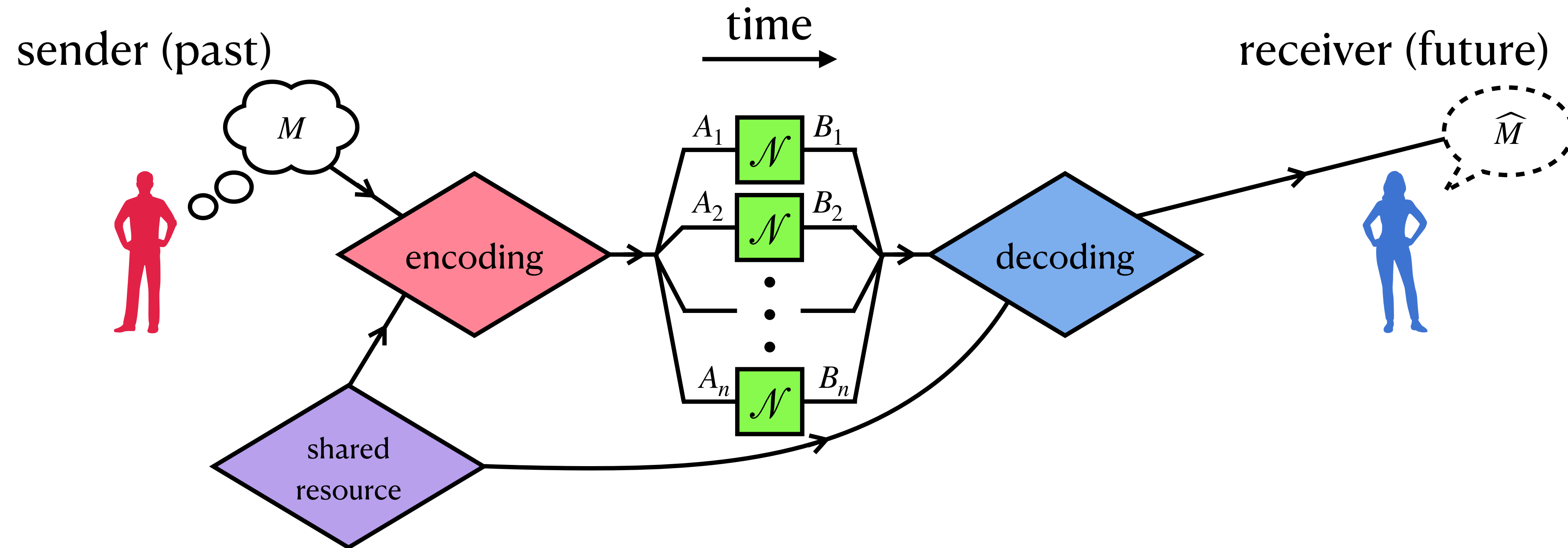
Communication over a noisy channel



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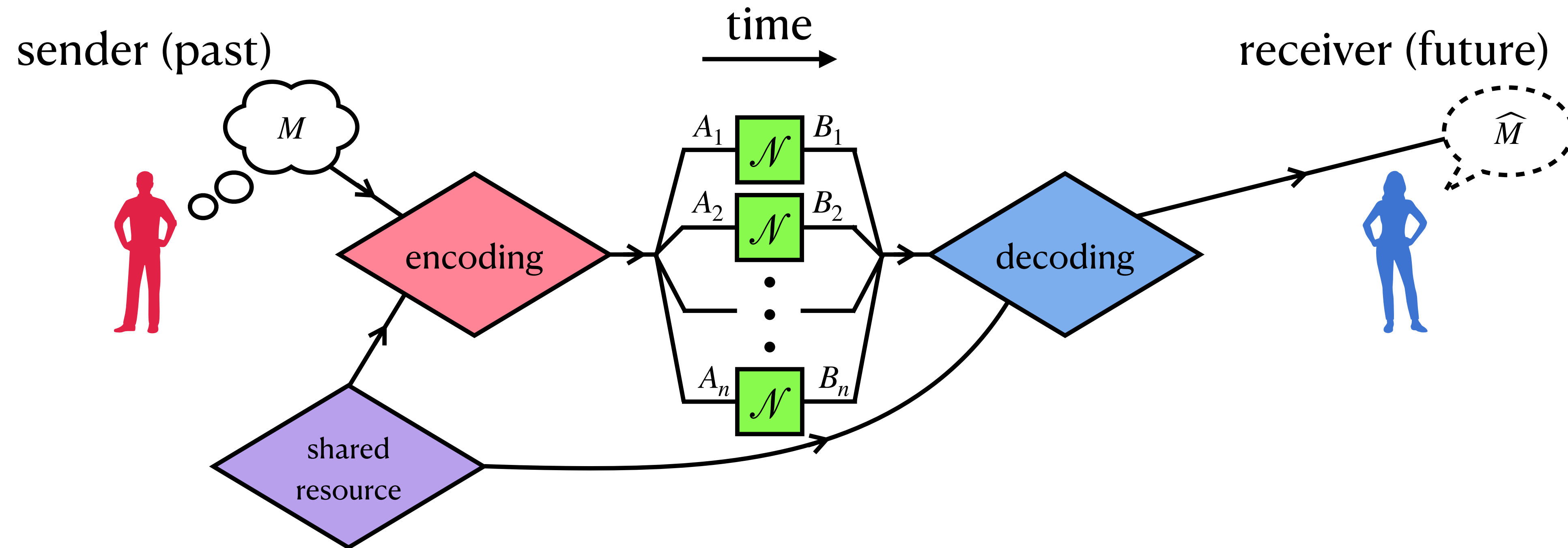
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Communication over a noisy channel



For quantum $\mathcal{N}_{A \rightarrow B}$:

Communication over a noisy channel



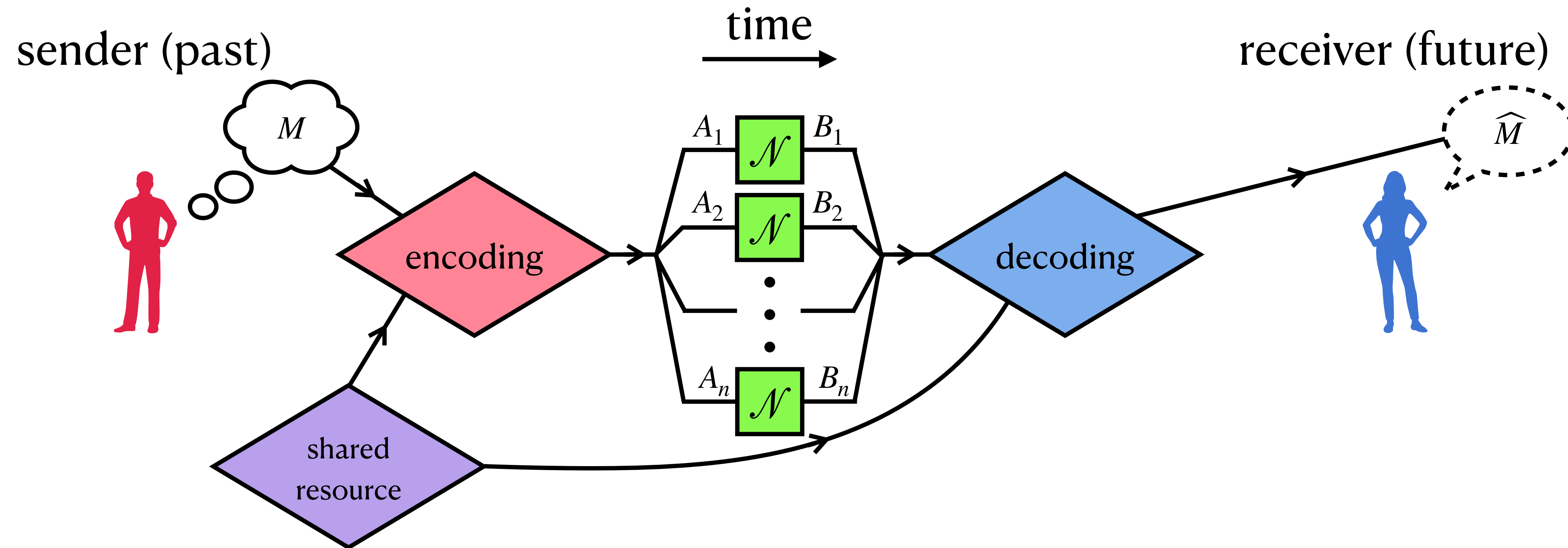
For quantum $\mathcal{N}_{A \rightarrow B}$:

- c. or q. message, entanglement assisted: $C_{\text{EA}}(\mathcal{N}) = 2Q_{\text{EA}}(\mathcal{N}) = I(\mathcal{N})$

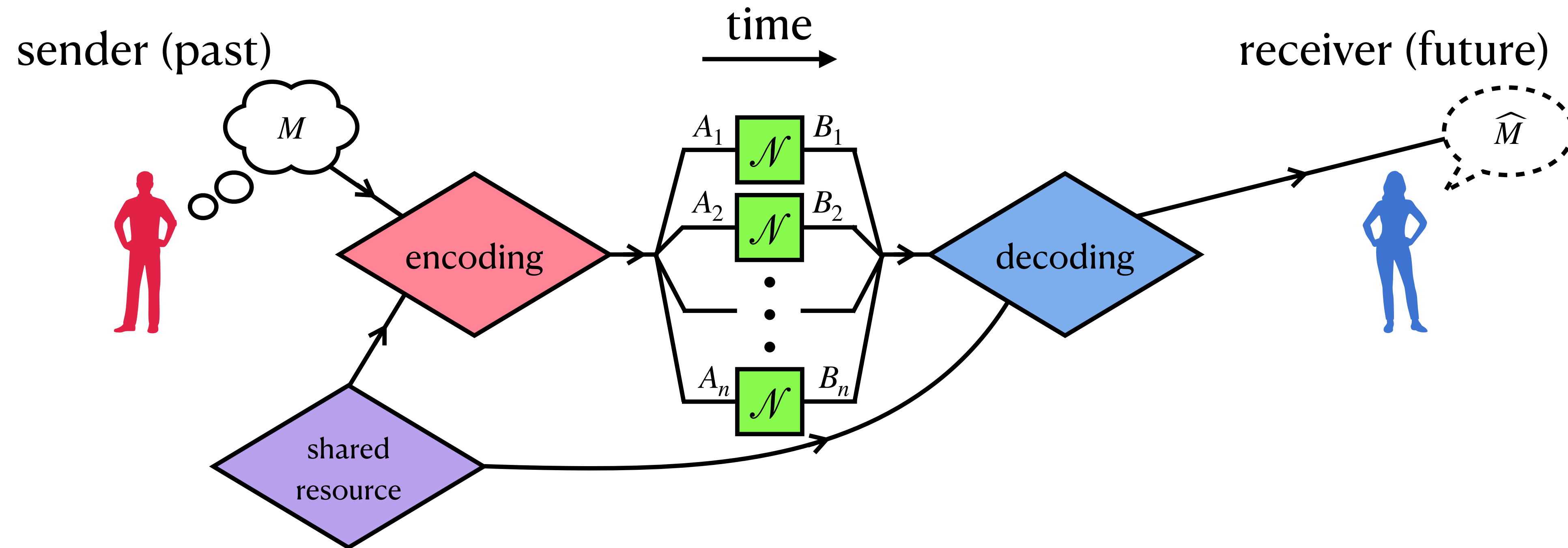
[Bennett 2002]

$$I(\mathcal{N}) := \sup_{\rho_{RA}} I(R; B)_{\mathcal{N}_{A \rightarrow B}[\rho_{RA}]}$$

Communication over a noisy channel

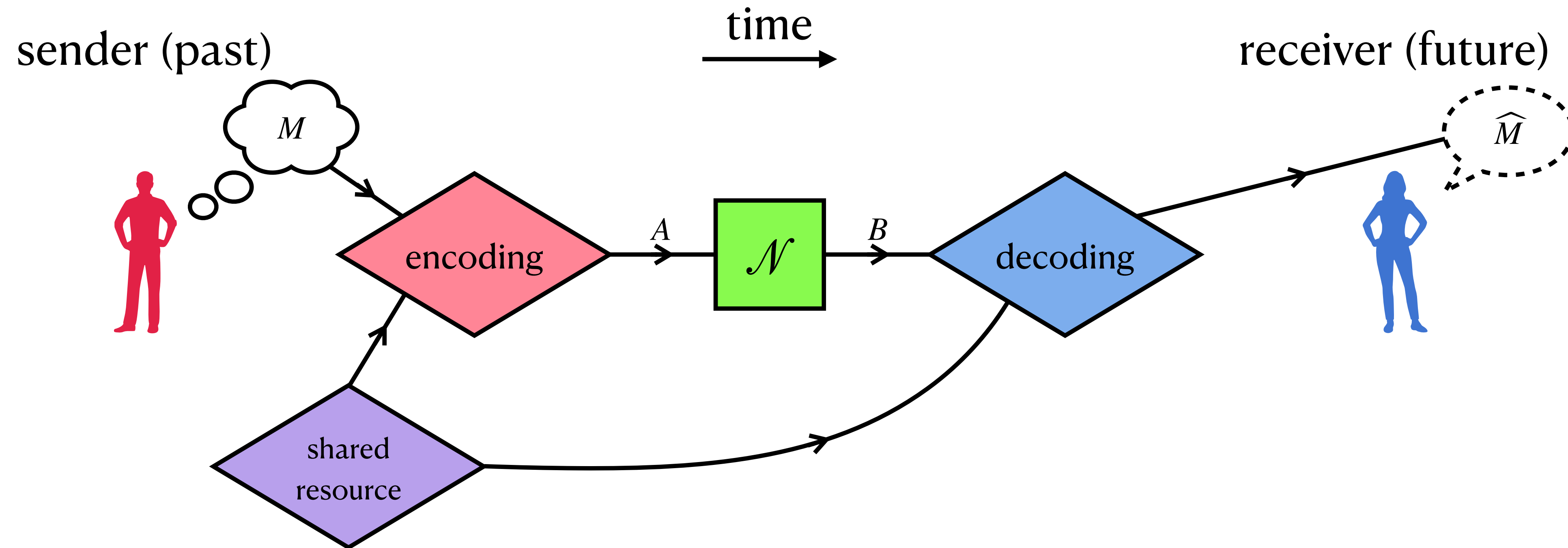


Communication over a noisy channel



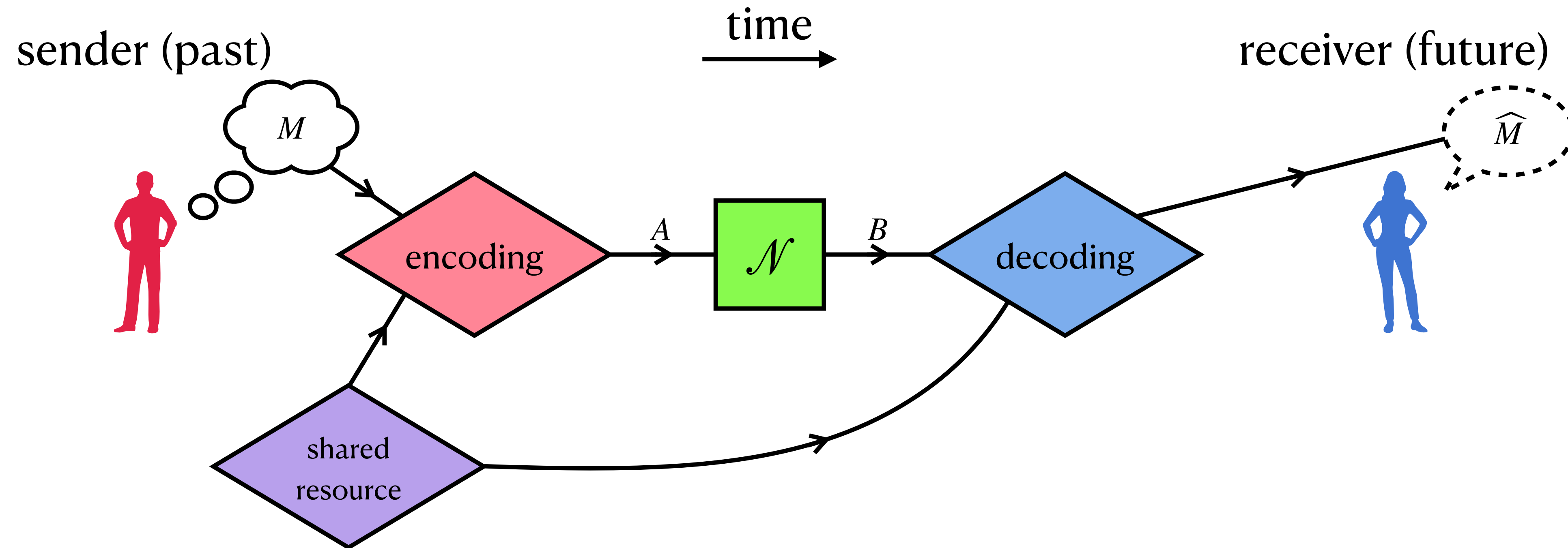
- (Asymptotic) capacity: maximum # bits reliably transmitted per channel use, $C(\mathcal{N}), Q(\mathcal{N}) \dots$

Communication over a noisy channel (one-shot)



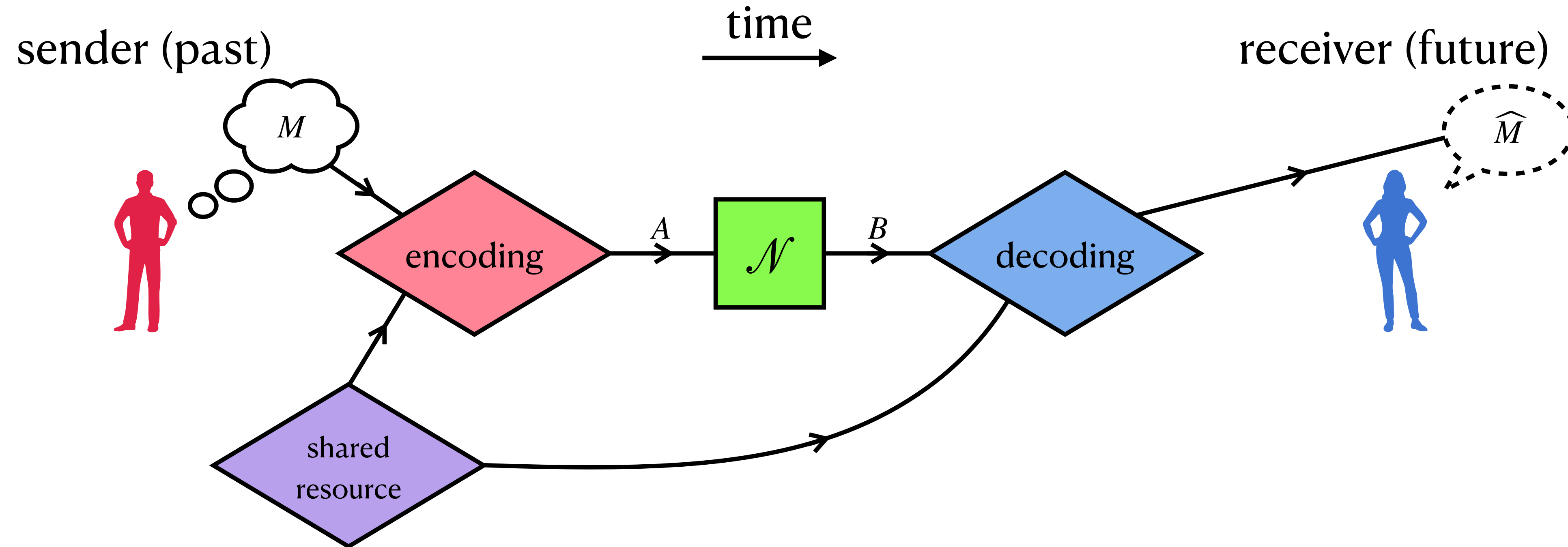
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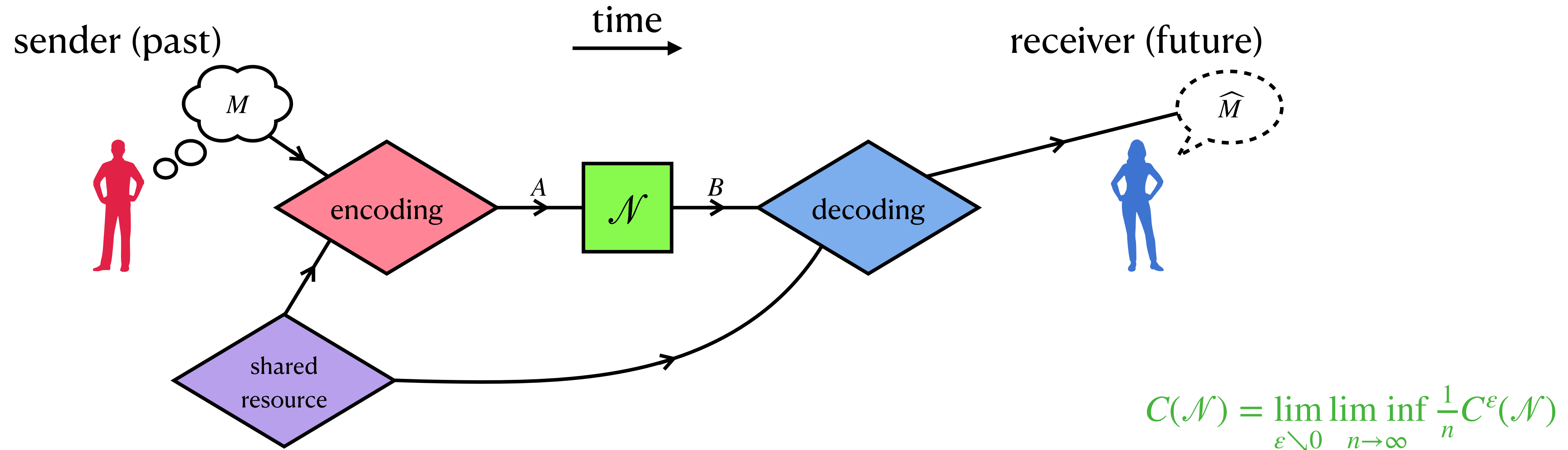
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$$C^\varepsilon(\mathcal{N}), Q^\varepsilon(\mathcal{N}) \dots$$

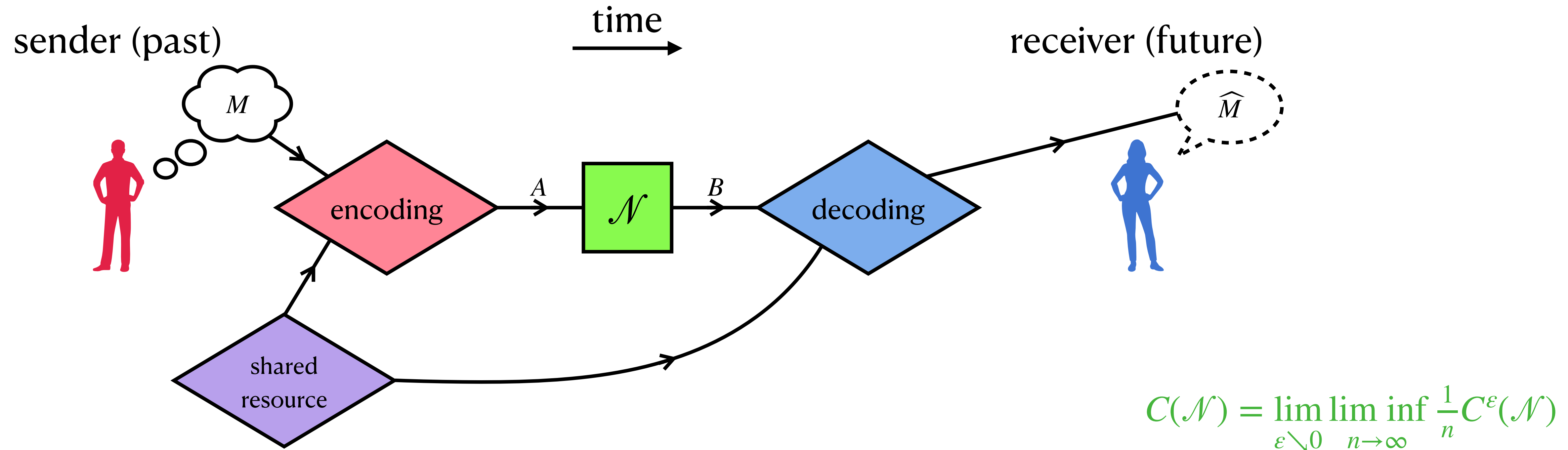
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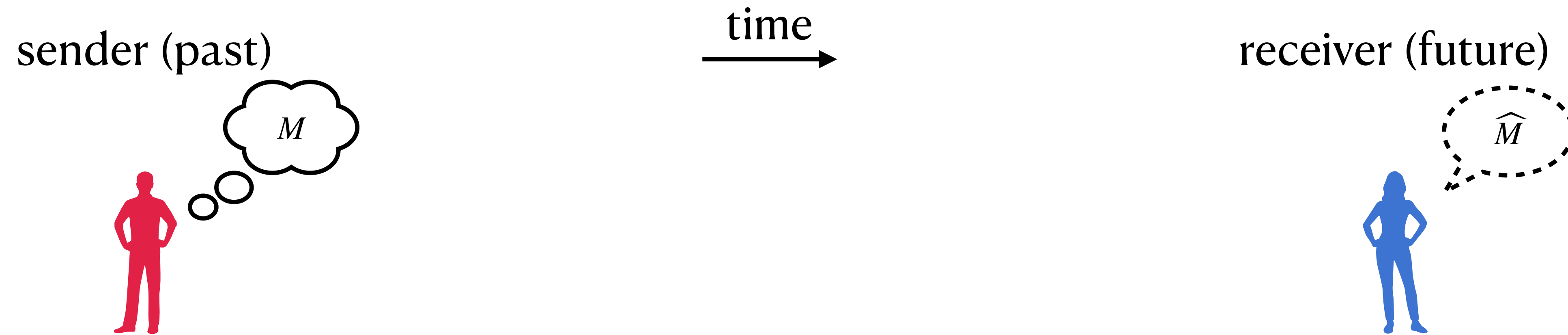


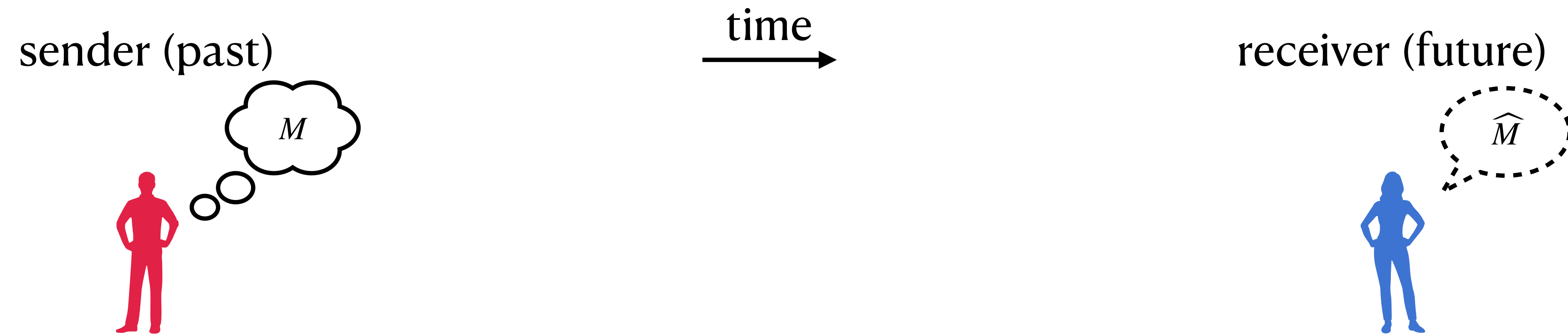
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Closed-form solution usually unknown

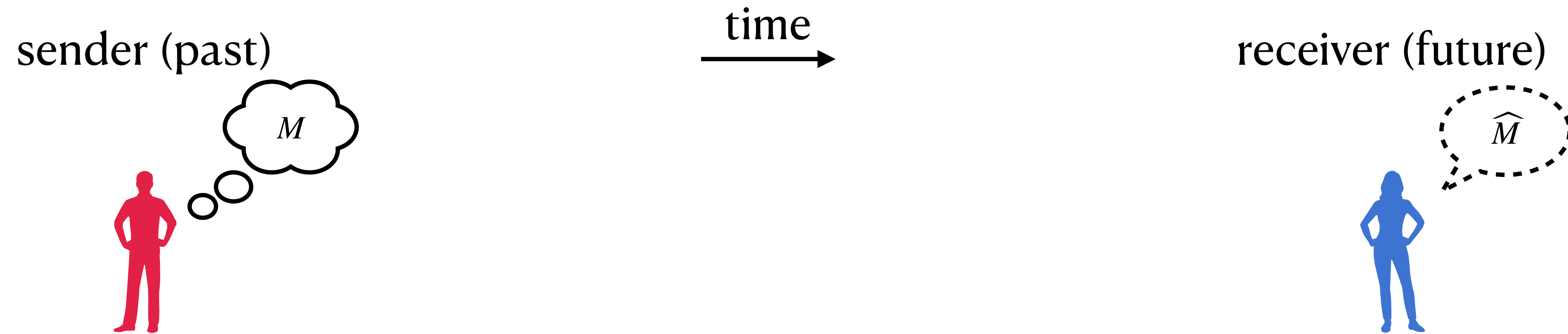
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Communication over a noisy channel (one-shot)

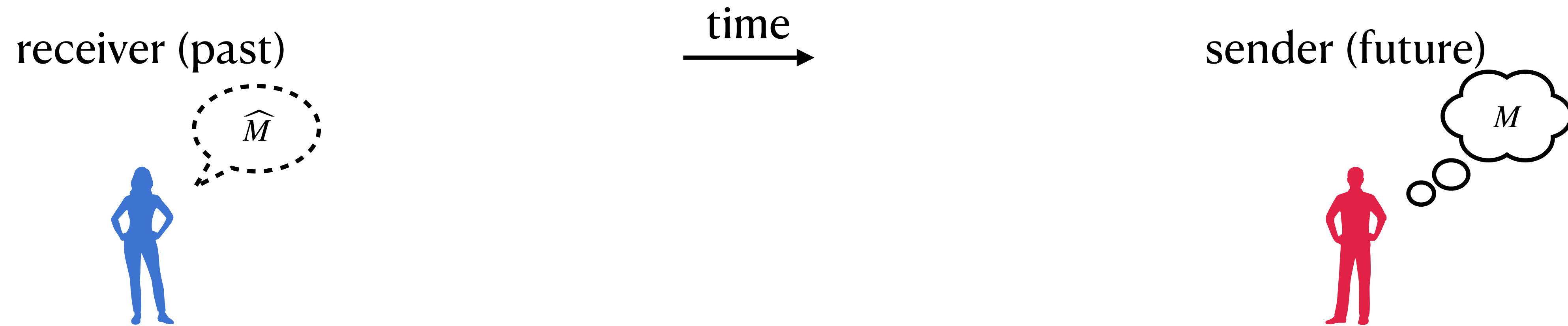




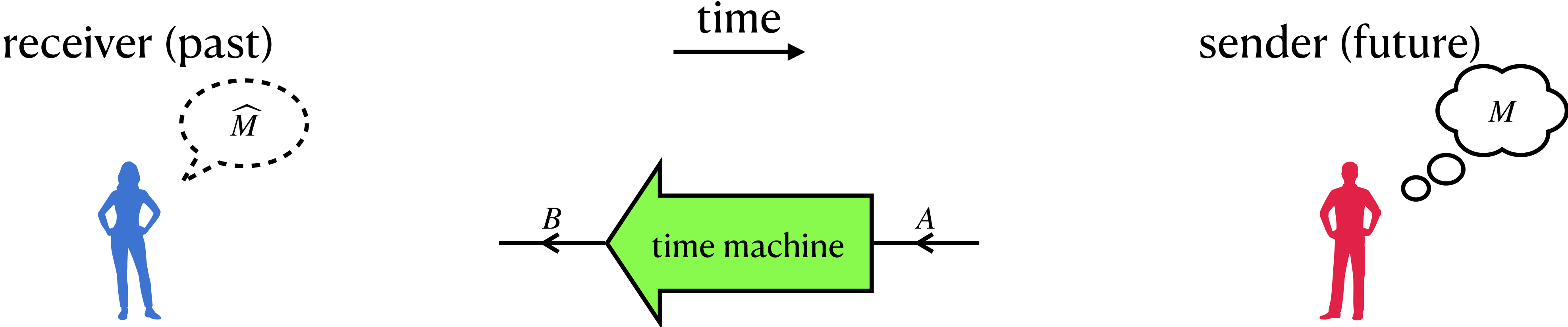
Communication, but backward in time!



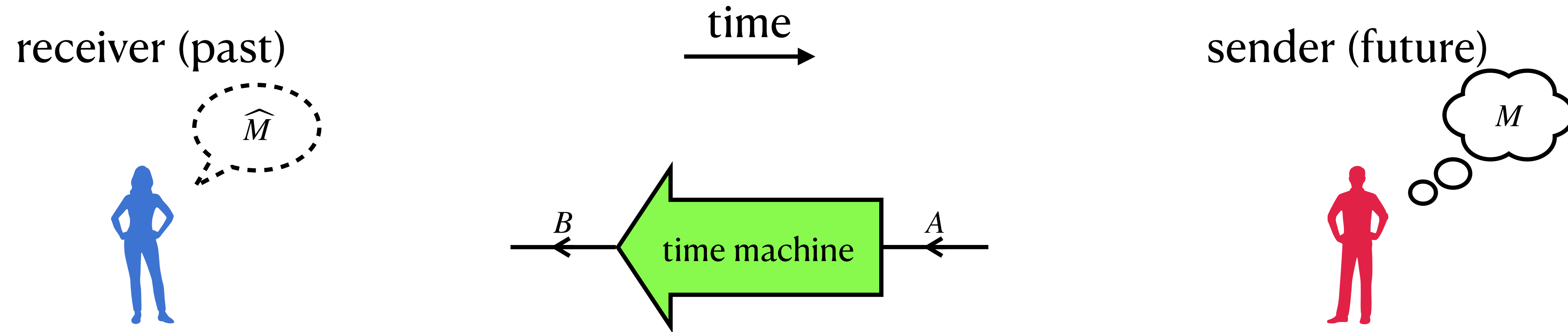
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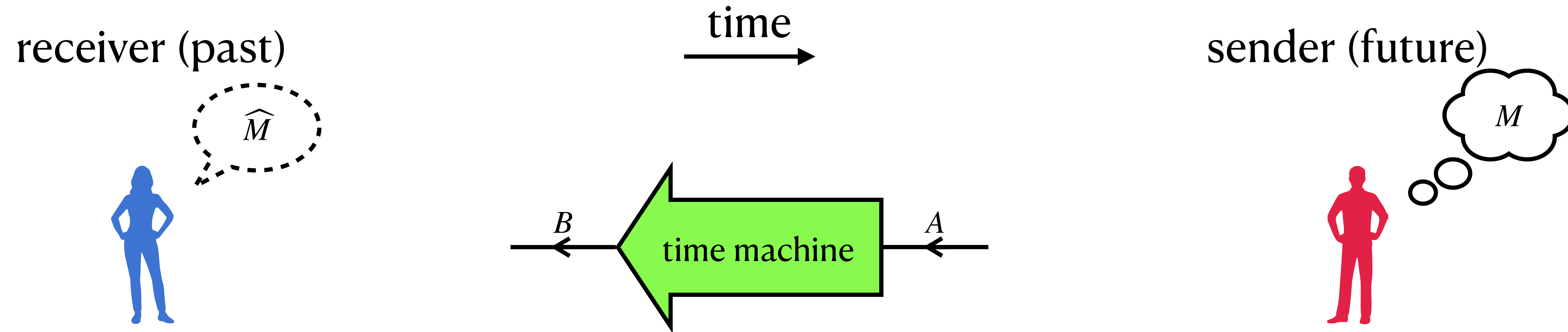


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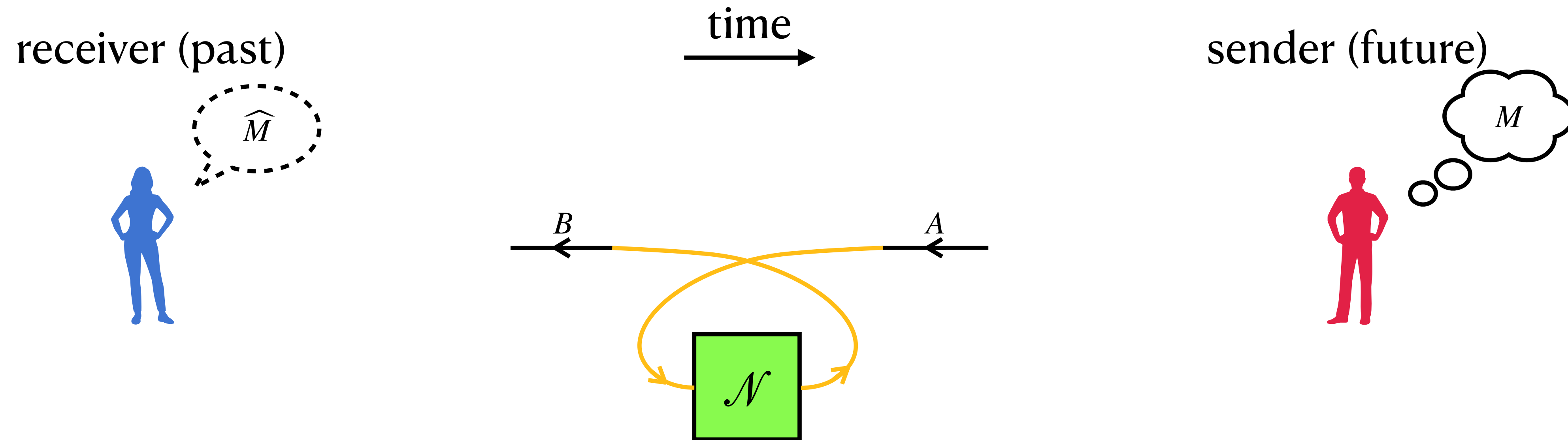
- What we study: communication, backward in time, over a given **noisy time machine**

Communication, but backward in time!



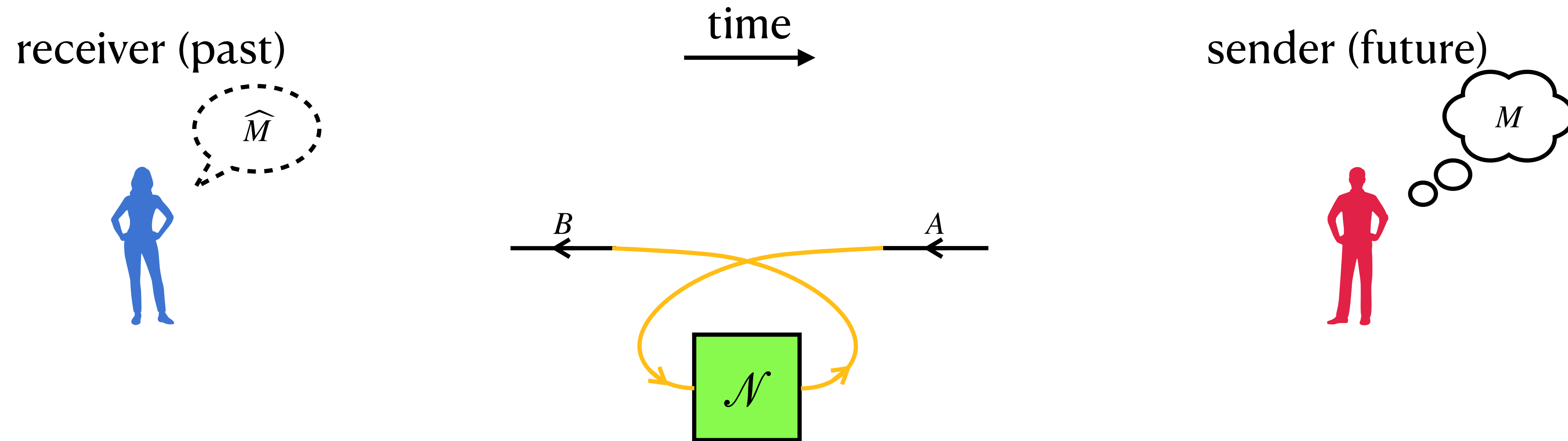
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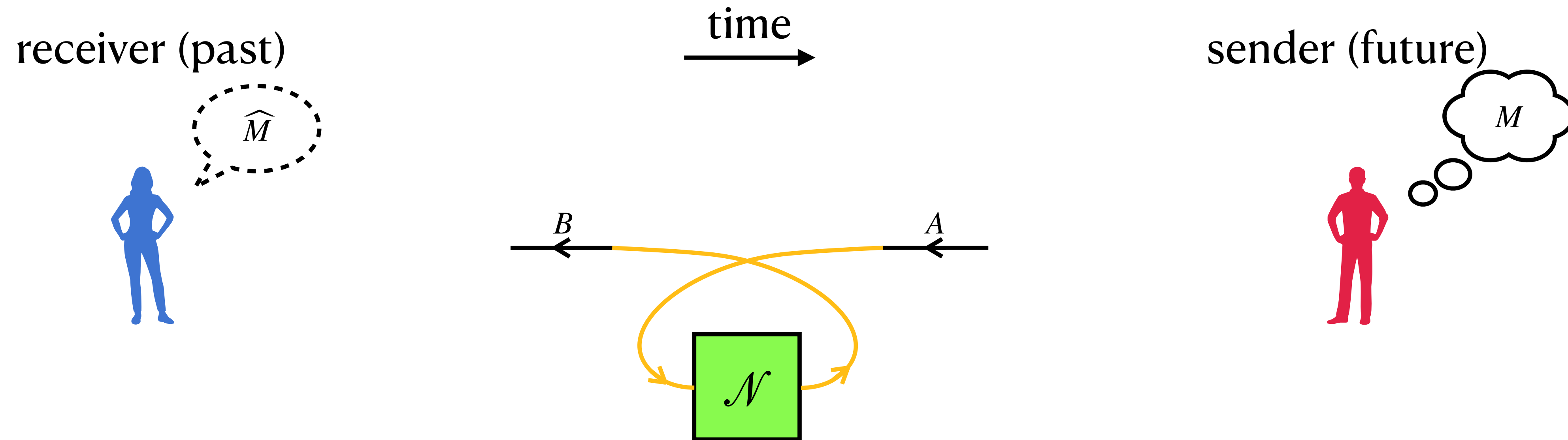
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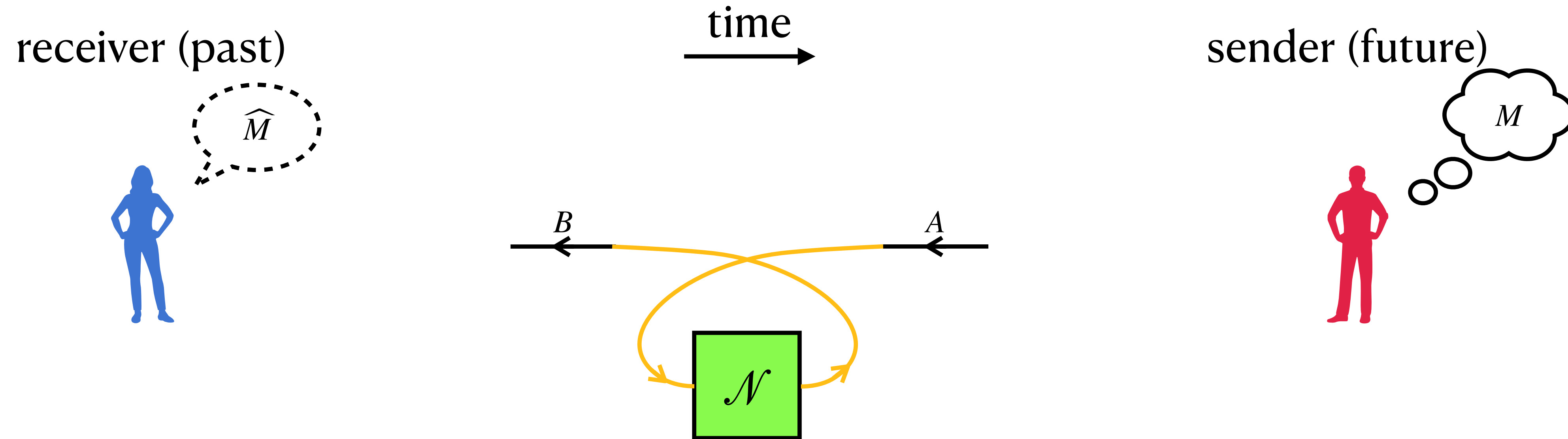
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Communication, but backward in time!



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- **Q: fundamental limit on efficiency? (“retrocausal” capacity)**

Communication, but backward in time!



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- Q: **fundamental limit on efficiency?** (“retrocausal” capacity)

We solve this precisely in the one-shot regime

What makes this an interesting problem?

What makes this an interesting problem?

1. The existence of time machines in our universe has not been ruled out theoretically

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1. The existence of time machines in our universe has not been ruled out theoretically
 - In principle admitted by general relativity

REVIEWS OF MODERN PHYSICS

VOLUME 21, NUMBER 3

JULY, 1949

An Example of a New Type of Cosmological Solutions of Einstein's Field Equations Gravitation

KURT GÖDEL

Institute for Advanced Study, Princeton, New Jersey

VOLUME 66, NUMBER 9

PHYSICAL REVIEW LETTERS

4 MARCH 1991

Closed Timelike Curves Produced by Pairs of Moving Cosmic Strings: Exact Solutions

J. Richard Gott, III

Department of Astrophysical Sciences, Princeton University, Princeton, New Jersey 08544

(Received 18 October 1990)

Exact solutions of Einstein's field equations are presented for the general case of two moving straight cosmic strings that do not intersect. The solutions for parallel cosmic strings moving in opposite directions, each with $\gamma_s > (\sin 4\pi\mu)^{-1}$ in the laboratory frame show closed timelike curves (CTC's) that circle the two strings as they pass, allowing observers to visit their own past. Similar results occur for non-parallel strings, and for masses in (2+1)-dimensional spacetime. For finite string loops the possibility that black-hole formation may prevent the formation of CTC's is discussed.

PACS numbers: 04.20.Jb, 95.30.Sf, 98.80.Cq

What makes this an interesting problem?

- 1. The existence of time machines in our universe has not been ruled out
 - May arise from black holes!

The black hole final state

Gary T. Horowitz

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Santa Barbara, CA 93106, U.S.A.
E-mail: gary@physics.ucsb.edu*

Juan Maldacena

*Institute for Advanced Study
Princeton, New Jersey 08540, U.S.A.
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ABSTRACT: We propose that in quantum gravity one need not impose a final-state boundary condition at black hole singularities. This resolves the tension between string theory and semiclassical arguments over whether information is lost.

KEYWORDS: Black Holes in String Theory, AdS-CFT Correspondence, Black Holes.

Comment on “The black hole final state”

Daniel Gottesman

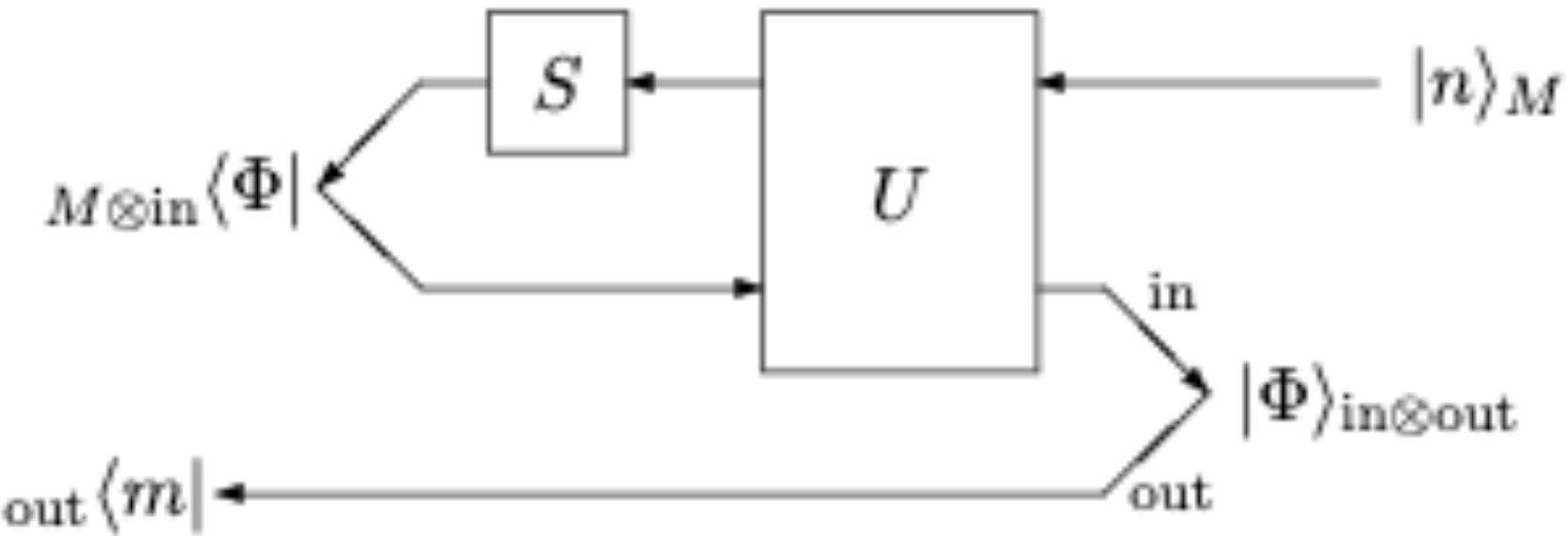
*Perimeter Institute
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E-mail: dgottesman@perimeterinstitute.ca*

John Preskill

*Institute for Quantum Information, California Institute of Technology
Pasadena, CA 91125, U.S.A.
E-mail: preskill@theory.caltech.edu*

ABSTRACT: Horowitz and Maldacena have suggested that the unitarity of the black hole S -matrix can be reconciled with Hawking’s semiclassical arguments if a final-state boundary condition is imposed at the spacelike singularity inside the black hole. We point out that, in this scenario, departures from unitarity can arise due to interactions between the collapsing body and the infalling Hawking radiation inside the event horizon. The amount of information lost when a black hole evaporates depends on the extent to which these interactions are entangling.

KEYWORDS: Black Holes in String Theory, Black Holes.



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2. Relaxation of a theory sometimes provides new insights into the theory

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2. Relaxation of a theory sometimes provides new insights into the theory
 - in foundations (nonsignaling correlations)

Quantum Nonlocality as an Axiom

Sandu Popescu¹ and Daniel Rohrlich²

Received July 2, 1993; revised July 19, 1993

In the conventional approach to quantum mechanics, indeterminism is an axiom and nonlocality is a theorem. We consider inverting the logical order, making nonlocality an axiom and indeterminism a theorem. Nonlocal "superquantum" correlations, preserving relativistic causality, can violate the CHSH inequality more strongly than any quantum correlations.

What makes this an interesting problem?

2. Relaxation of a theory sometimes provides new insights into the theory
- in complexity theory (postselection; nonlinearity; closed timelike curves)

PROCEEDINGS
— OF —
THE ROYAL
SOCIETY **A**

Quantum computing, and probabilistic po

BY SCOTT AARONSON
*Department of Combinatorics and Optimization,
University of Waterloo, Waterloo, Ontario N2L 3G1,
(aaronson@uwaterloo.ca)*

I study the class of problems efficiently solvable with quantum computers, and show that the ability to ‘postselect’ on the outcomes of measurements increases the computational power of quantum computers. In particular, I show that several simple changes to the standard quantum circuit model allow us to solve PP-complete problems efficiently. The result generalizes the celebrated theorem of Beigel, Reingold and Spielmann, as well as a generalization of that theorem by Aaronson. This illustrates that quantum computing can yield new insights into classical computation.

Keywords: quantum computers; computational complexity; Born’s rule

VOLUME 81, NUMBER 18

PHYSICAL REVIEW LETTERS

Nonlinear Quantum Mechanics Implies Polynomial-Time Solution for NP-Complete and #P Problems

Daniel S. Abrams*
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Seth Lloyd†
*d’Arbeloff Laboratory for Information Sciences and Technology, Department of Mechanical Engineering,
MIT, Cambridge, Massachusetts 02139*
(Received 27 January 1998)

If quantum states exhibit small nonlinearities during time evolution, then quantum computers can be used to solve NP-complete and #P problems in polynomial time. We provide algorithms for solving NP-complete and #P oracle problems by exploiting nonlinear quantum logic gates. Using the standard quantum circuit model as a simple example, the explicit construction of these gates is derived from the principles of quantum physics. Nonlinear quantum algorithms are also presented using Polchinski type nonlinearities. These results do not allow for superluminal communication. [S0031-9007(98)07489-4]

PACS numbers: 03.67.–a

PROCEEDINGS
— OF —
THE ROYAL
SOCIETY **A**

Proc. R. Soc. A (2009) **465**, 631–647
doi:10.1098/rspa.2008.0350
Published online 11 November 2008

Closed timelike curves make quantum and classical computing equivalent

BY SCOTT AARONSON^{1,*} AND JOHN WATROUS²
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²*School of Computer Science, University of Waterloo, Waterloo, Ontario,
Canada N2L 3G1*

While closed timelike curves (CTCs) are not known to exist, studying their consequences has led to non-trivial insights into general relativity, quantum information and other areas. In this paper, we show that, if CTCs existed, quantum computers would be no more powerful than classical computers: both would have the (extremely large) power of the complexity class polynomial space (PSPACE), consisting of all problems solvable by a conventional computer using a polynomial amount of memory. This solves an open problem proposed by one of us in 2005, and gives an essentially complete understanding of computational complexity in the presence of CTCs. Following the work of Deutsch, we treat a CTC as simply a region of spacetime where a ‘causal consistency’ condition is imposed, meaning that nature has to produce a (probabilistic or quantum) fixed point of some evolution operator. Our conclusion is then a consequence of the following theorem: given any quantum circuit (not necessarily unitary), a fixed point of the circuit can be (implicitly) computed in PSPACE. This theorem might have independent applications in quantum information.

Keywords: quantum computing; closed timelike curves; computational complexity

What makes this an interesting problem?

2. Relaxation of a theory sometimes provides new insights into the theory
- in metrology (protocols inspired closed timelike curves)

PHYSICAL REVIEW LETTERS **131**, 150202 (2023)

Nonclassical Advantage in Metrology Established via Quantum Simulations of Hypothetical Closed Timelike Curves

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²Cavendish Laboratory, Department of Physics, University of Cambridge, Cambridge CB3 0HE, United Kingdom

³Laboratory for X-ray Nanoscience and Technologies, Paul Scherrer Institut, 5232 Villigen, Switzerland

⁴Department of Physics and Quantum Center, Eidgenössische Technische Hochschule Zürich, CH-8093 Zürich, Switzerland

⁵Joint Center for Quantum Information and Computer Science, NIST and University of Maryland, College Park, Maryland 20742, USA

⁶Institute for Physical Science and Technology, University of Maryland, College Park, Maryland 20742, USA

 (Received 3 September 2022; revised 8 September 2023; accepted 11 September 2023; published 12 October 2023)

We construct a metrology experiment in which the metrologist can sometimes amend the input state by simulating a closed timelike curve, a worldline that travels backward in time. The existence of closed timelike curves is hypothetical. Nevertheless, they can be simulated probabilistically by quantum-teleportation circuits. We leverage such simulations to pinpoint a counterintuitive nonclassical advantage achievable with entanglement. Our experiment echoes a common information-processing task: A metrologist must prepare probes to input into an unknown quantum interaction. The goal is to infer as much information per probe as possible. If the input is optimal, the information gained per probe can exceed any value achievable classically. The problem is that, only after the interaction does the metrologist learn which input would have been optimal. The metrologist can attempt to change the input by effectively teleporting the optimal input back in time, via entanglement manipulation. The effective time travel sometimes fails but ensures that, summed over trials, the metrologist's winnings are positive. Our Gedankenexperiment demonstrates that entanglement can generate operational advantages forbidden in classical chronology-respecting theories.

DOI: [10.1103/PhysRevLett.131.150202](https://doi.org/10.1103/PhysRevLett.131.150202)

PHYSICAL REVIEW LETTERS **132**, 260801 (2024)

Featured in Physics

Agnostic Phase Estimation

Xingrui Song¹, Flavio Salvati², Chandrashekhhar Gaikwad¹, Nicole Yunger Halpern^{3,4}

David R. M. Arvidsson-Shukur⁵ and Kater Murch¹

¹Department of Physics, Washington University, St. Louis, Missouri 63130, USA

²Cavendish Laboratory, Department of Physics, University of Cambridge, Cambridge, CB3 0HE, United Kingdom

³Joint Center for Quantum Information and Computer Science, NIST and University of Maryland, College Park, Maryland 20742, USA

⁴Institute for Physical Science and Technology, University of Maryland, College Park, Maryland 20742, USA

⁵Hitachi Cambridge Laboratory, J. J. Thomson Avenue, Cambridge CB3 0HE, United Kingdom

 (Received 11 March 2024; accepted 13 May 2024; published 27 June 2024)

The goal of quantum metrology is to improve measurements' sensitivities by harnessing quantum resources. Metrologists often aim to maximize the quantum Fisher information, which bounds the measurement setup's sensitivity. In studies of fundamental limits on metrology, a paradigmatic setup features a qubit (spin-half system) subject to an unknown rotation. One obtains the maximal quantum Fisher information about the rotation if the spin begins in a state that maximizes the variance of the rotation-inducing operator. If the rotation axis is unknown, however, no optimal single-qubit sensor can be prepared. Inspired by simulations of closed timelike curves, we circumvent this limitation. We obtain the maximum quantum Fisher information about a rotation angle, regardless of the unknown rotation axis. To achieve this result, we initially entangle the probe qubit with an ancilla qubit. Then, we measure the pair in an entangled basis, obtaining more information about the rotation angle than any single-qubit sensor can achieve. We demonstrate this metrological advantage using a two-qubit superconducting quantum processor. Our measurement approach achieves a quantum advantage, outperforming every entanglement-free strategy.

DOI: [10.1103/PhysRevLett.132.260801](https://doi.org/10.1103/PhysRevLett.132.260801)

What makes this an interesting problem?

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3. Unusual public interest

What makes this an interesting problem?

3. Unusual public interest

Inquiry regarding your research on retrocausal quantum communication

External

Inbox x



[Redacted sender name]

to me ▼

Tue 23 Jun, 06:31



Dear Mr. Kaiyuan Ji,

[Redacted email body content]

Do you believe the time will soon come when I might be able to send a message to my past self—specifically to May 2023? There are such critical and defining moments in our lives where a message from our future self could provide the exact, necessary guidance to make the right choice. I will follow the progress of your scientific research with great interest.

What makes this an interesting problem?

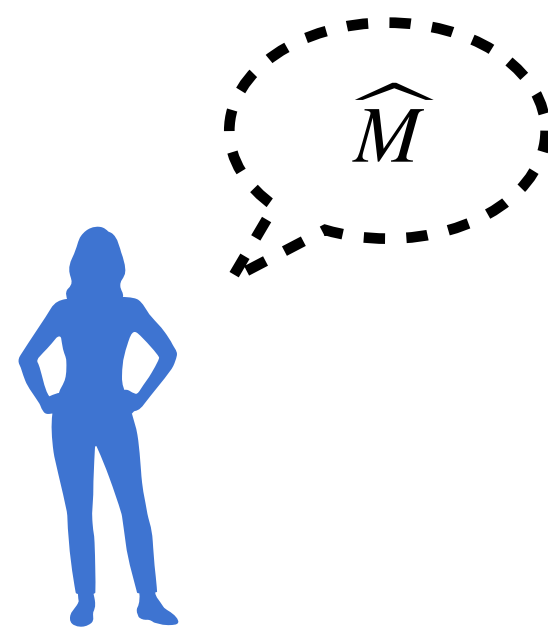
What makes this an interesting problem?

4. Fictional inspiration

What makes this an interesting problem?

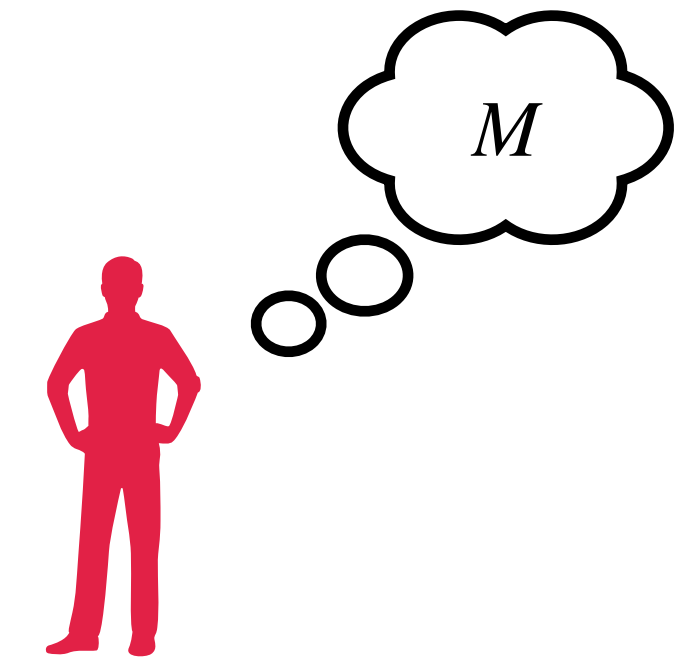
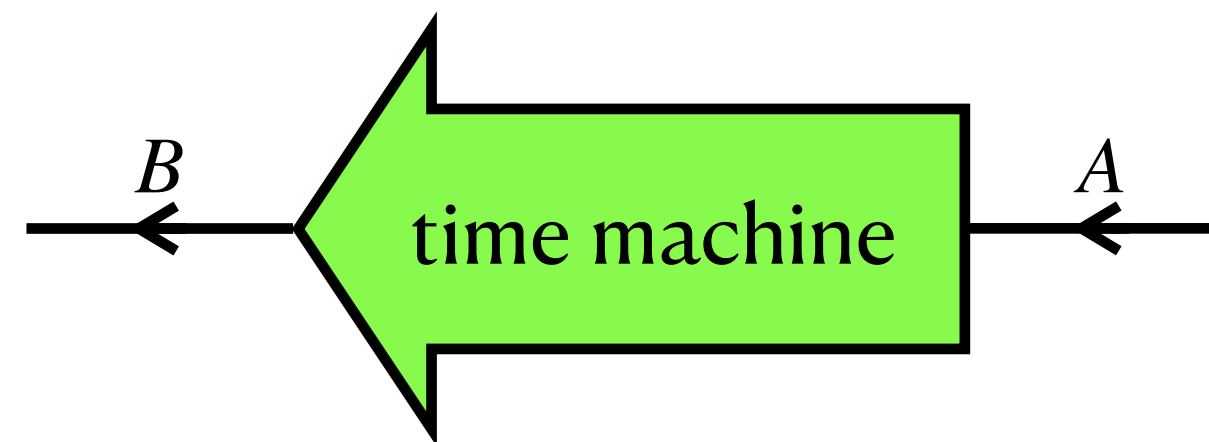
4. Fictional inspiration

receiver (past)



time
→

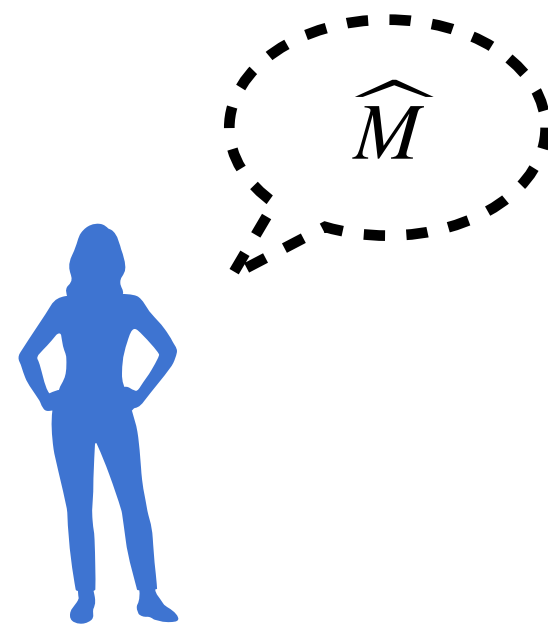
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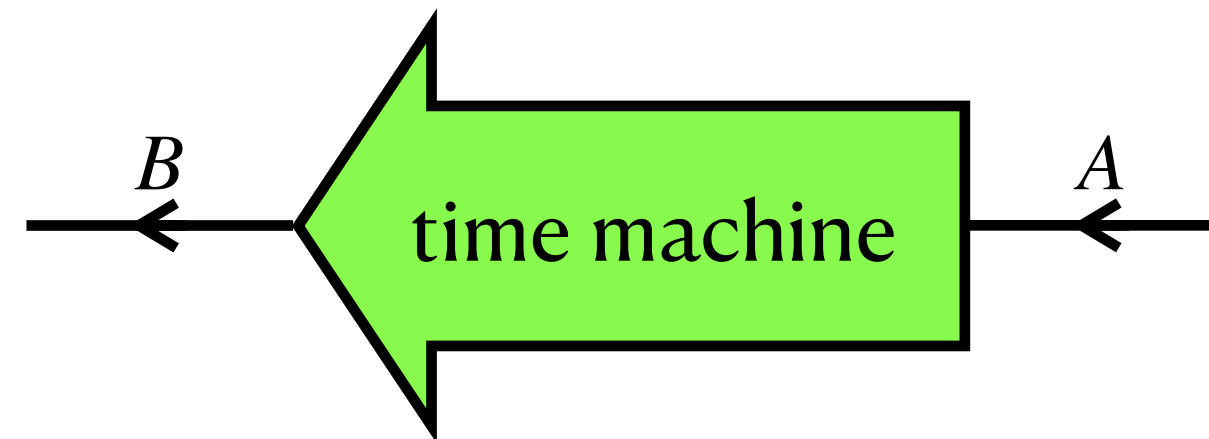
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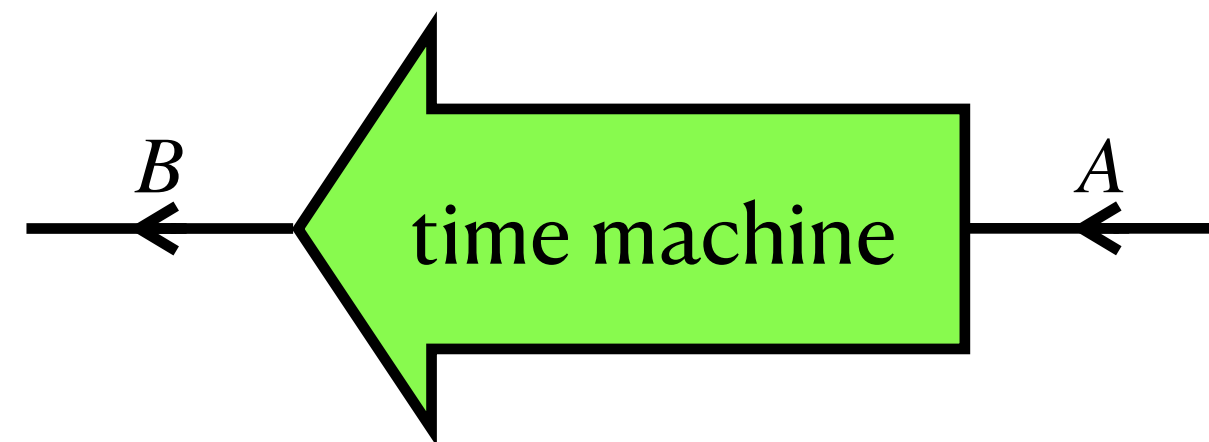
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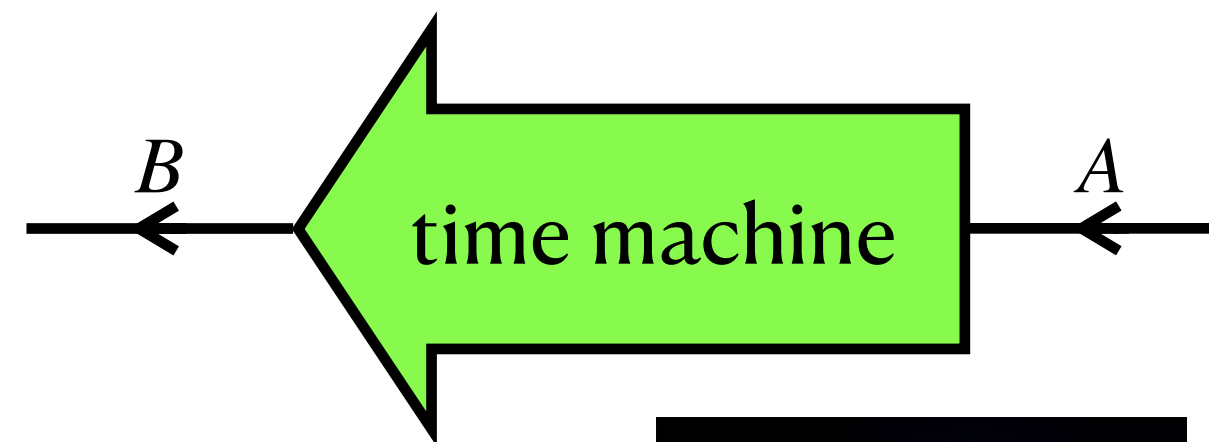
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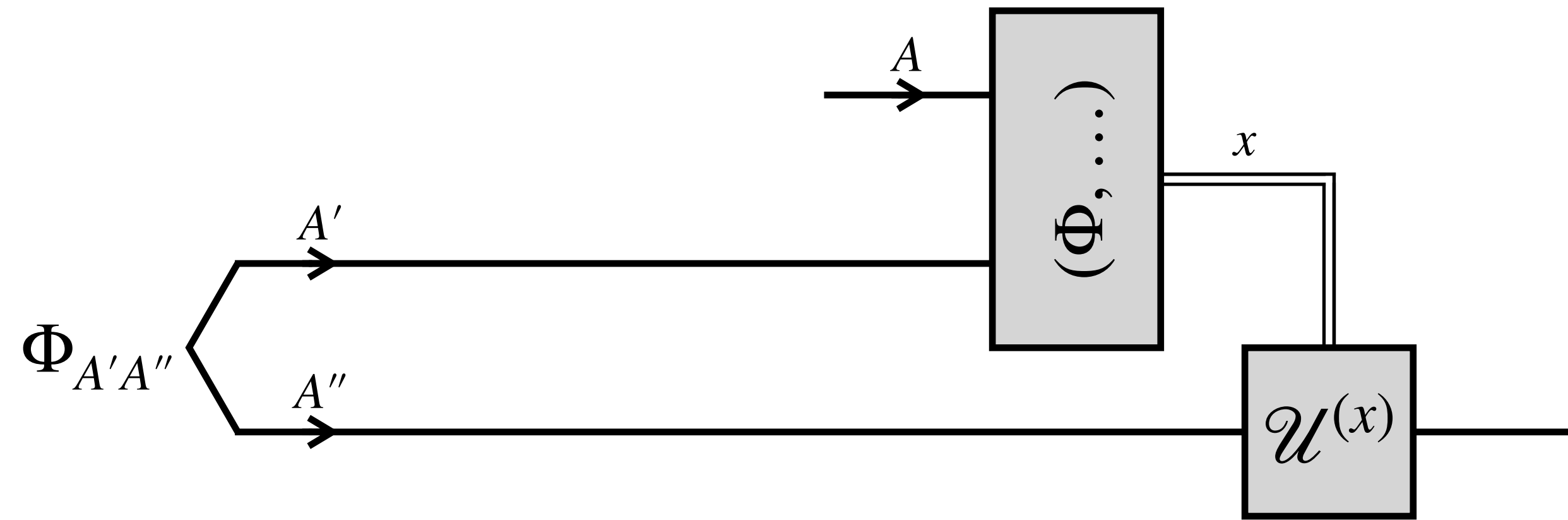
Preliminaries

Time machine - 101

How to describe a noisy time machine mathematically?

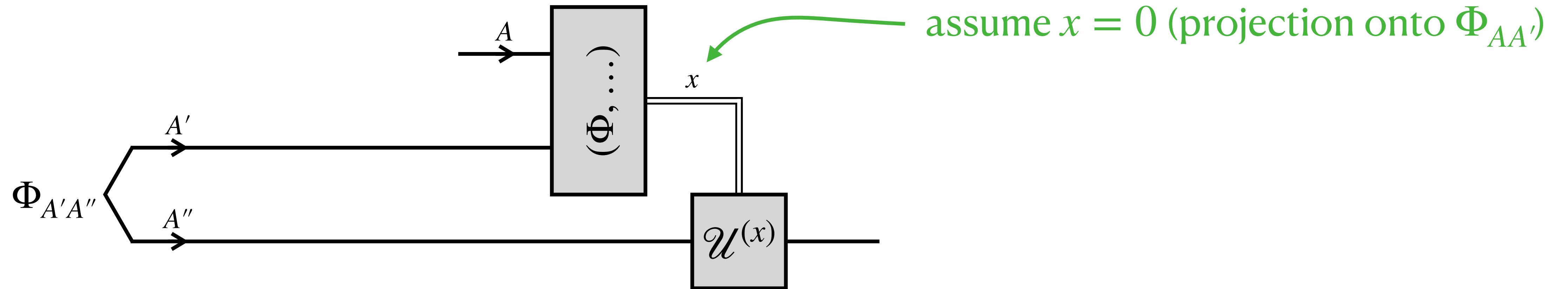
How to describe a noisy time machine mathematically?

Teleportation

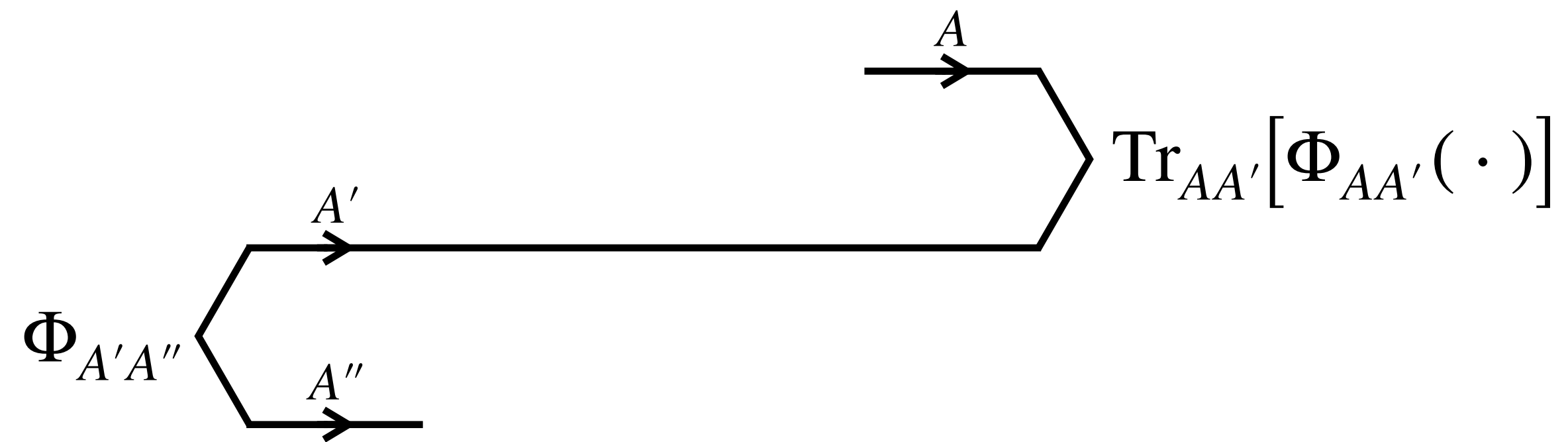


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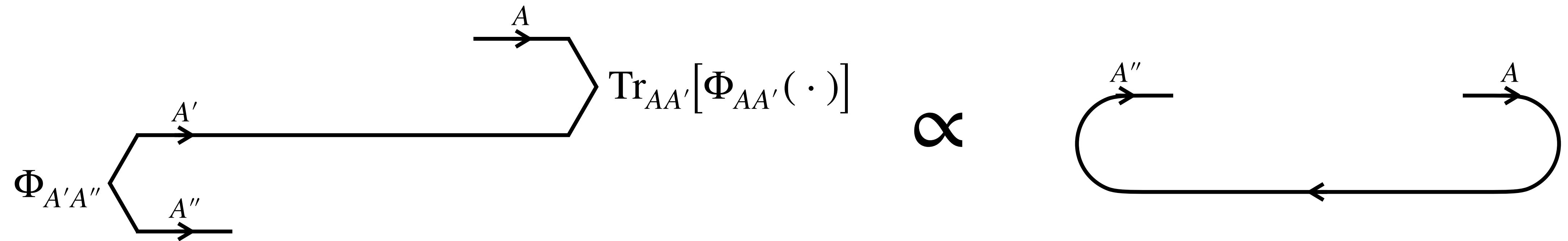
Teleportation



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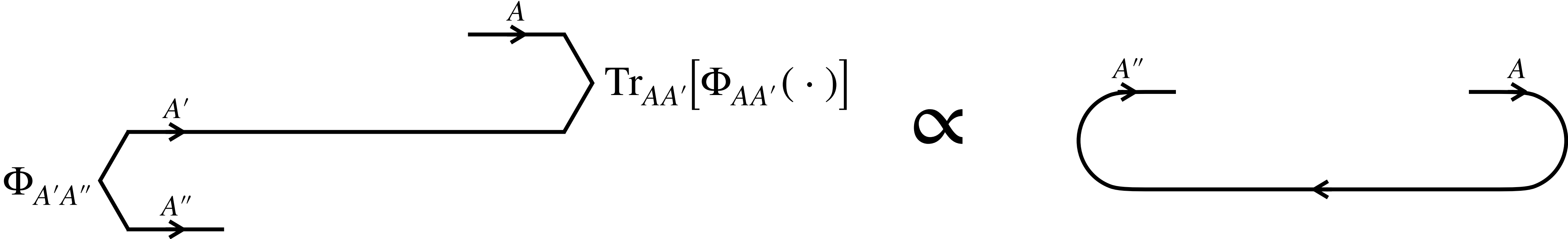
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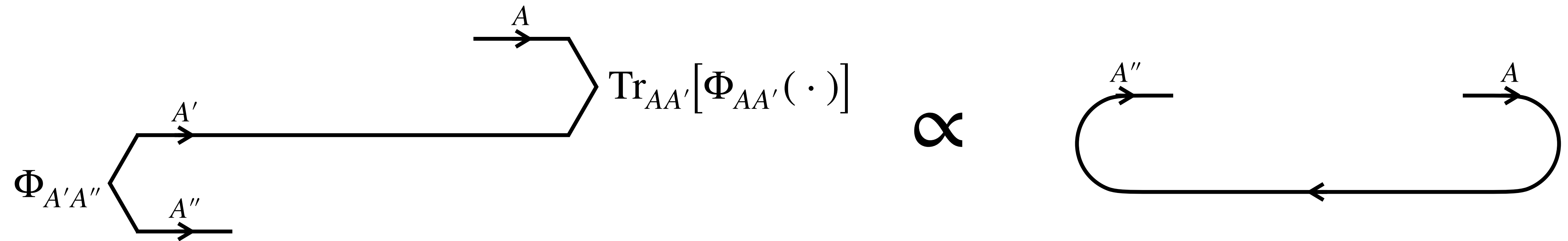
How to describe a noisy time machine mathematically?

“Postselected” closed timelike curve (P-CTC)

[Bennett & Schumacher 2002 (unpublished)]
[Svetlichny 2009]
[Lloyd et al. 2011]

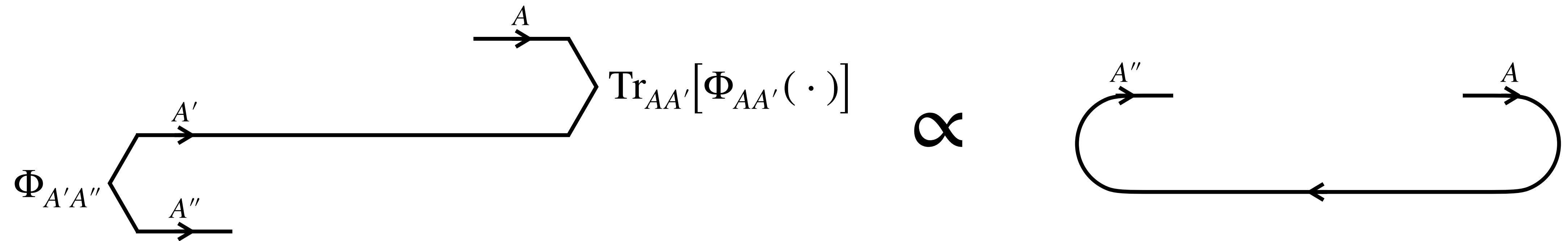


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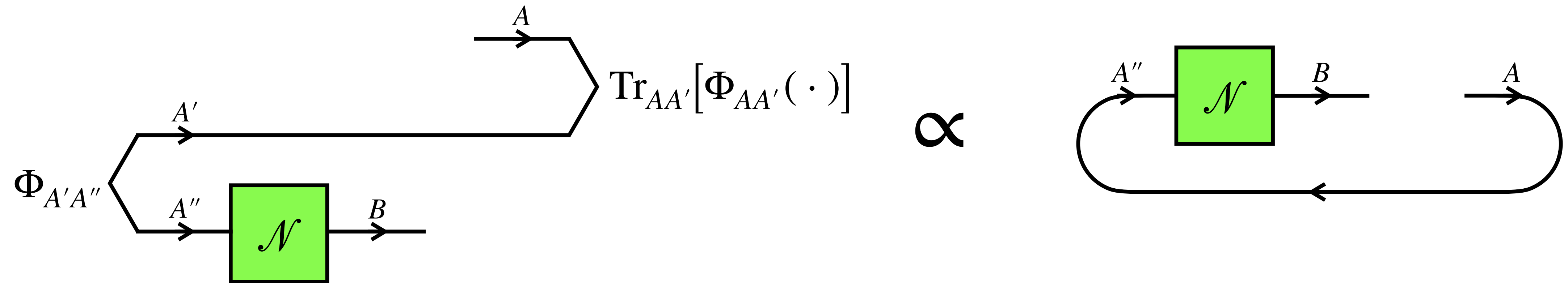
“Noisy” P-CTC?



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“Noisy” P-CTC?

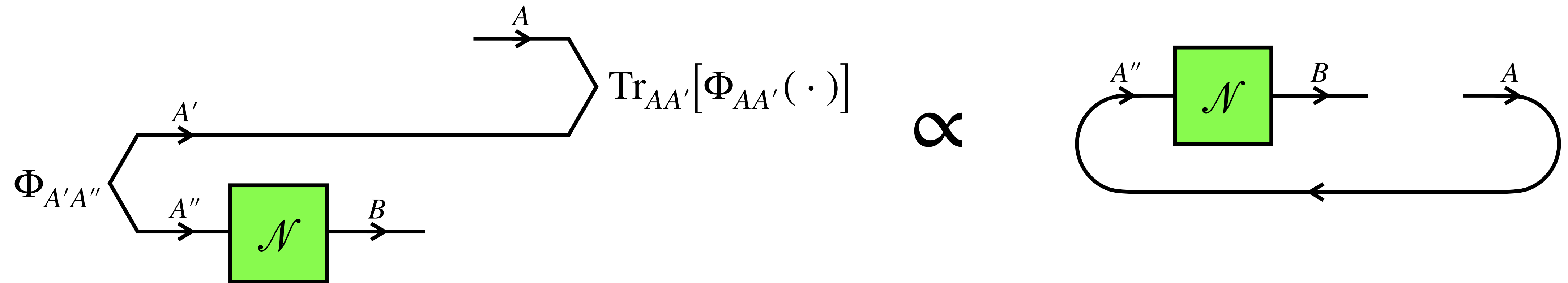
[Genkina et al. 2012]



How to describe a noisy time machine mathematically?

“Noisy” P-CTC?

[Genkina et al. 2012]

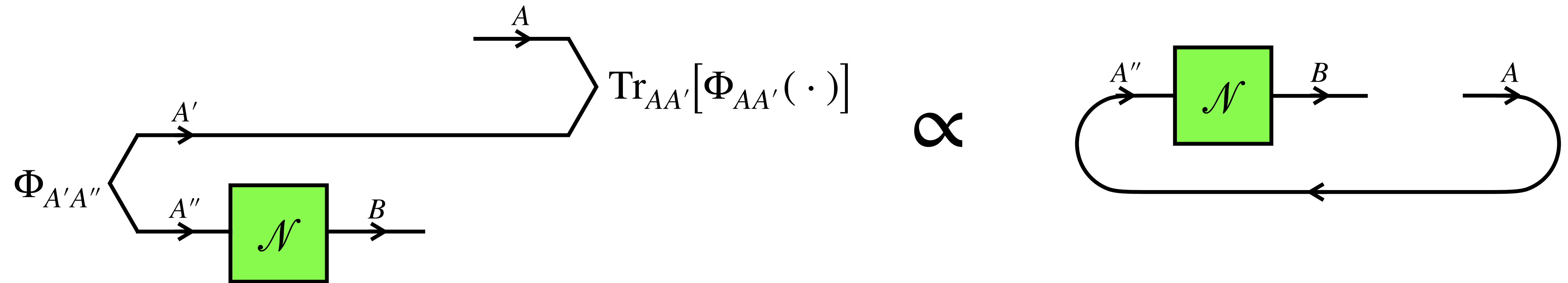


The effect of the noisy P-CTC represented by a CPTP map $\mathcal{N}_{A \rightarrow B}$:

How to describe a noisy time machine mathematically?

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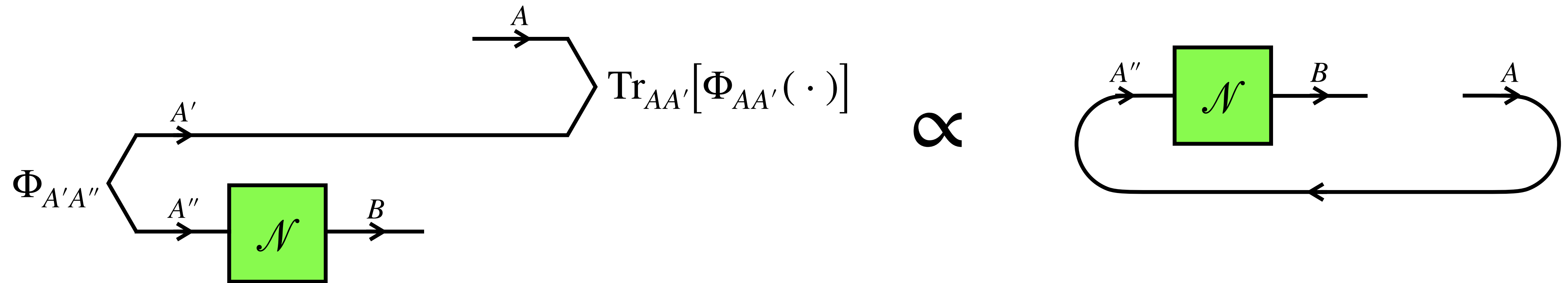
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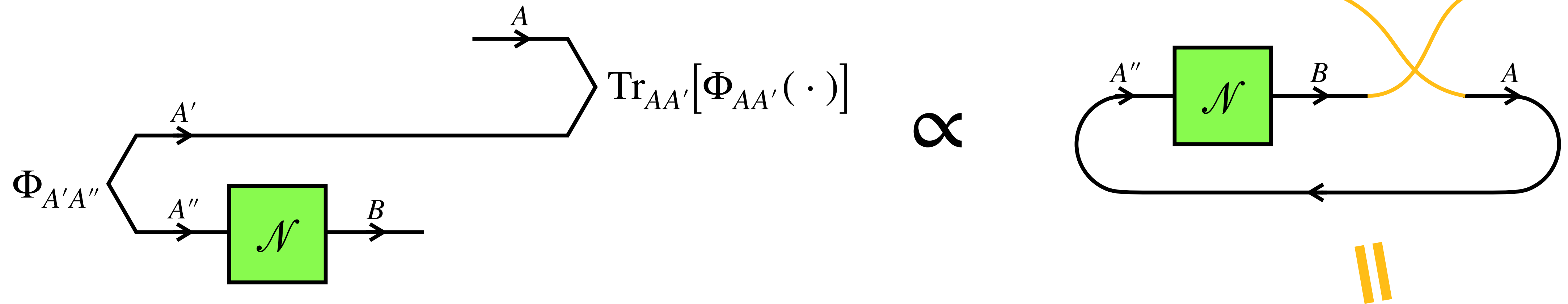
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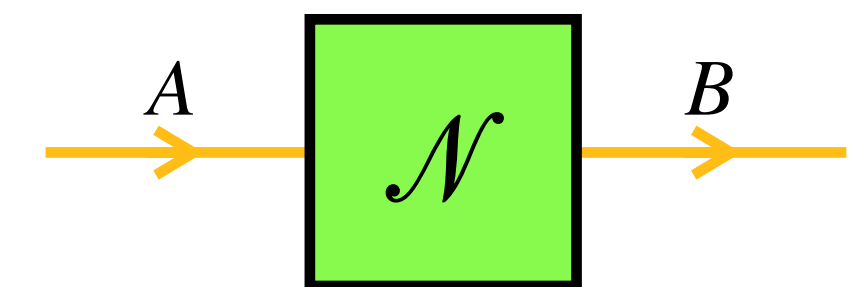
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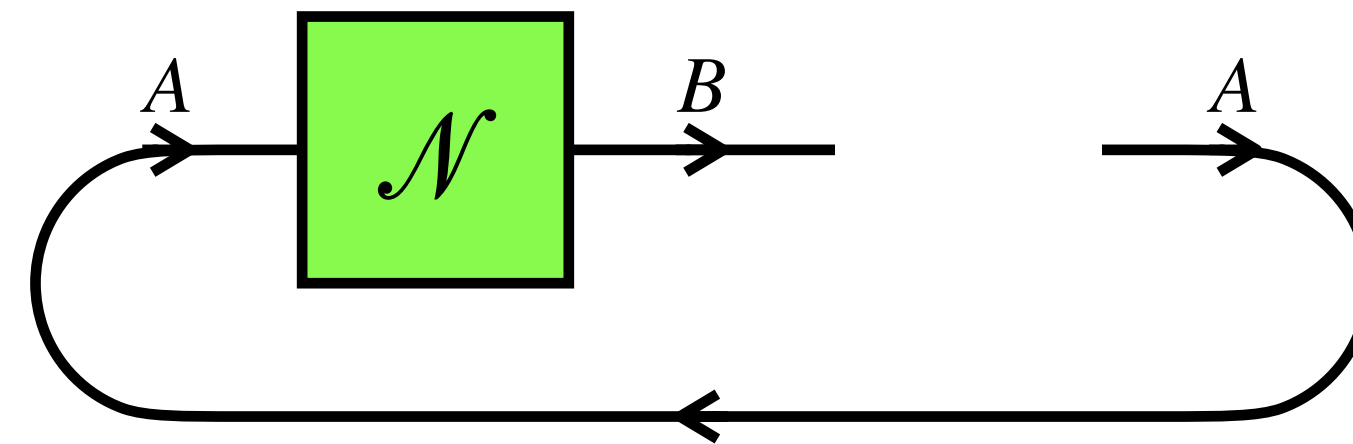
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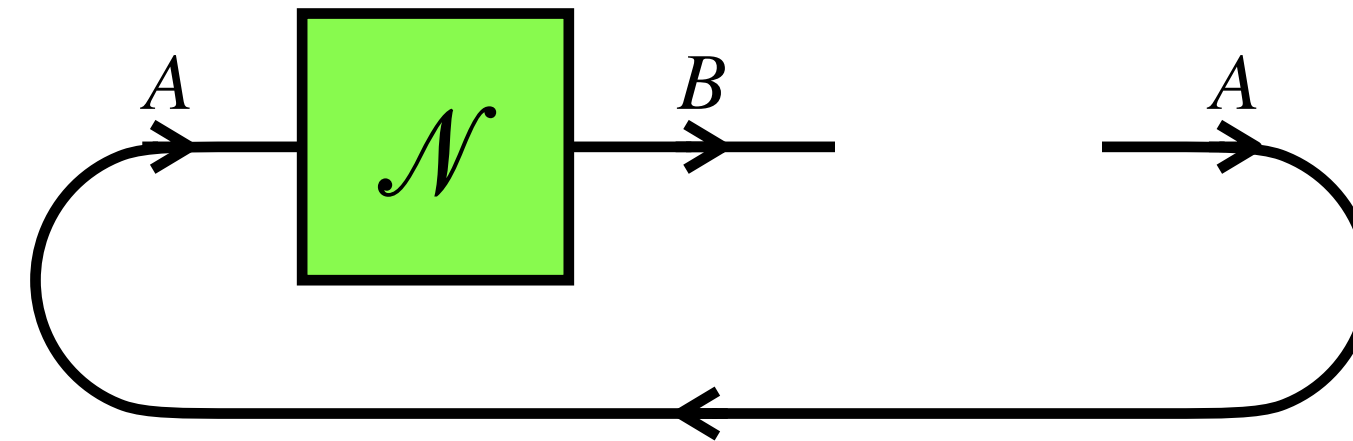
Combine a P-CTC with an ordinary channel

Combine a P-CTC with an ordinary channel



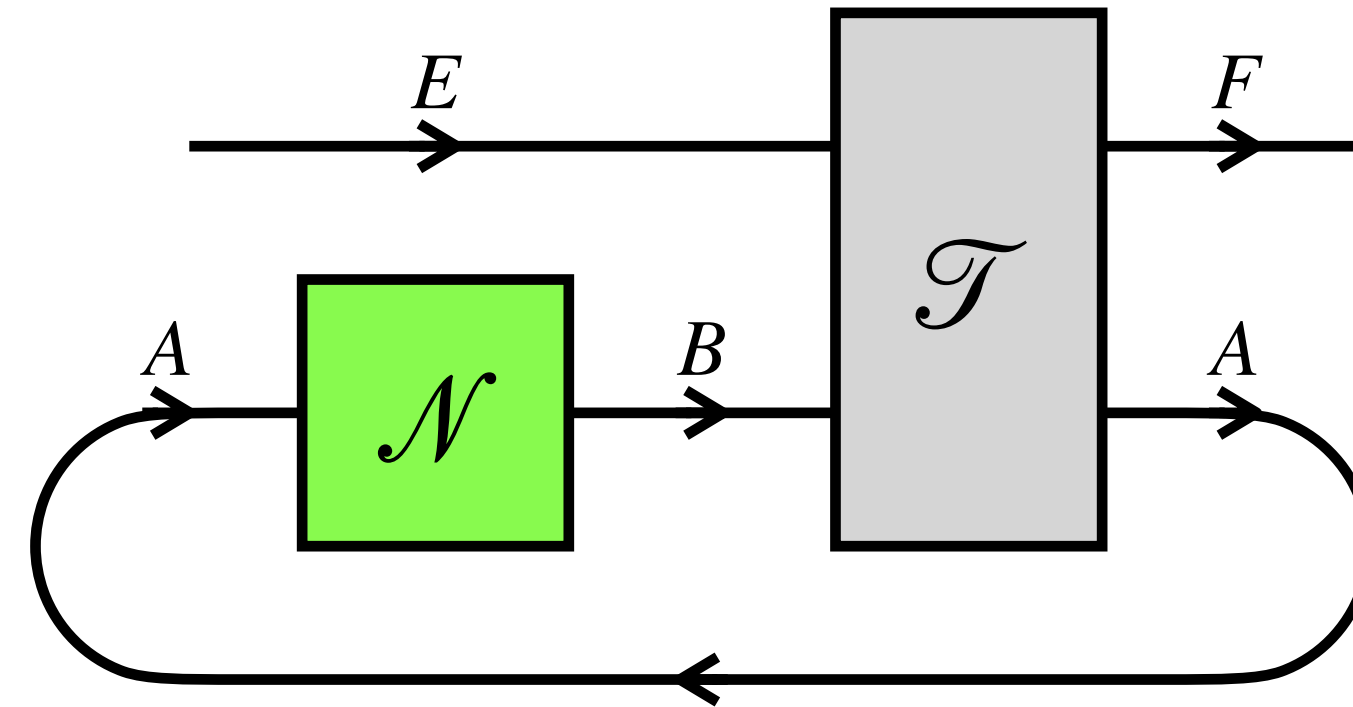
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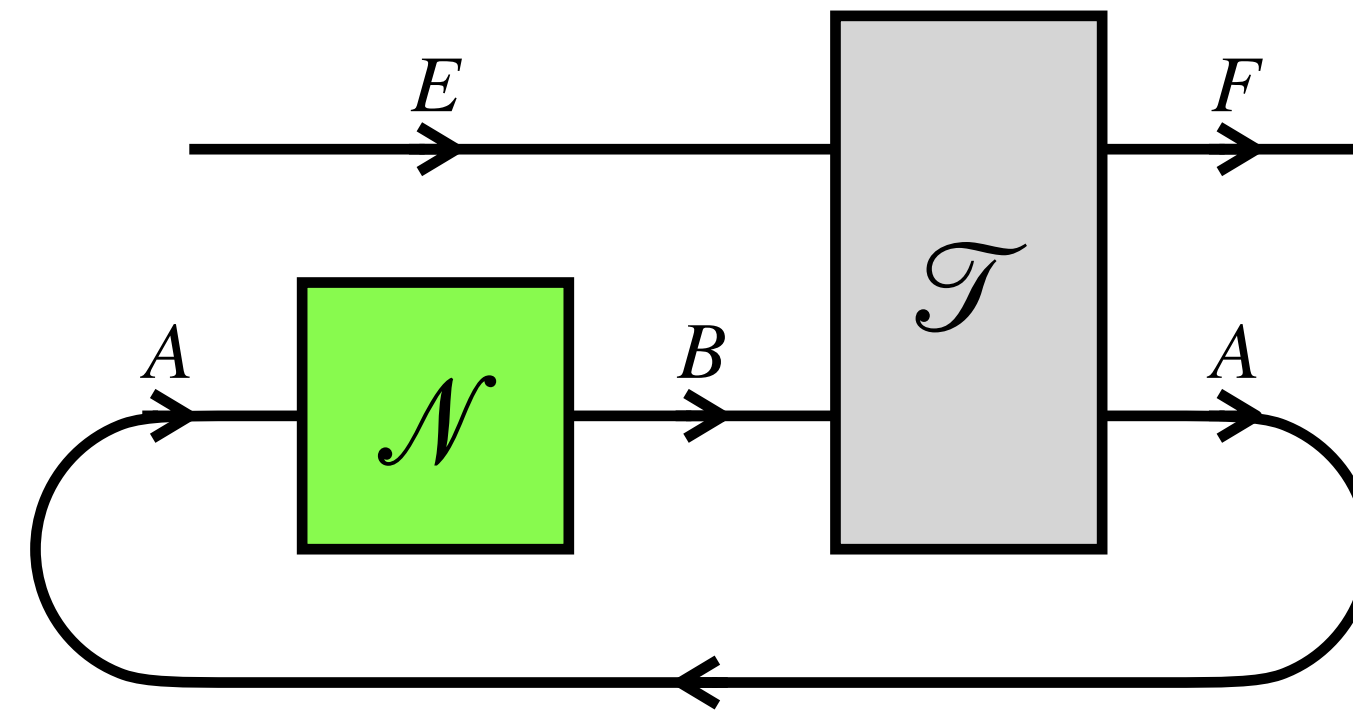
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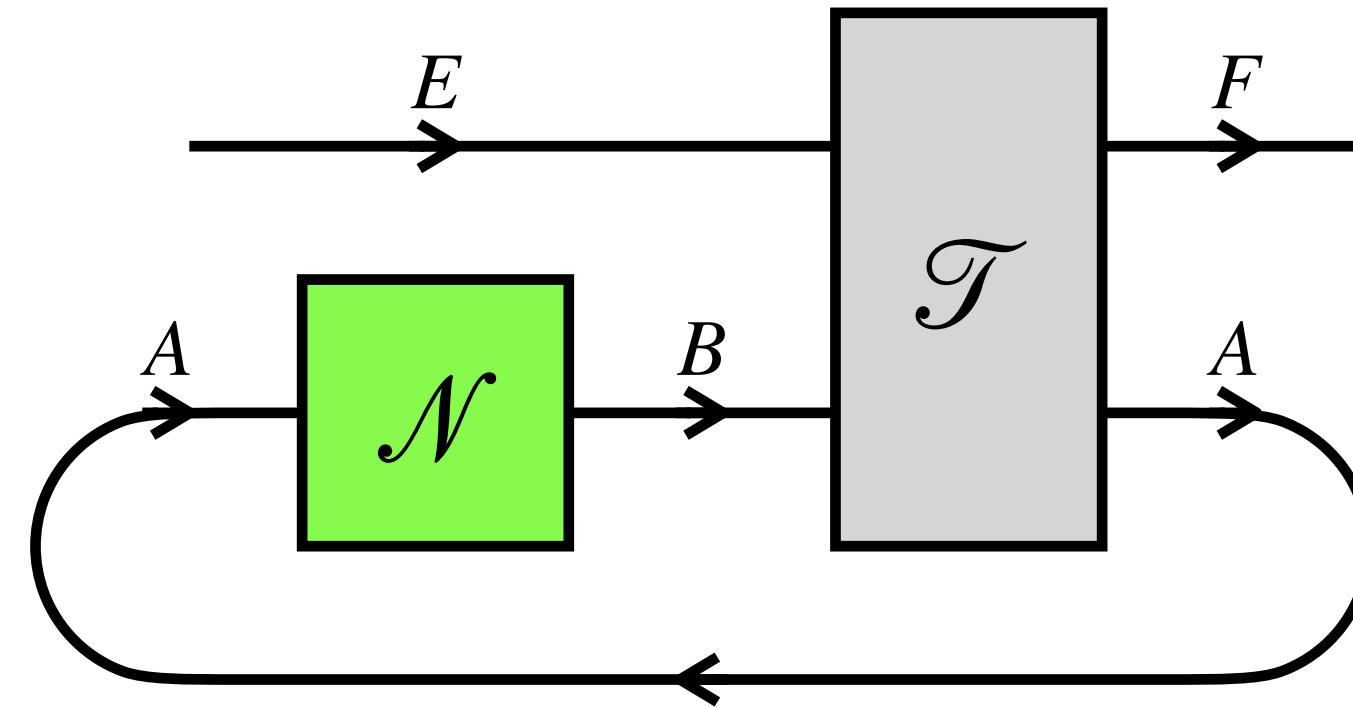


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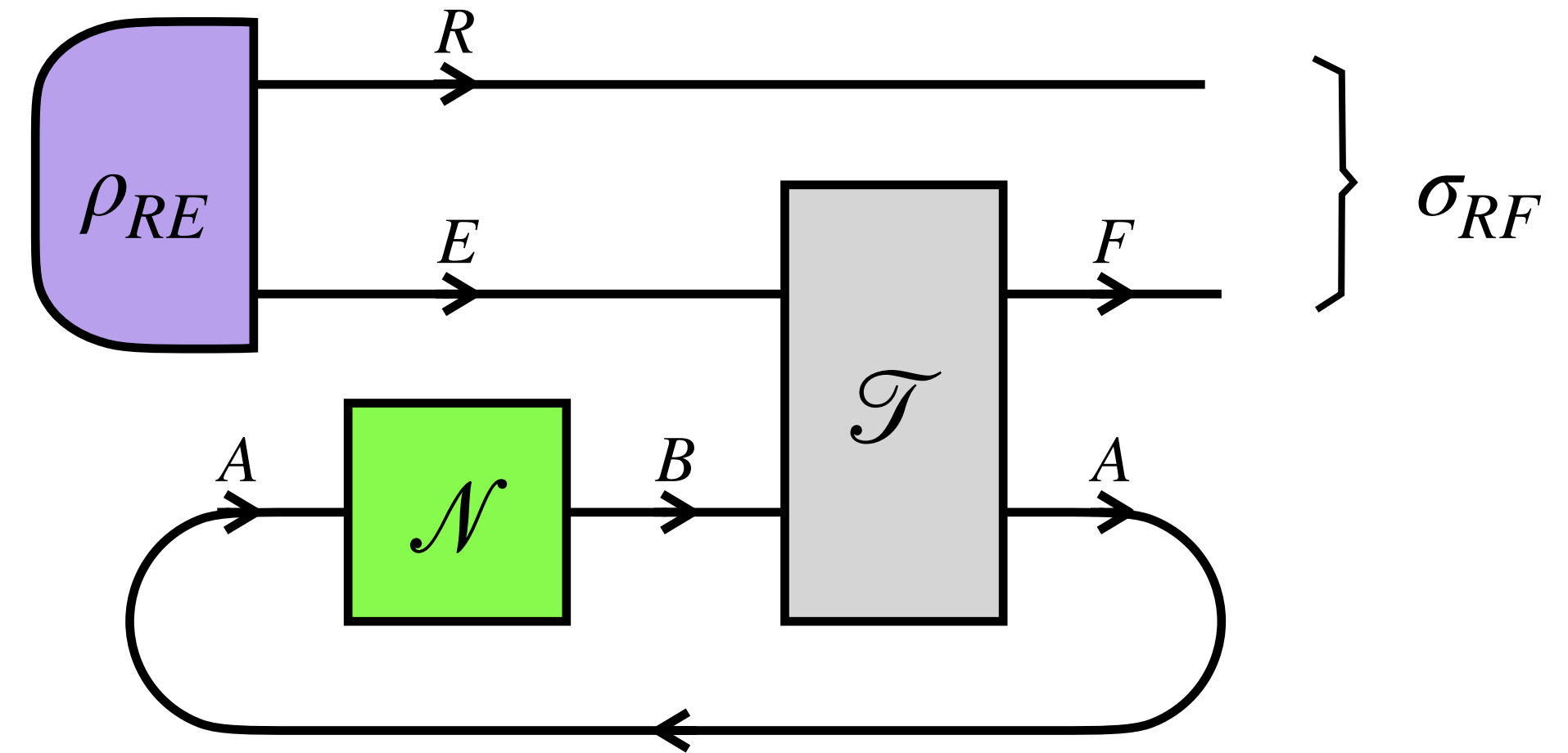


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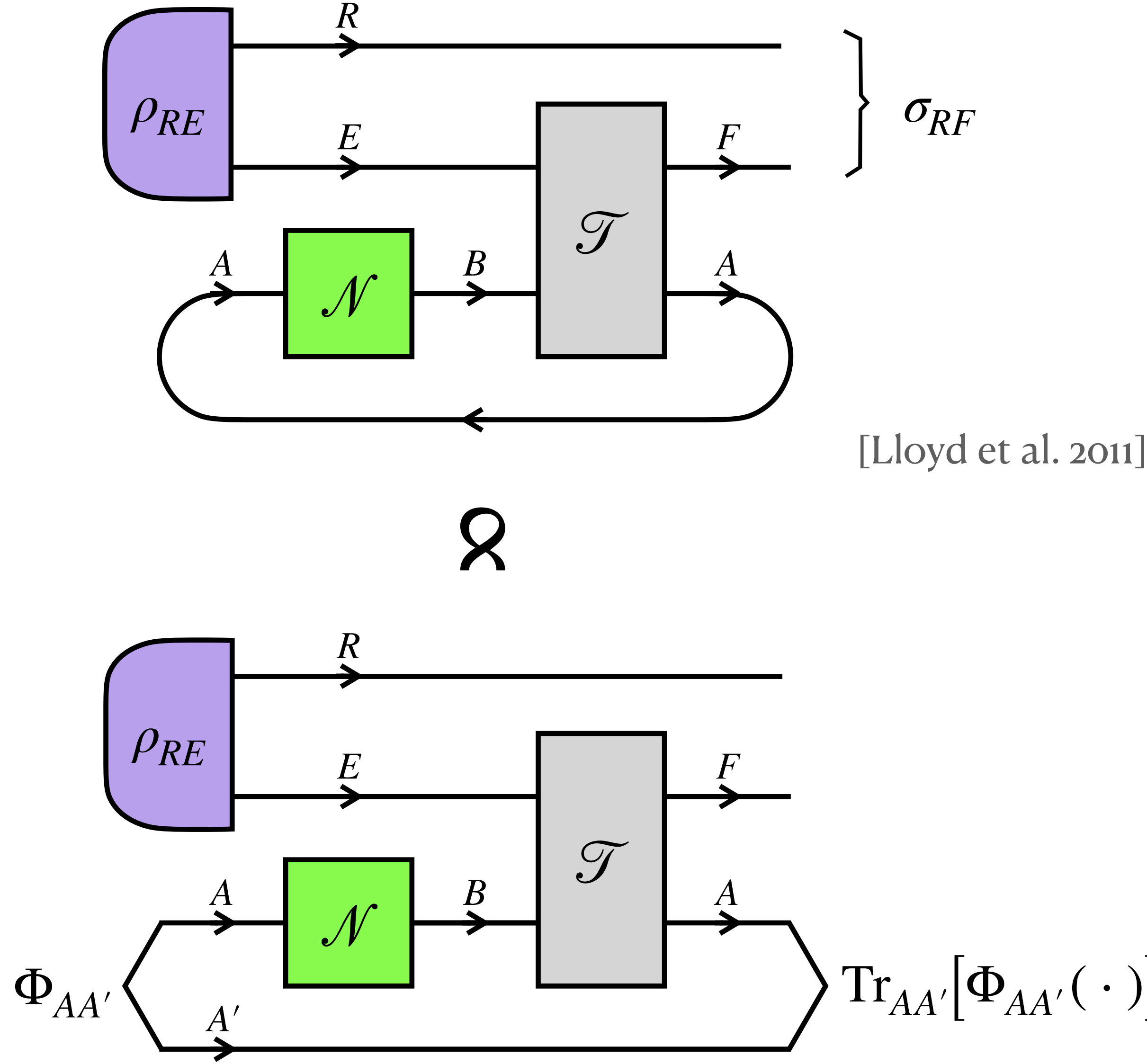


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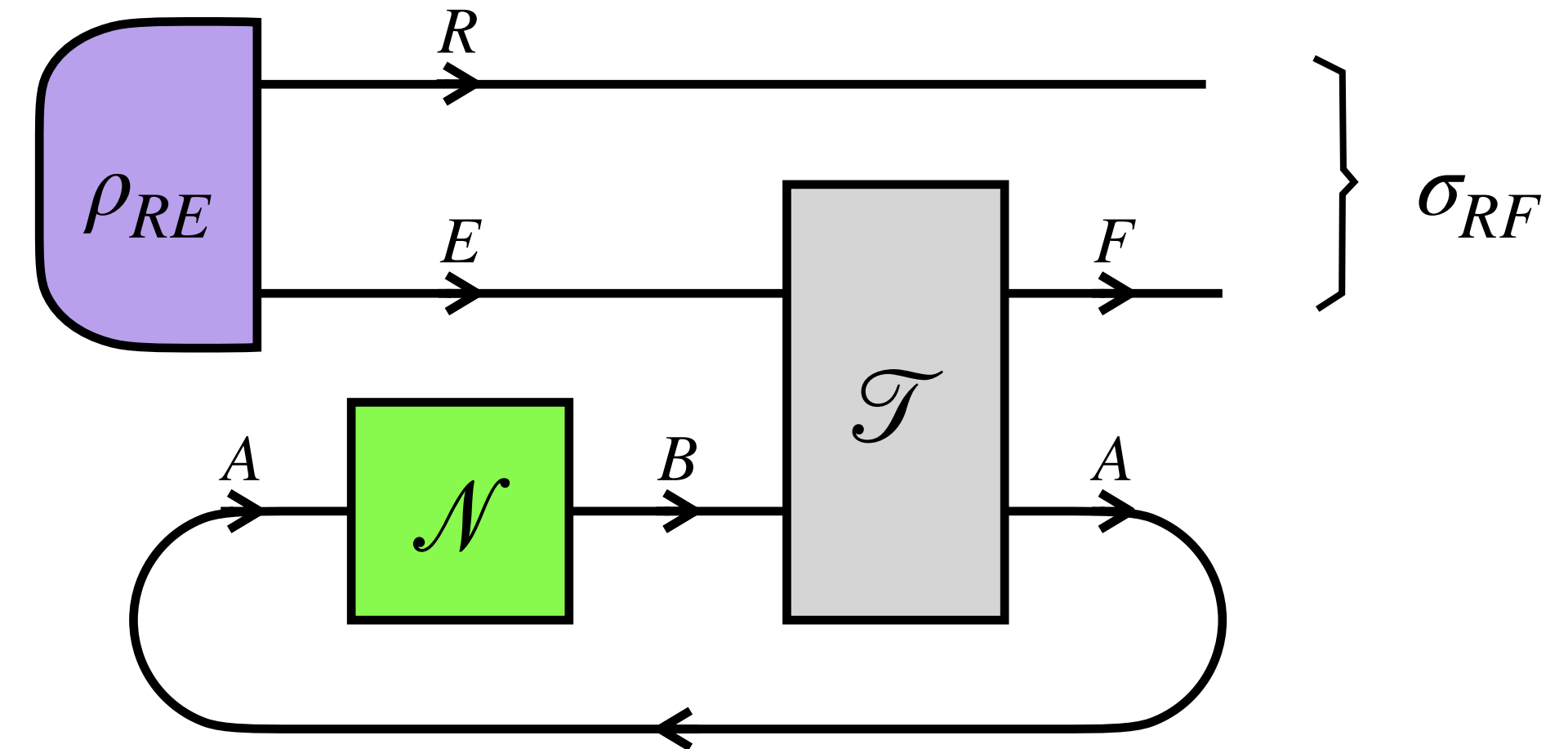


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Compose the noisy P-CTC with $\mathcal{T}_{EB \rightarrow FA}$

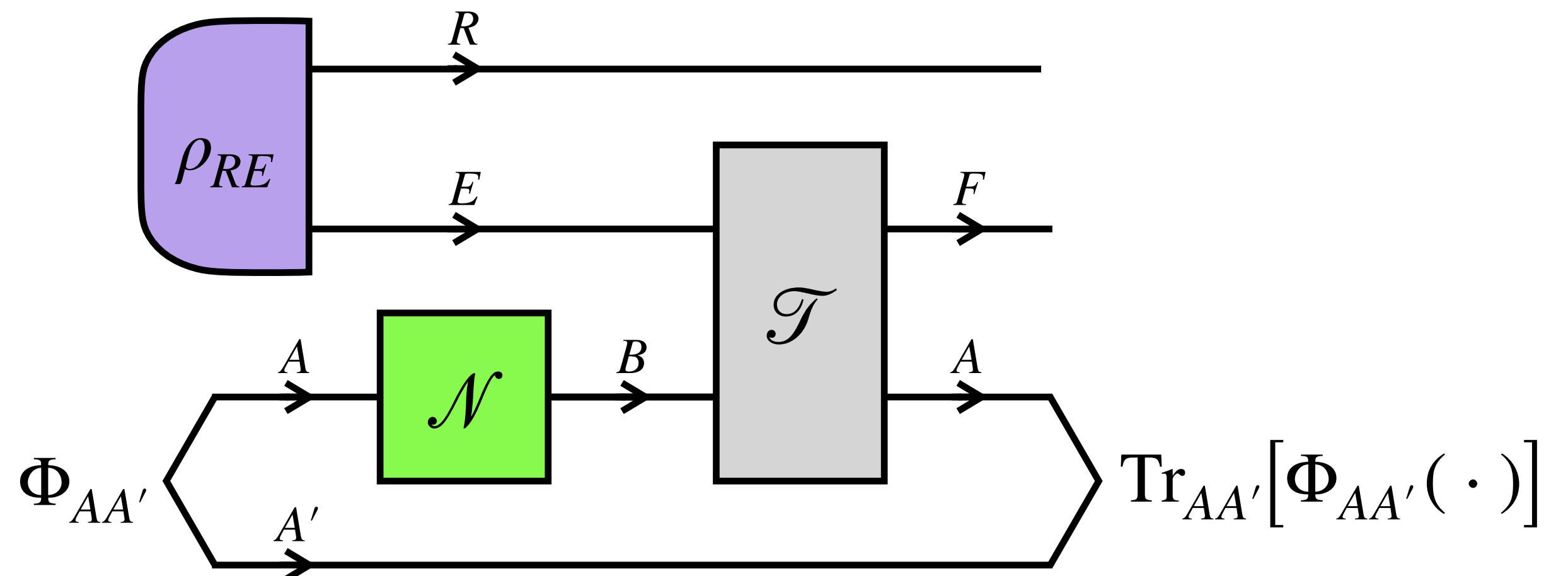
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[Lloyd et al. 2011]

$\bowtie$

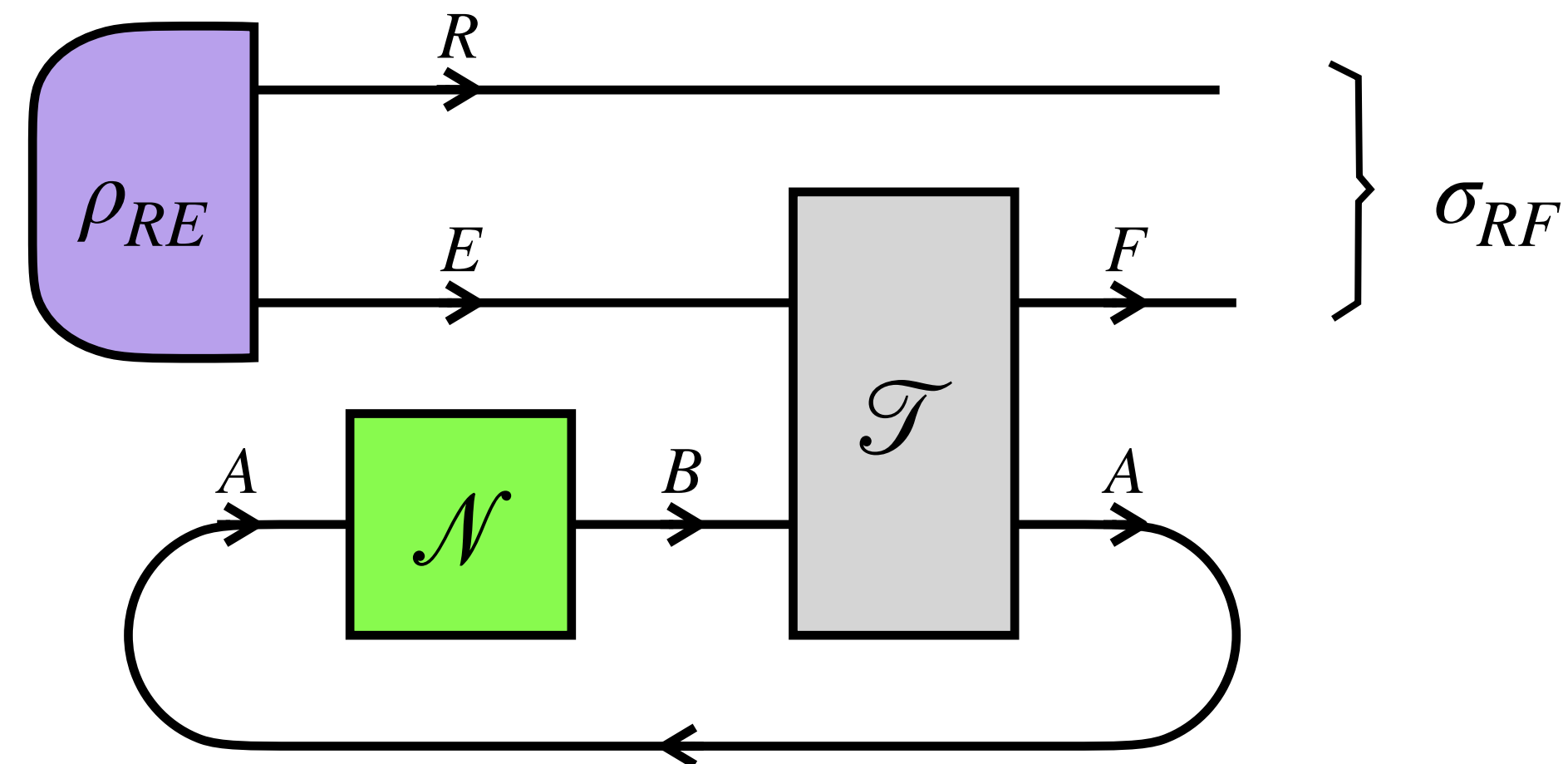


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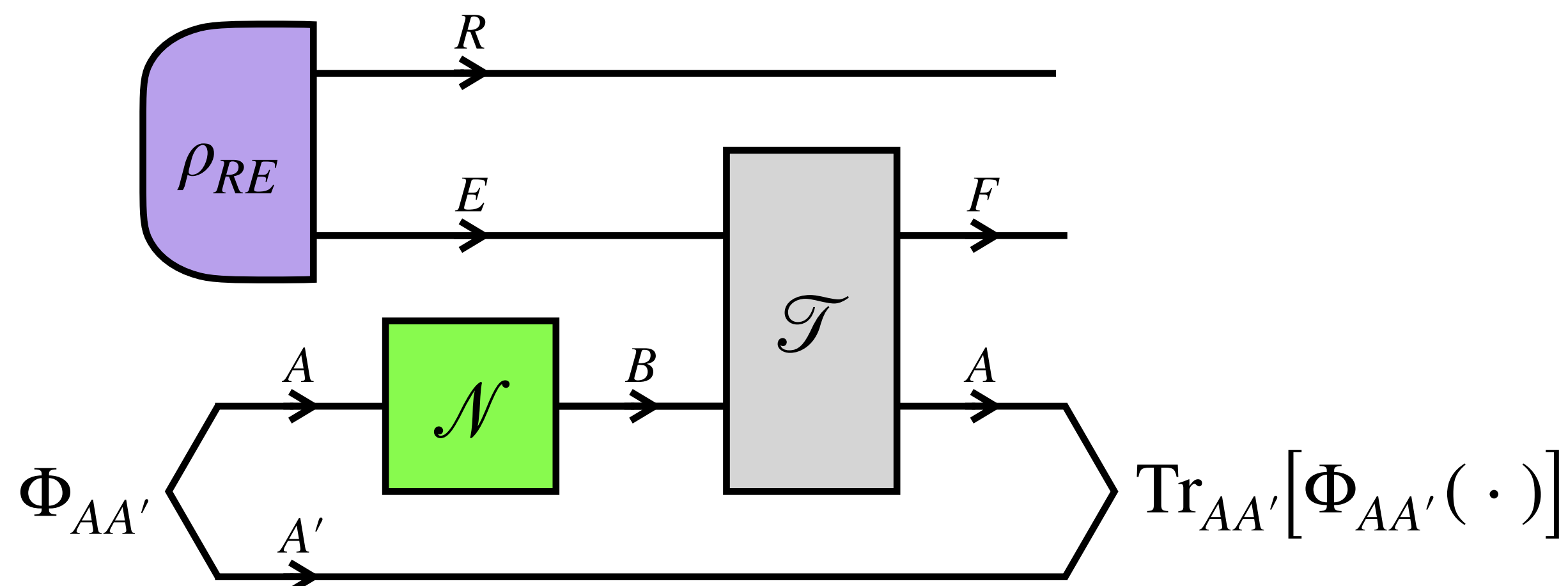
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- Renormalize to make σ_{RF} a state



[Lloyd et al. 2011]

8



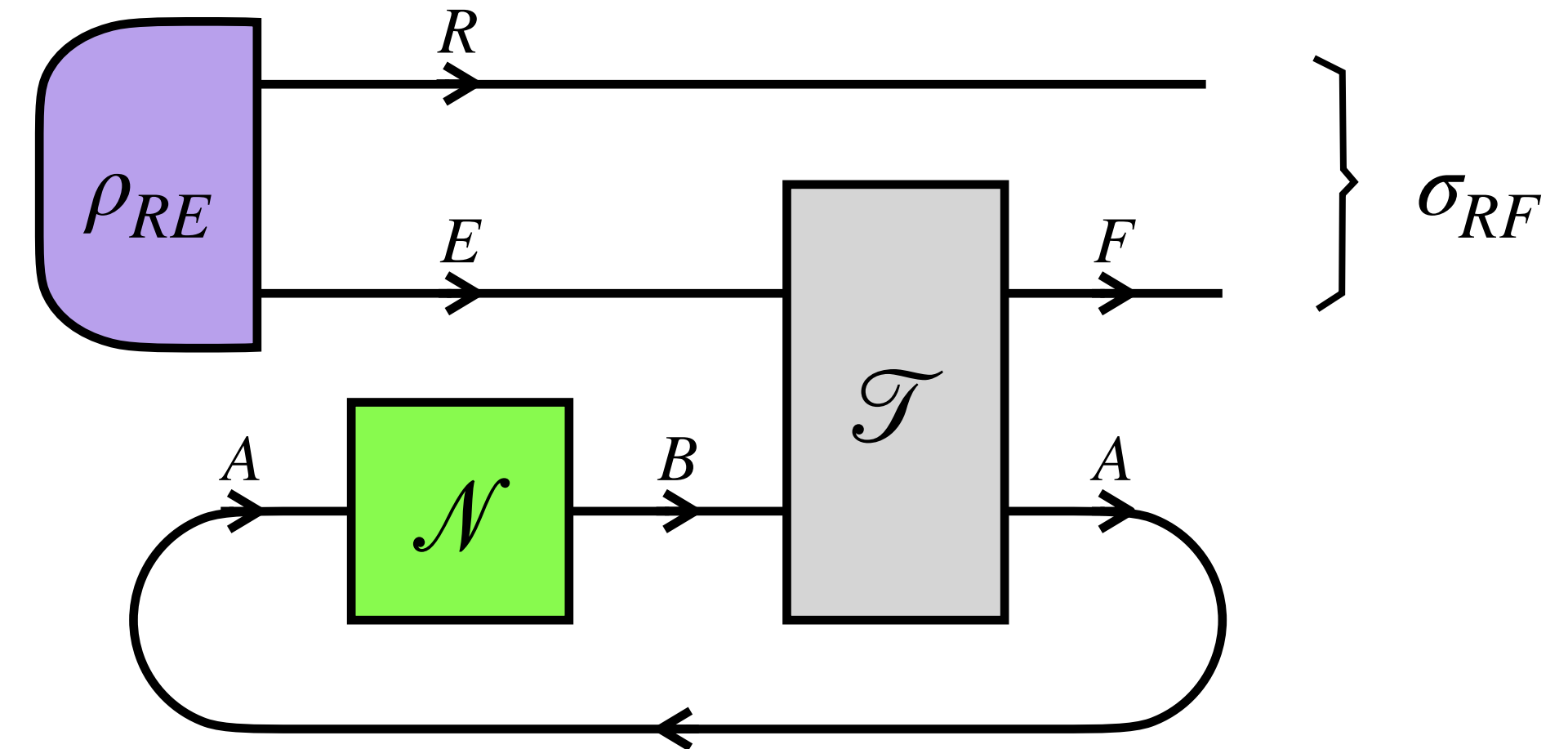
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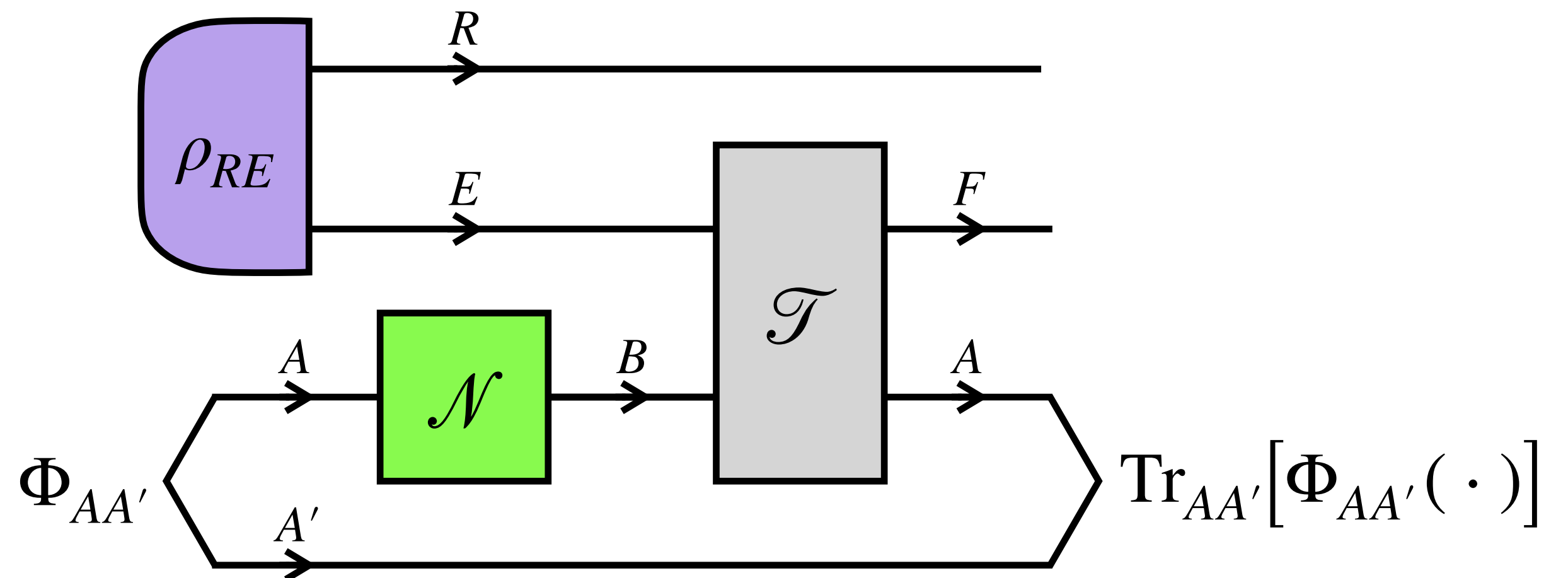
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- The map is **nonlinear**!



[Lloyd et al. 2011]

$\bowtie$



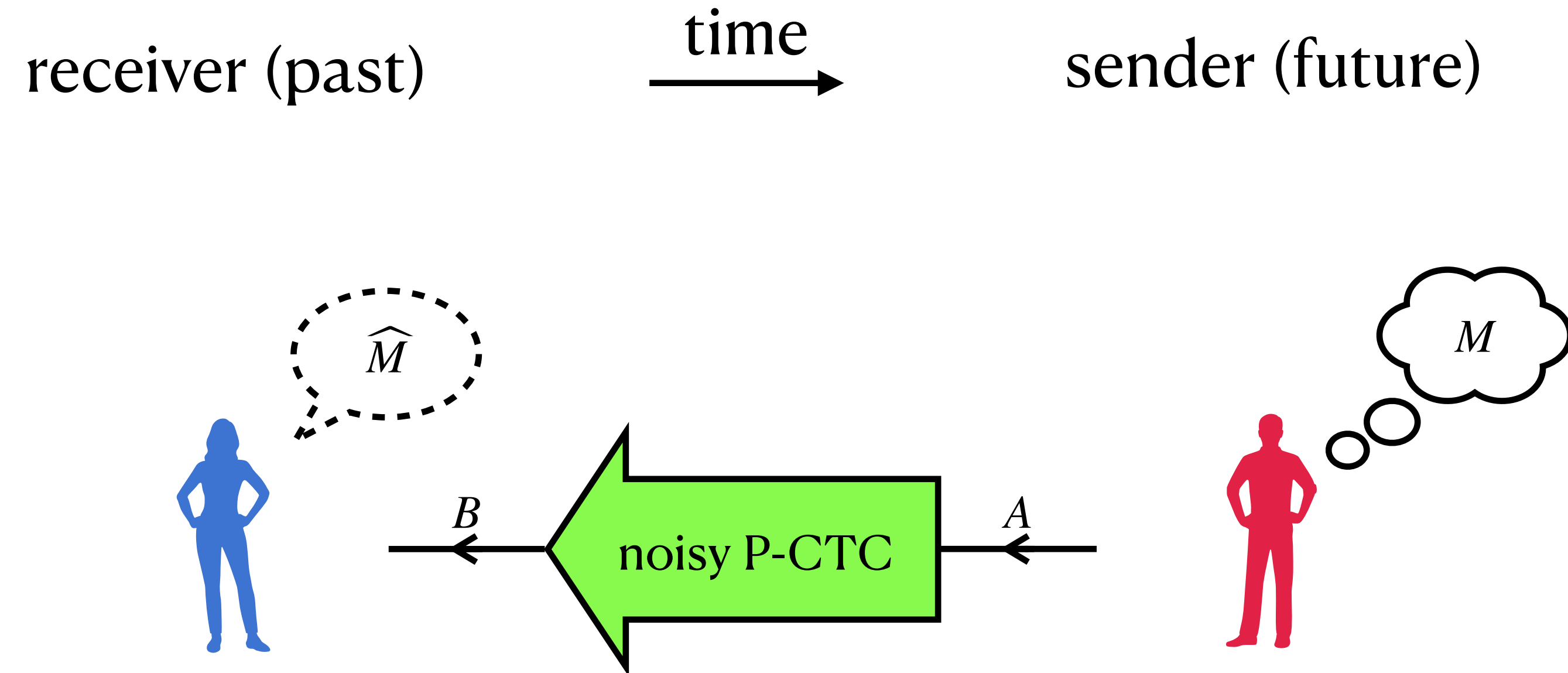
“Retrocausal” communication:

Messaging the past from the future

Retrocausal communication

Setting: a sender in the future is connected to a receiver in the past through a noisy P-CTC

Goal: transmit quantum or classical messages reliably from the sender to the receiver, as efficiently as possible, using whatever strategy **allowed by standard quantum physics**

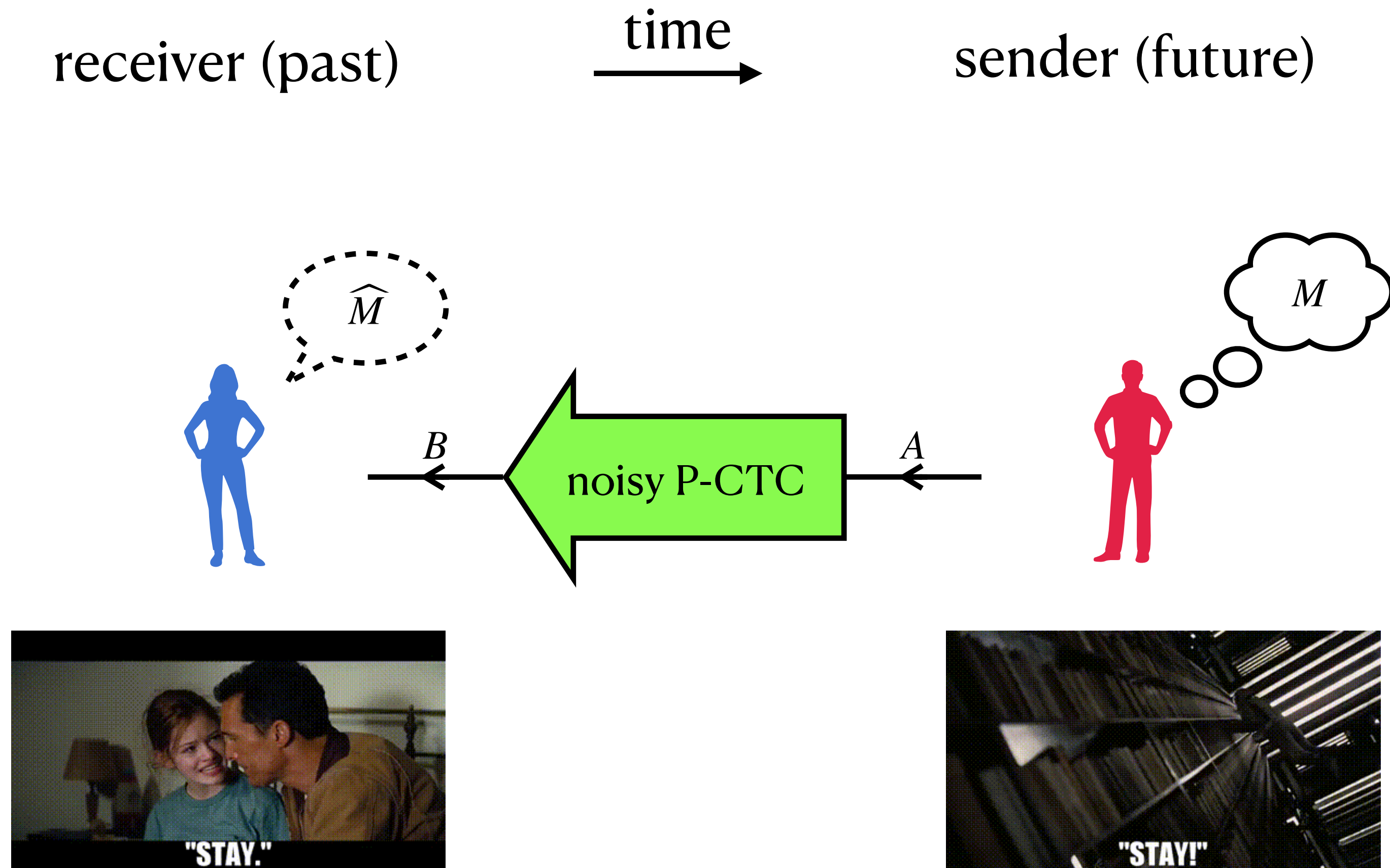


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Exactly as in the film *Interstellar*!

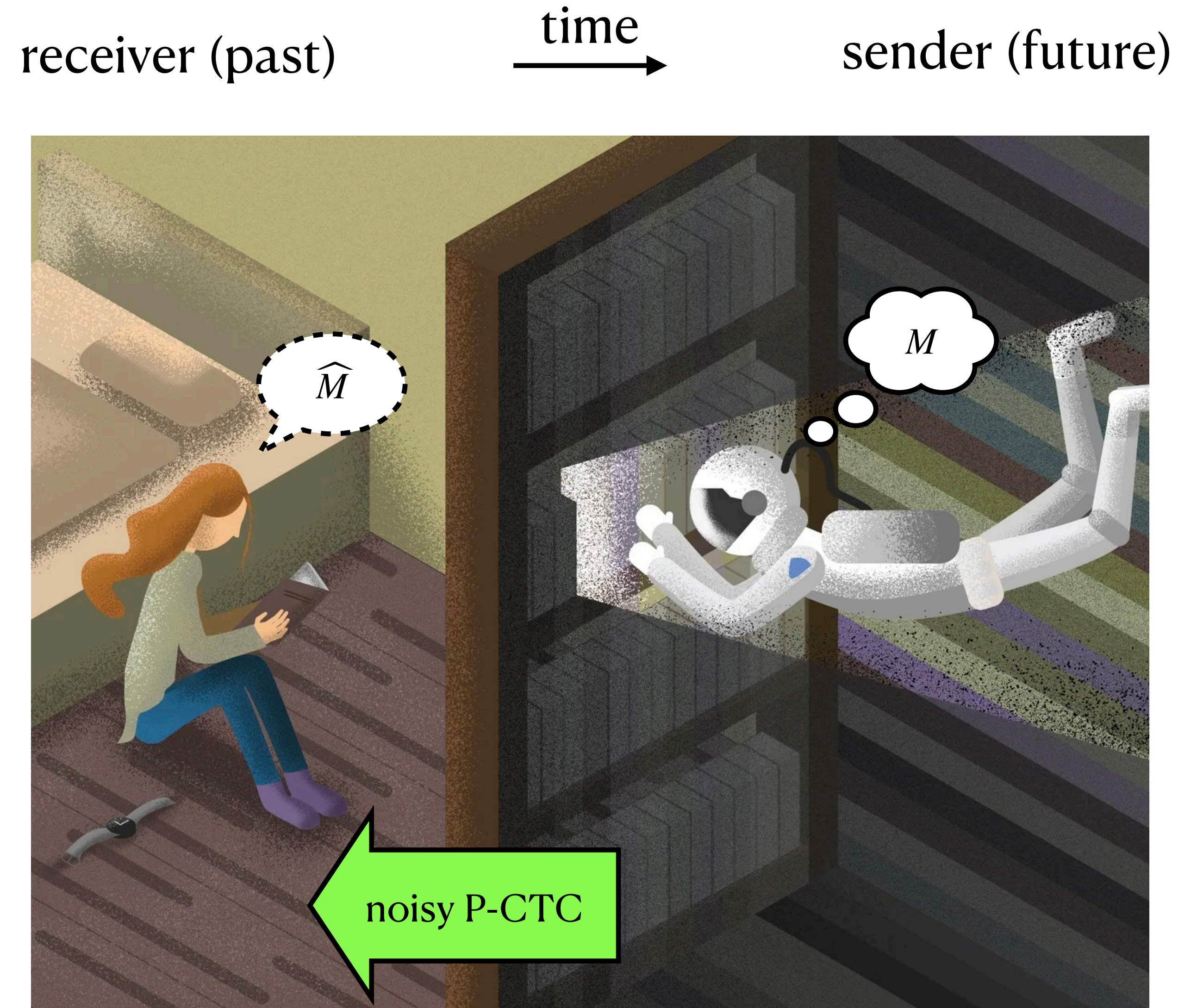


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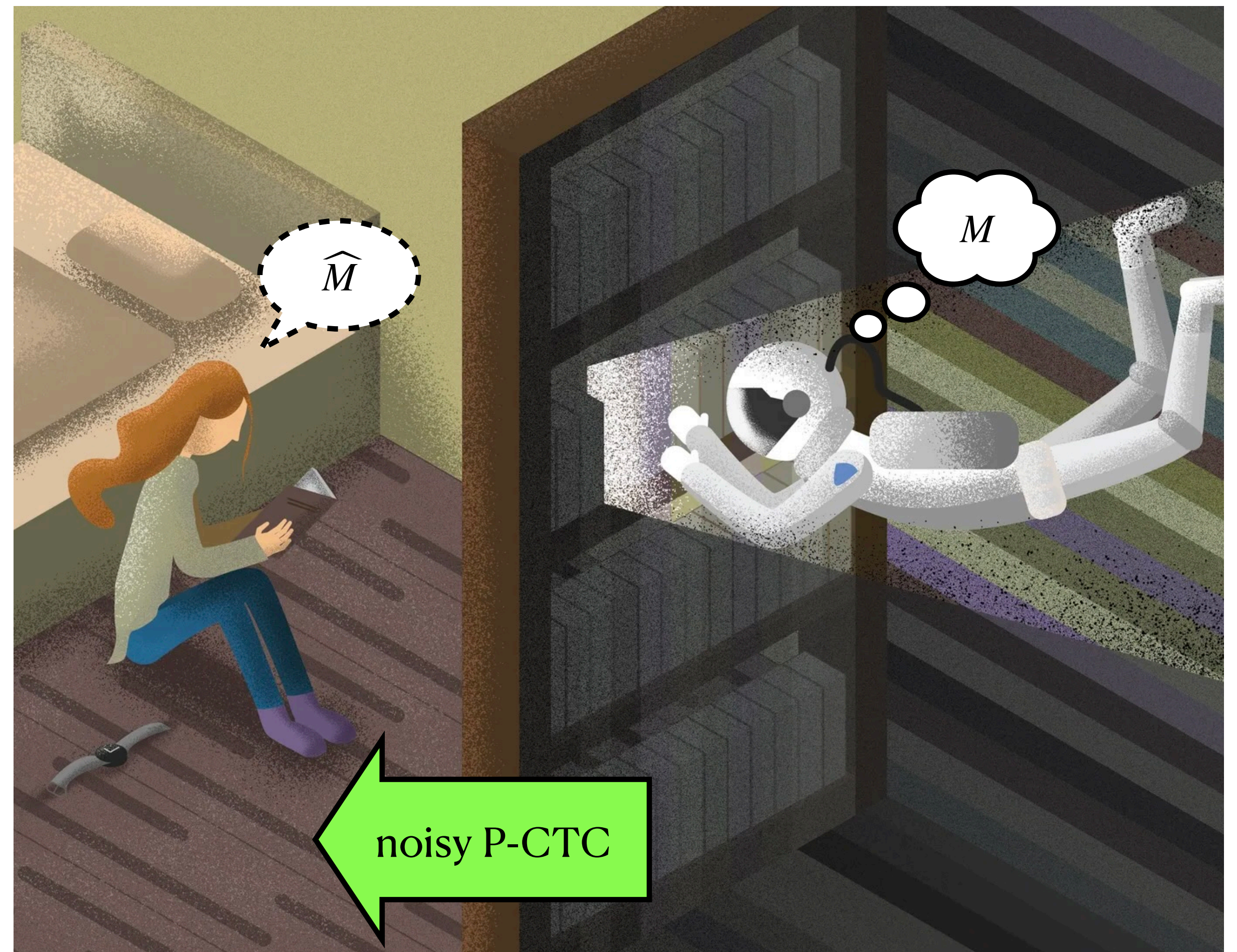
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receiver (past) $\xrightarrow{\text{time}}$ sender (future)



Retrocausal communication

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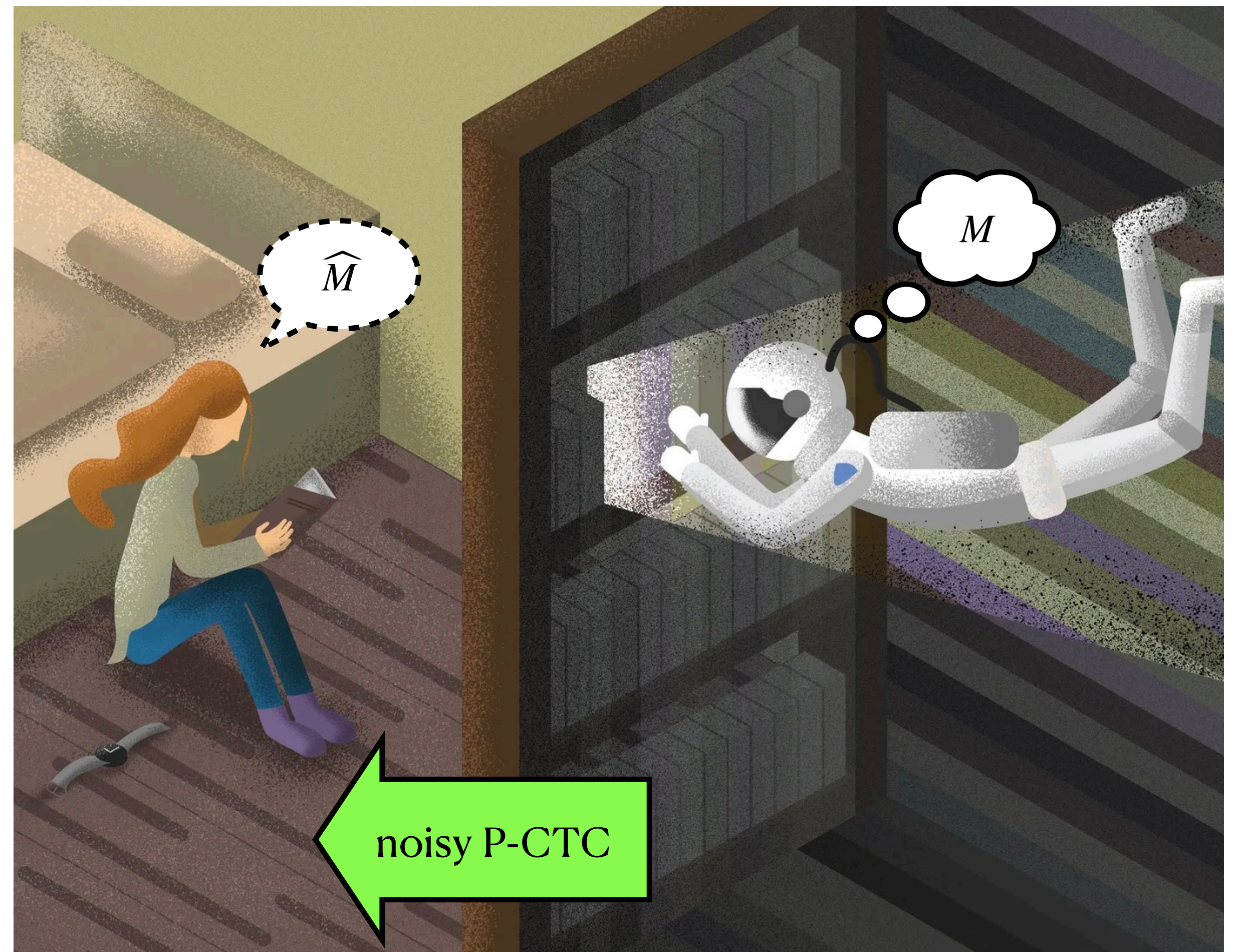
father

daughter

Goal: transmit quantum or classical messages reliably from the sender to the receiver, as efficiently as possible, using whatever strategy **allowed by standard quantum physics**

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receiver (past) $\xrightarrow{\text{time}}$ sender (future)



Retrocausal communication

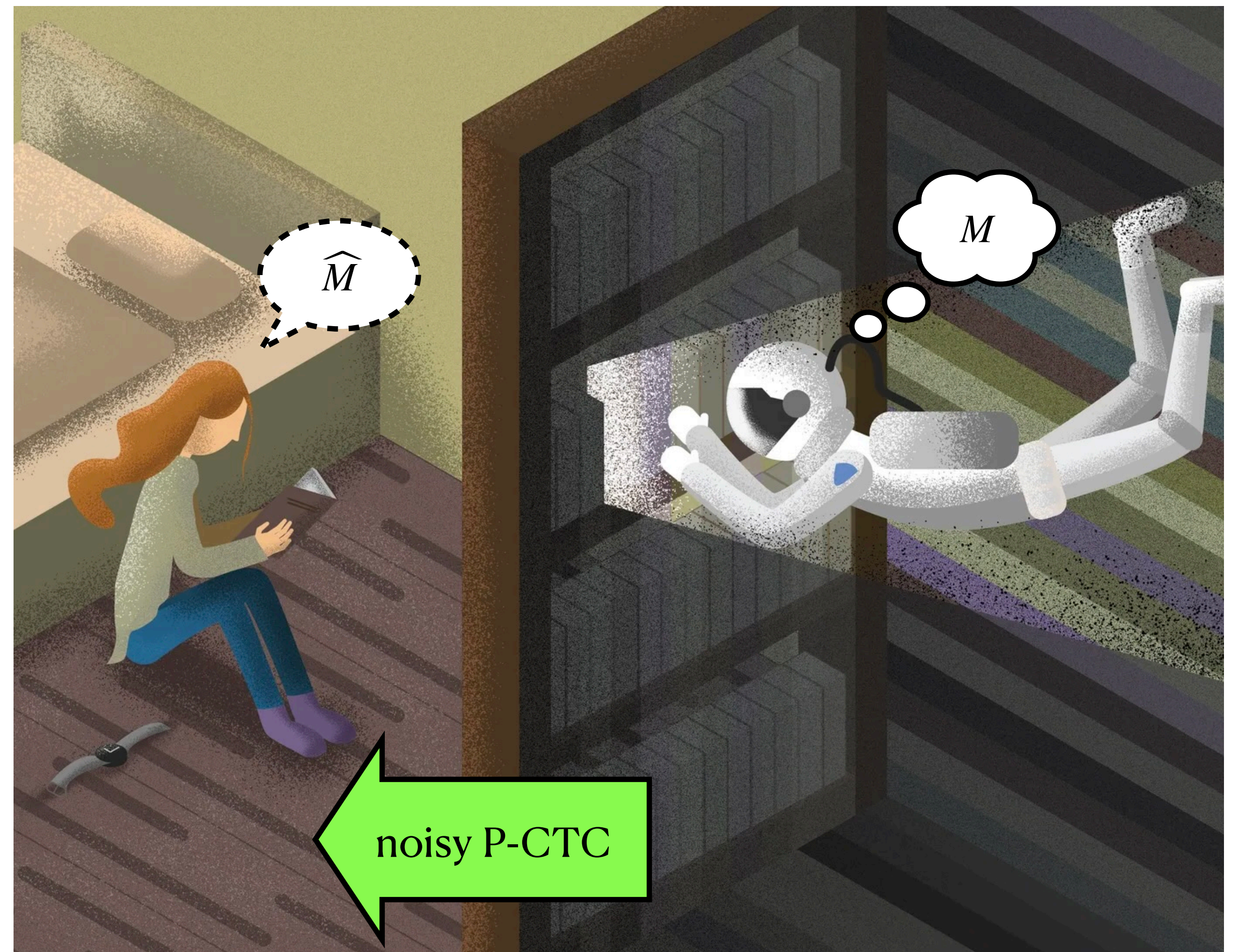
Setting: a **sender** in the future is connected to a **receiver** in the past through a **noisy P-CTC**

bookshelf

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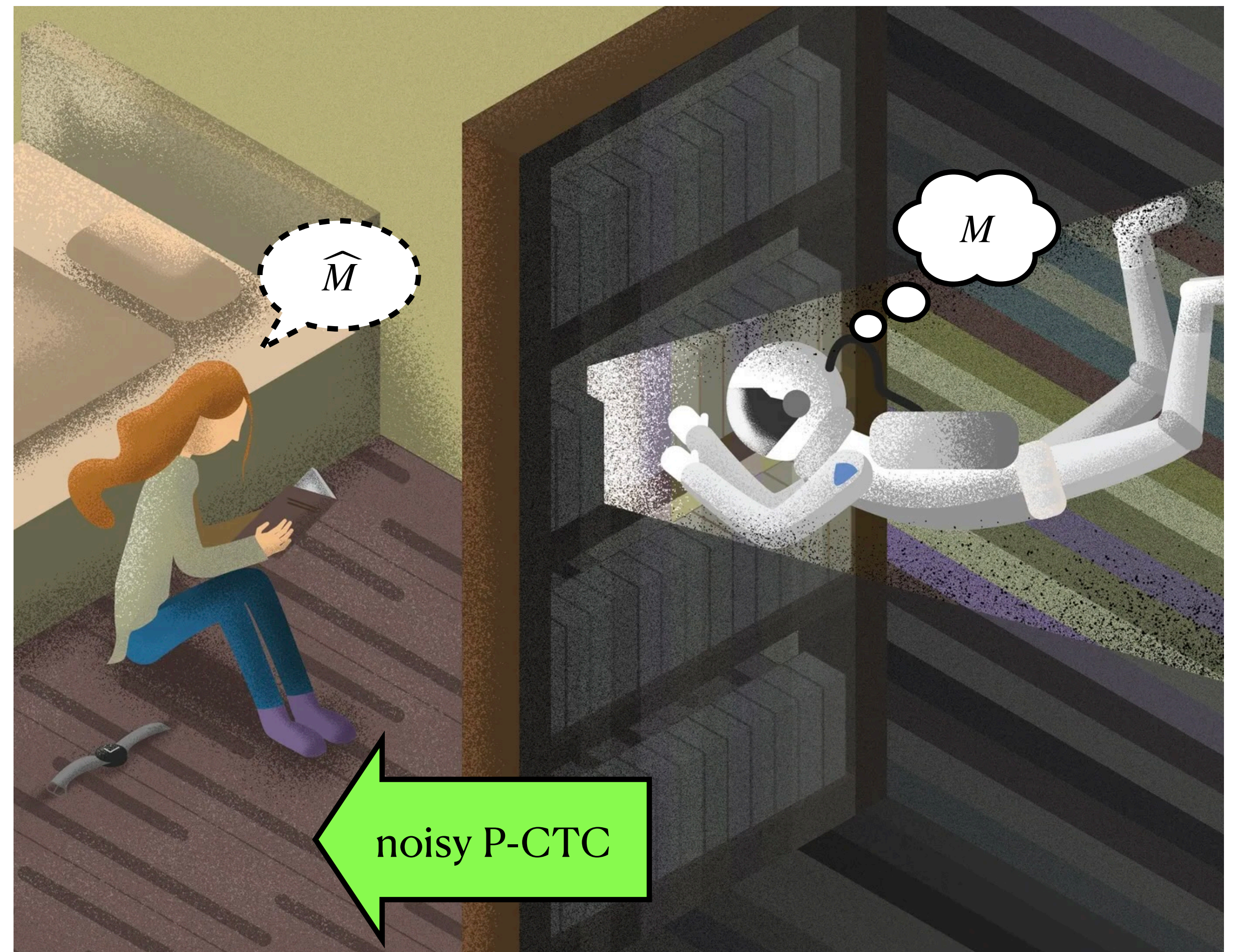
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General strategy

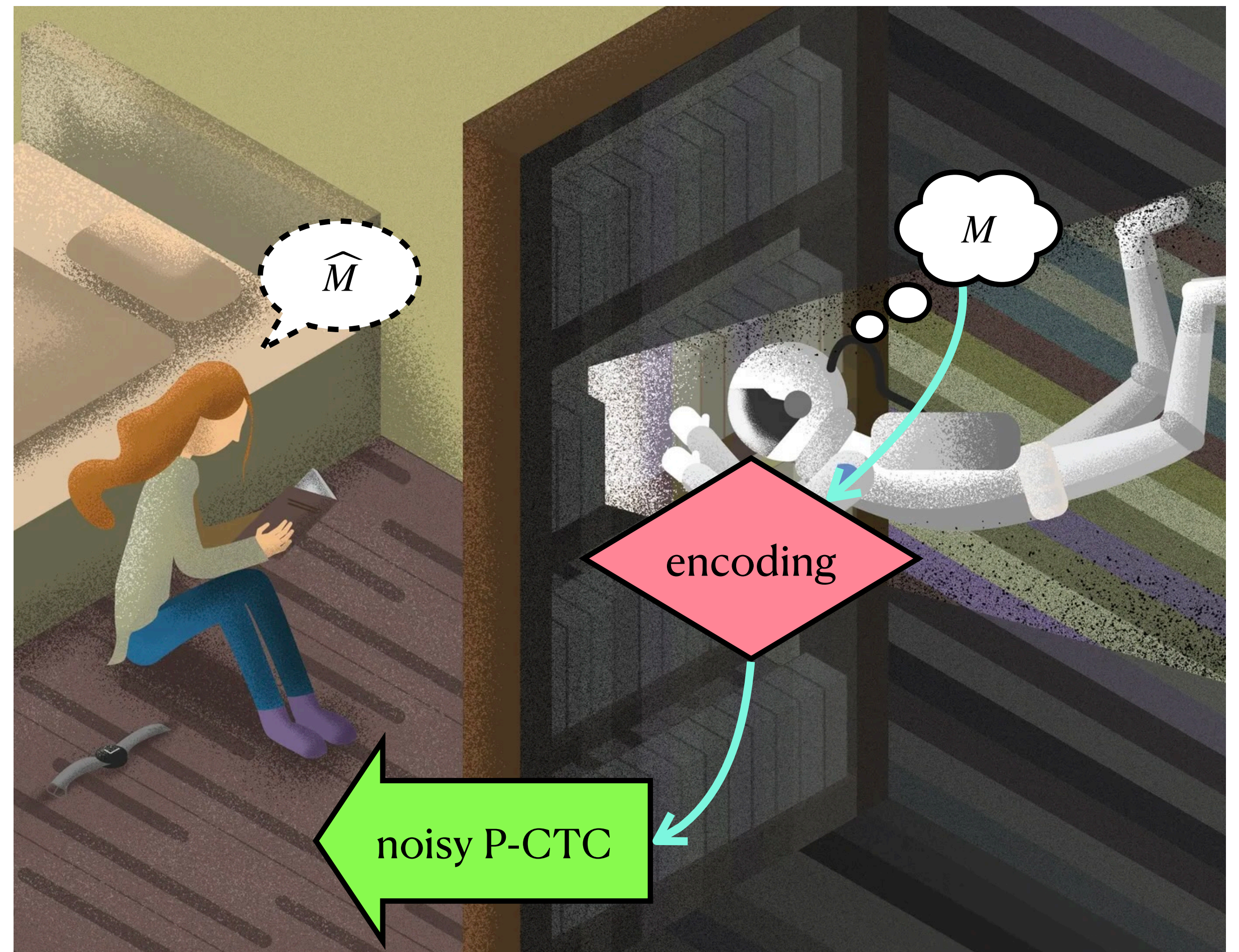
receiver (past) $\xrightarrow{\text{time}}$ sender (future)



General strategy

- Encoding

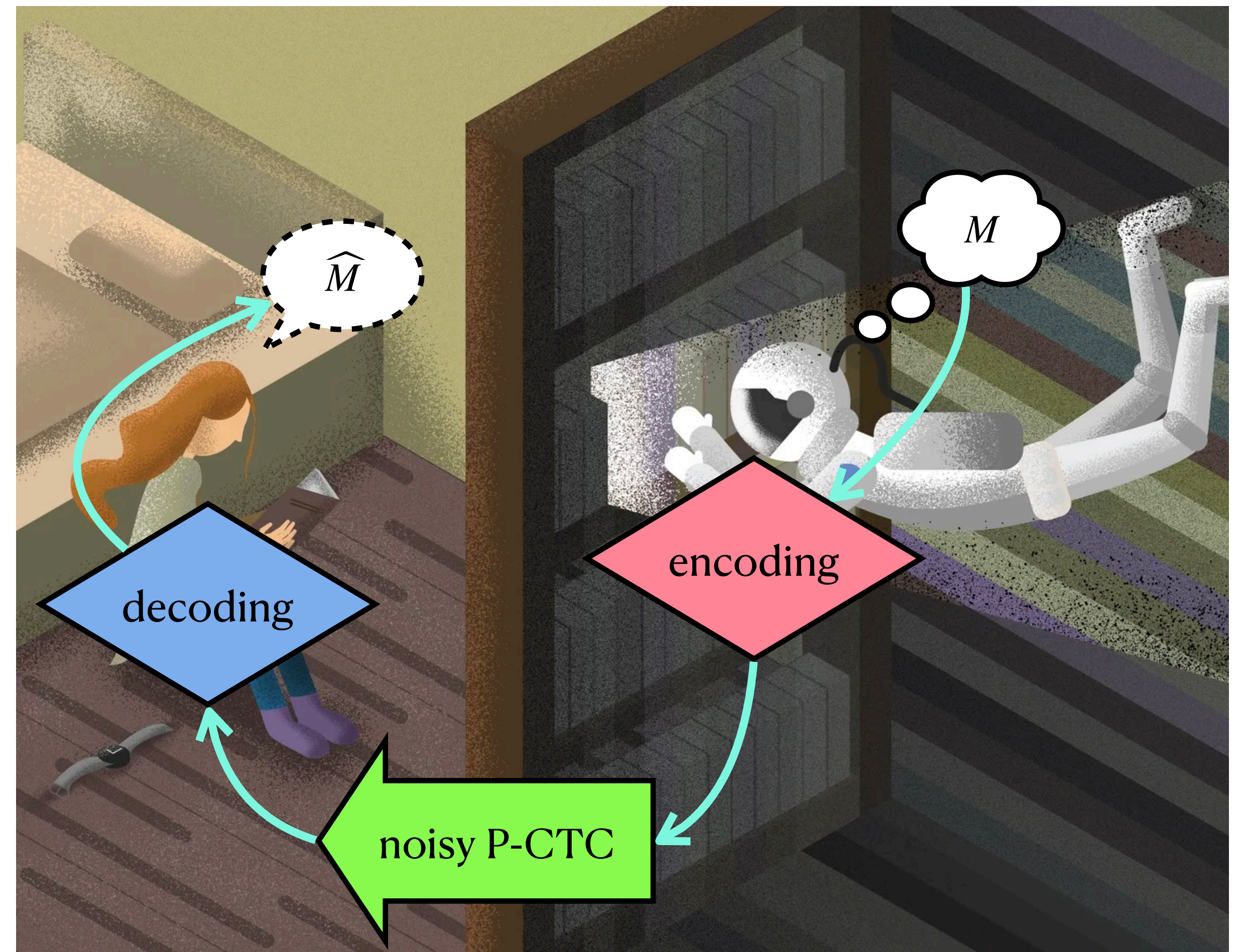
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General strategy

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- Decoding

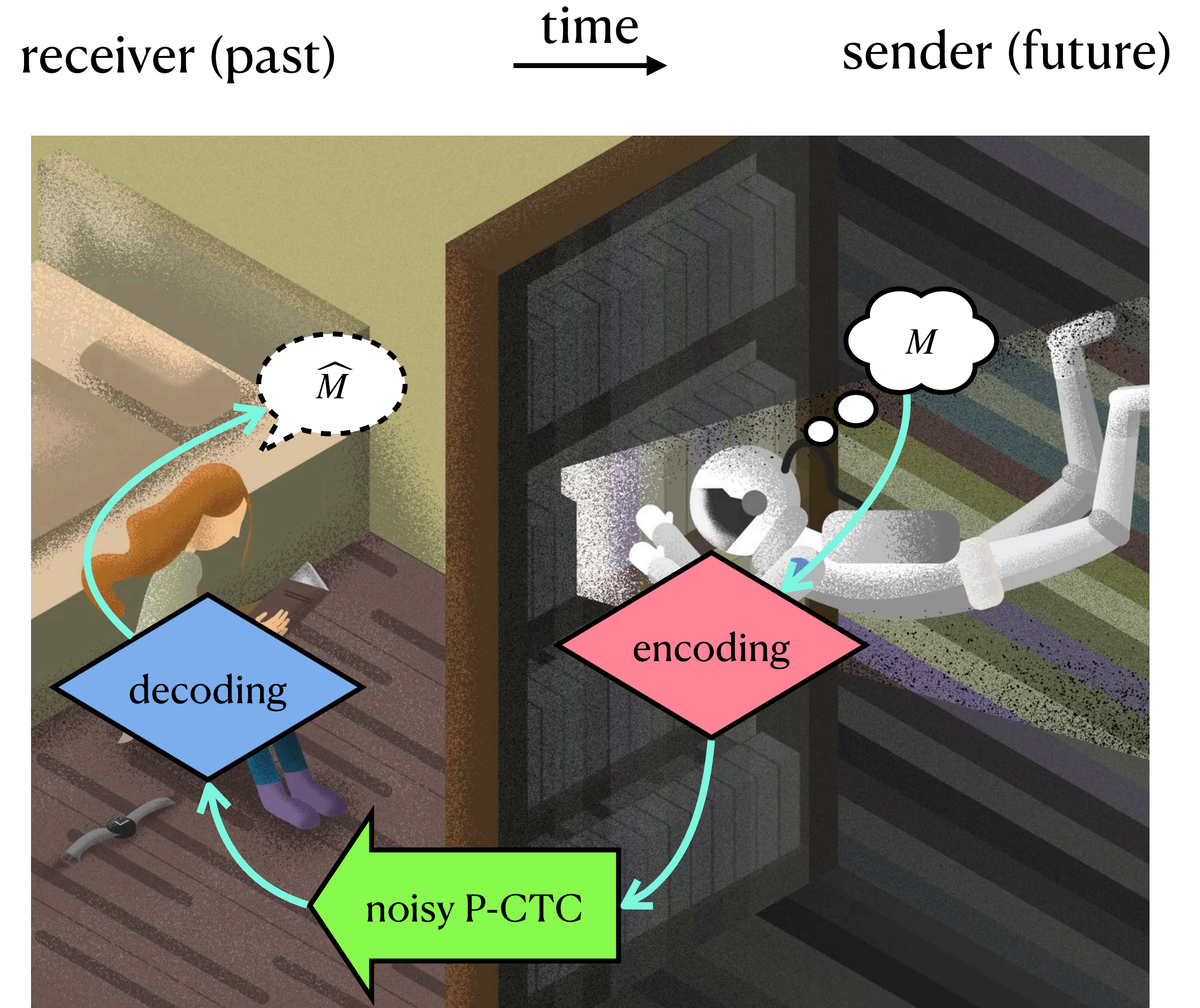
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General strategy

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- Decoding

Encoding (in the future) can in general be causally influenced by decoding (in the past)!

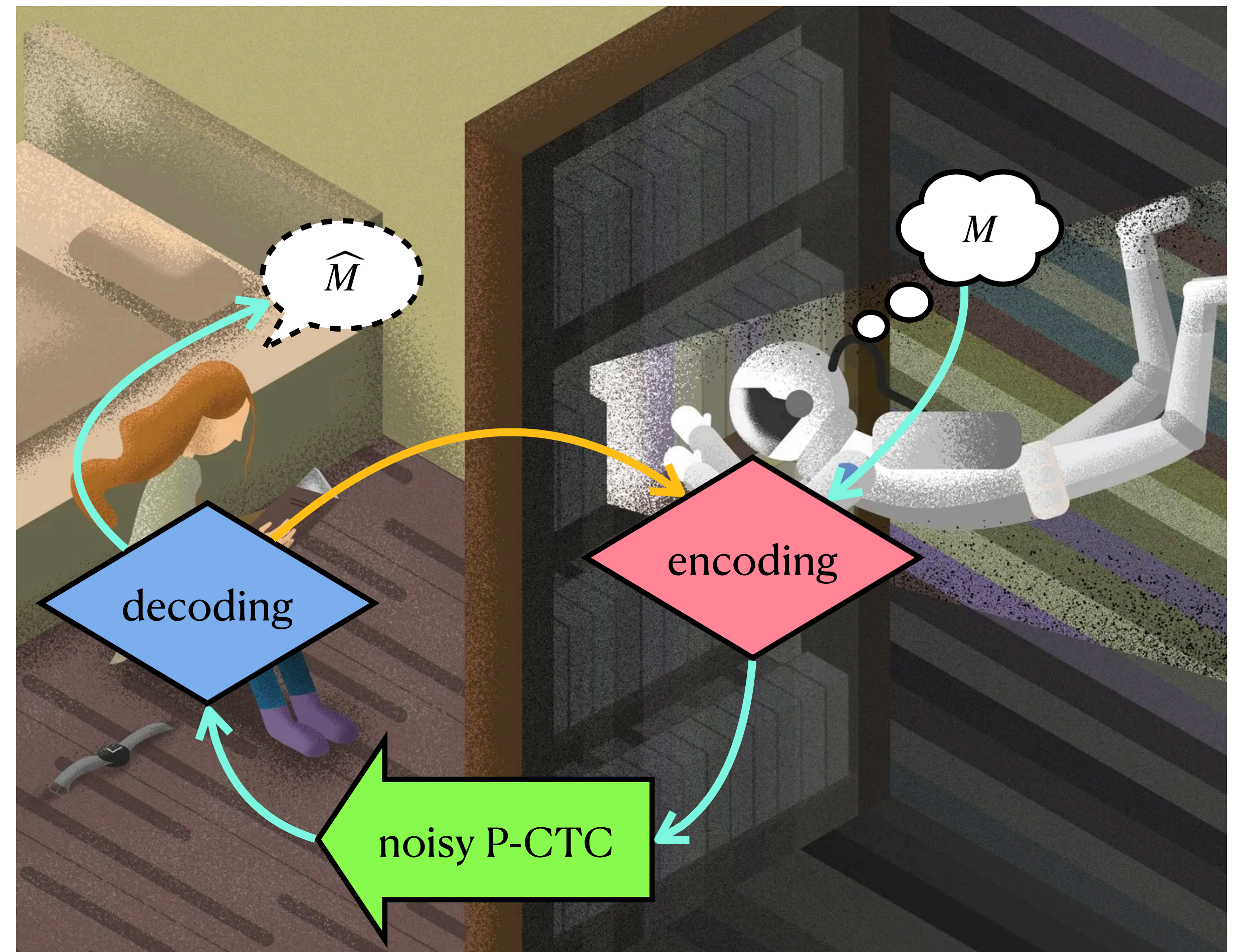


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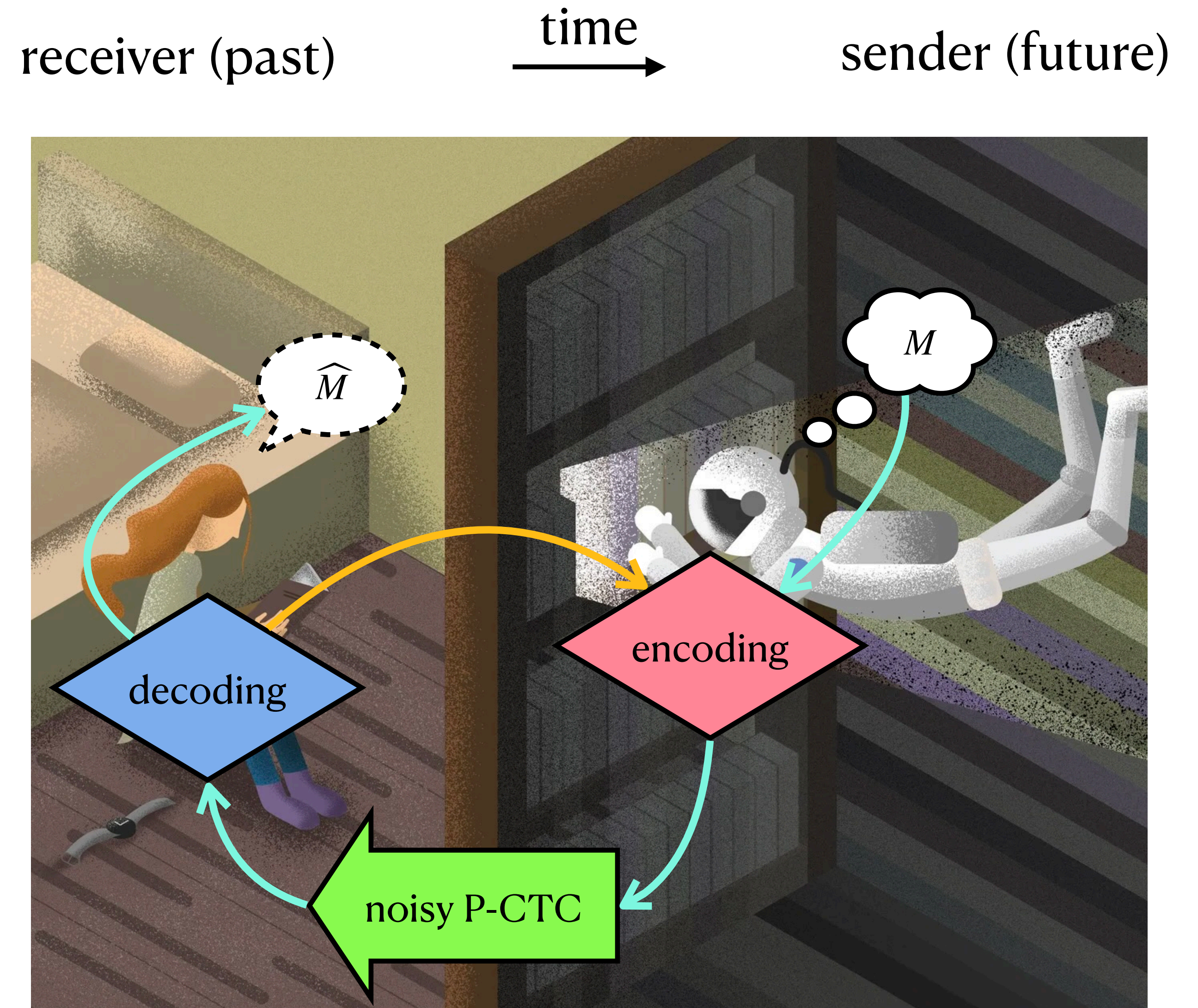


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- **Causal loop**, but perfectly reasonable

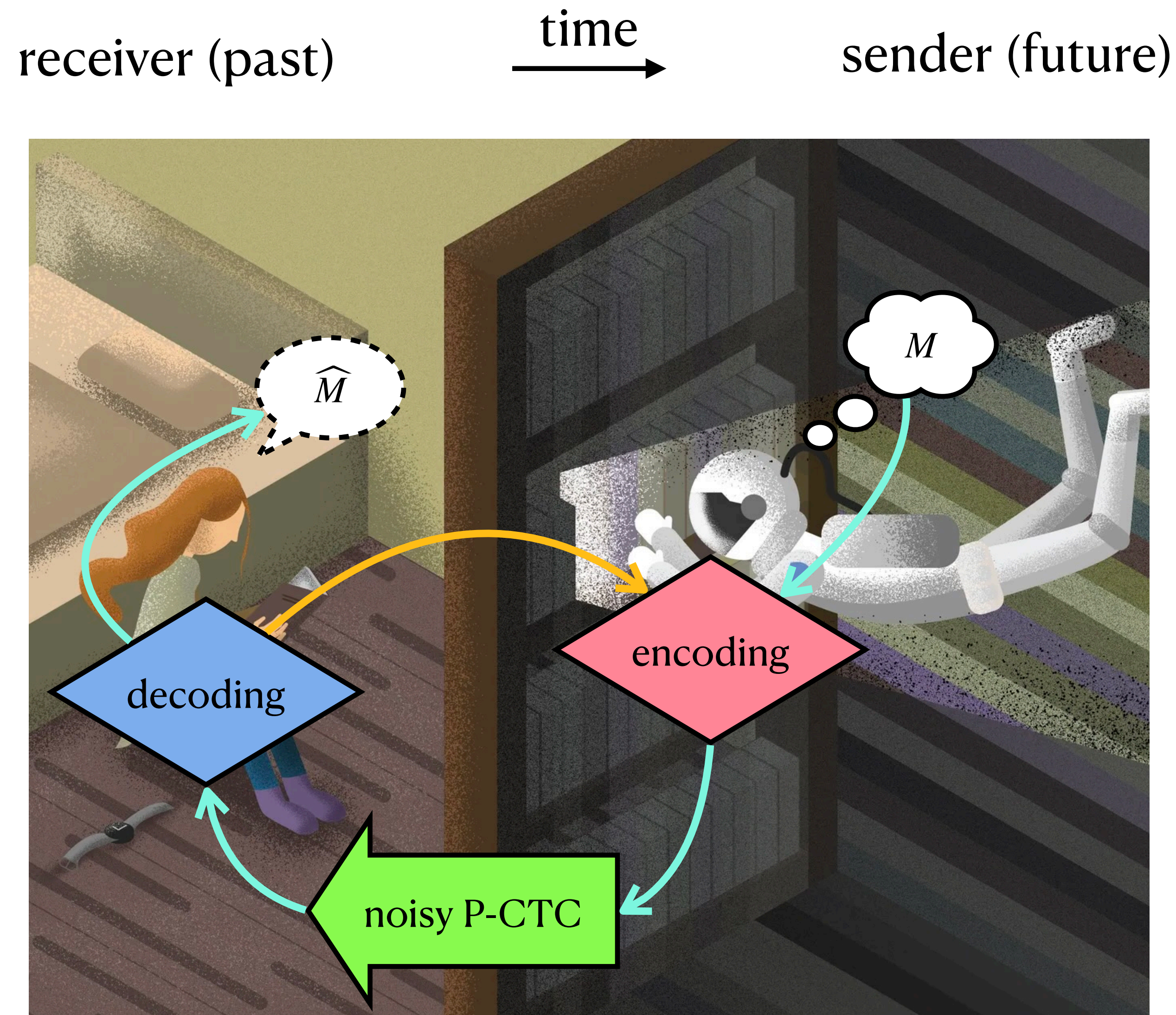


General strategy

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Encoding (in the future) can in general be causally influenced by decoding (in the past)!

- **Causal loop**, but perfectly reasonable
- More powerful than entanglement assistance

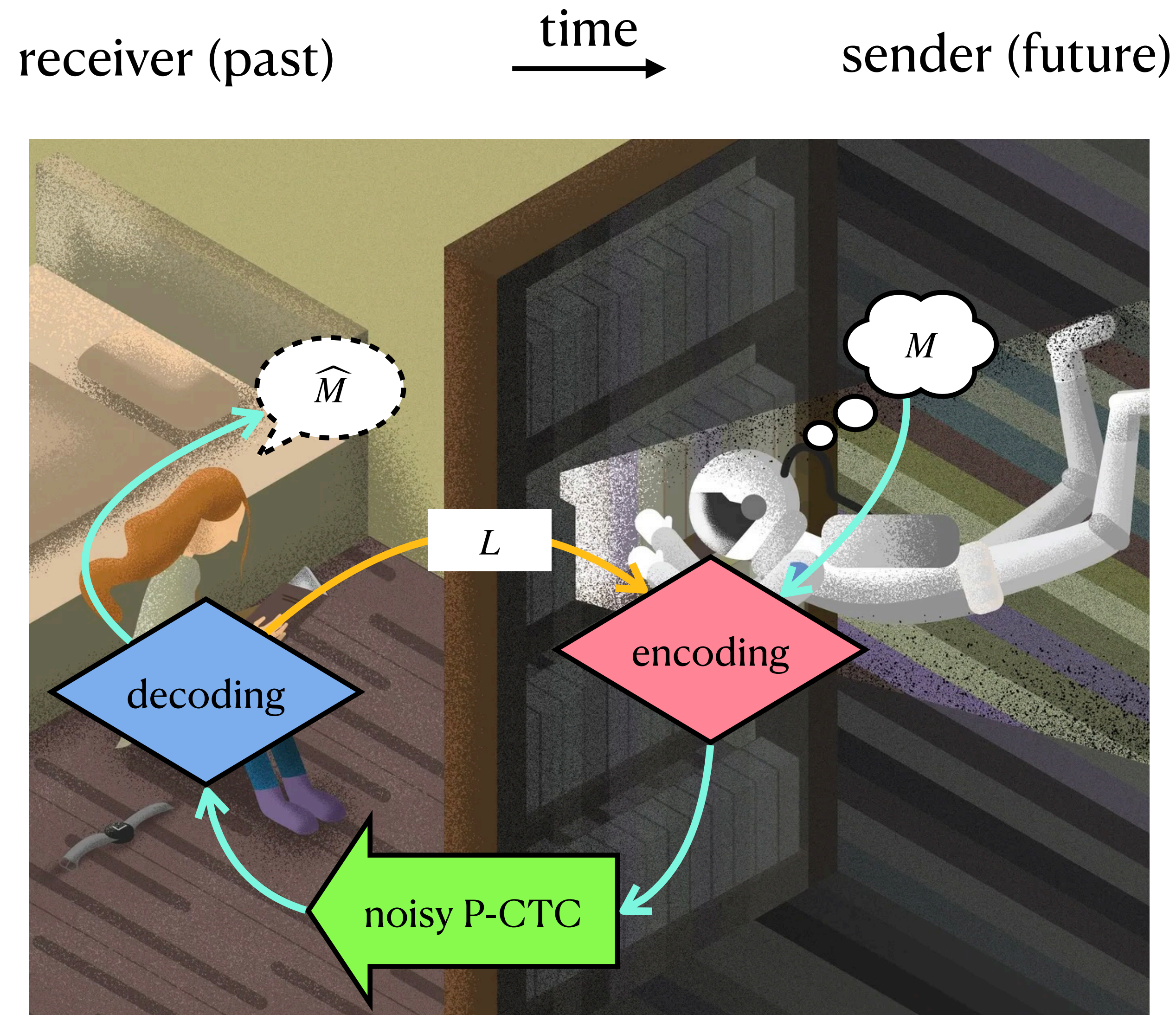


General strategy

- Encoding
- Decoding
- Daughter leaves q. memory L for father to retrieve and consult in the future

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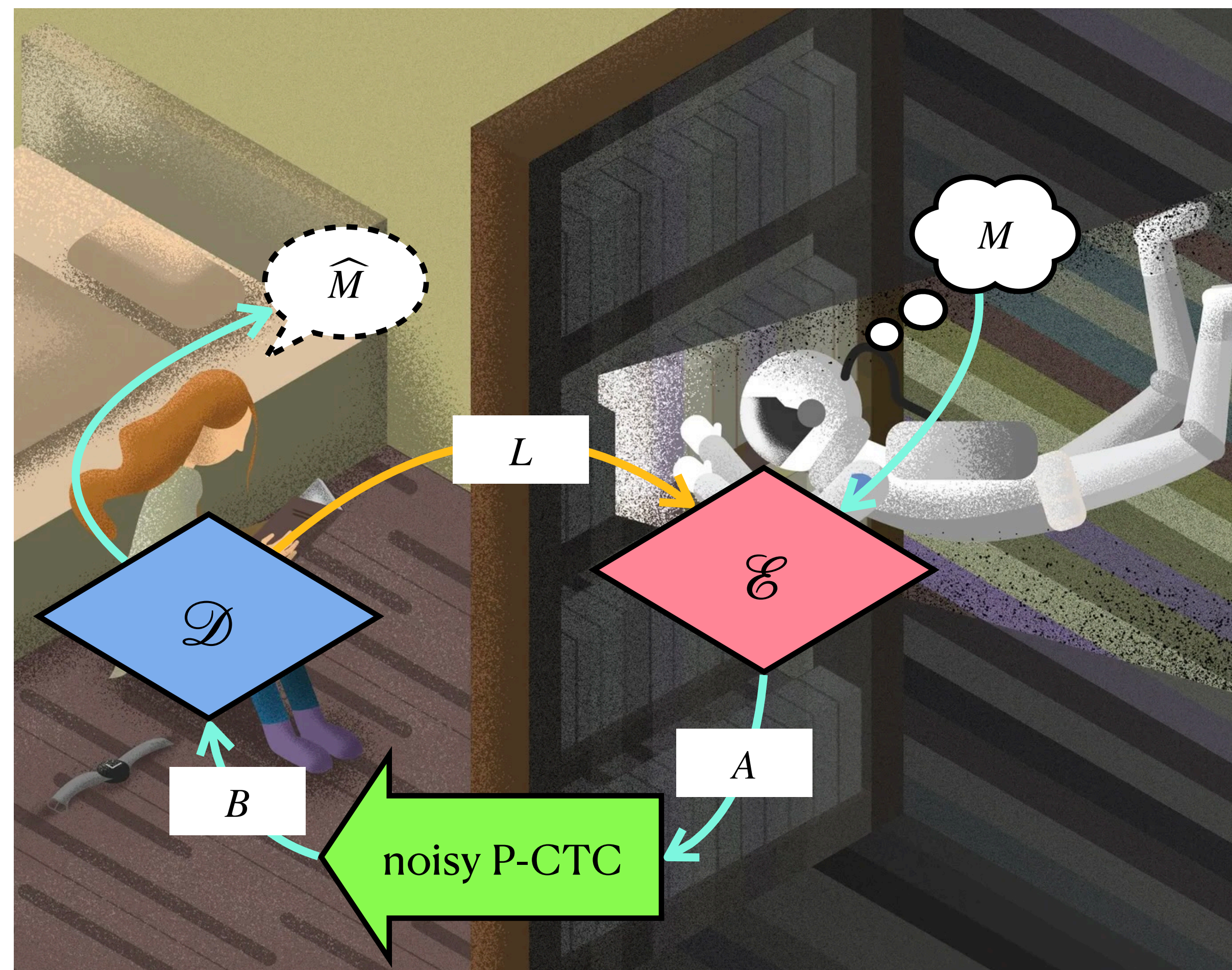
General strategy

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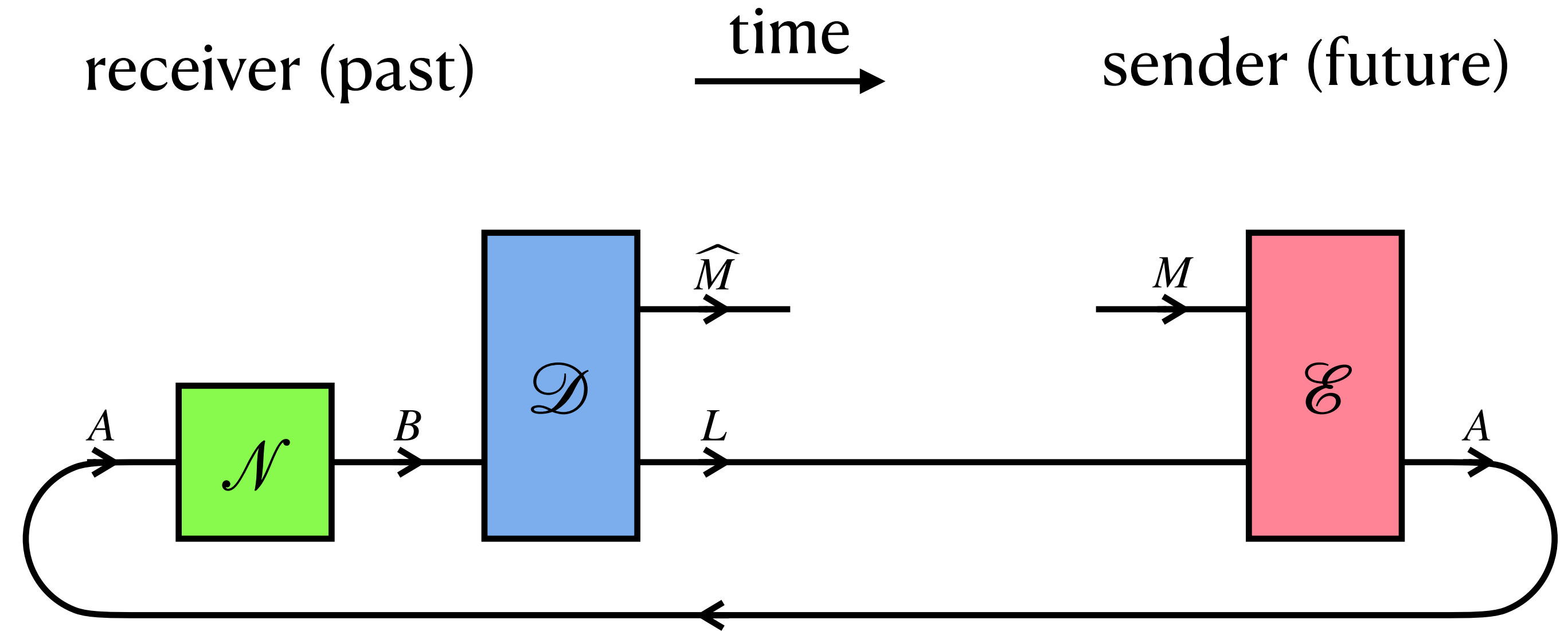


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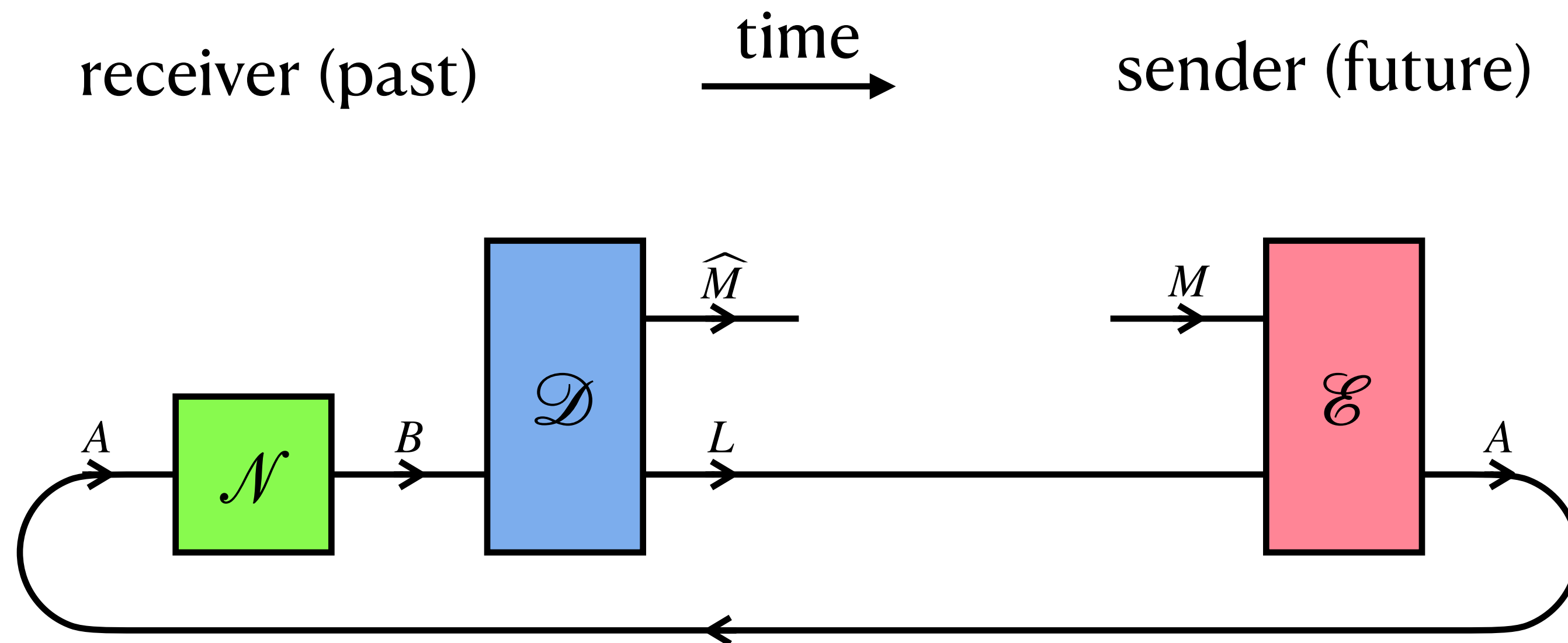


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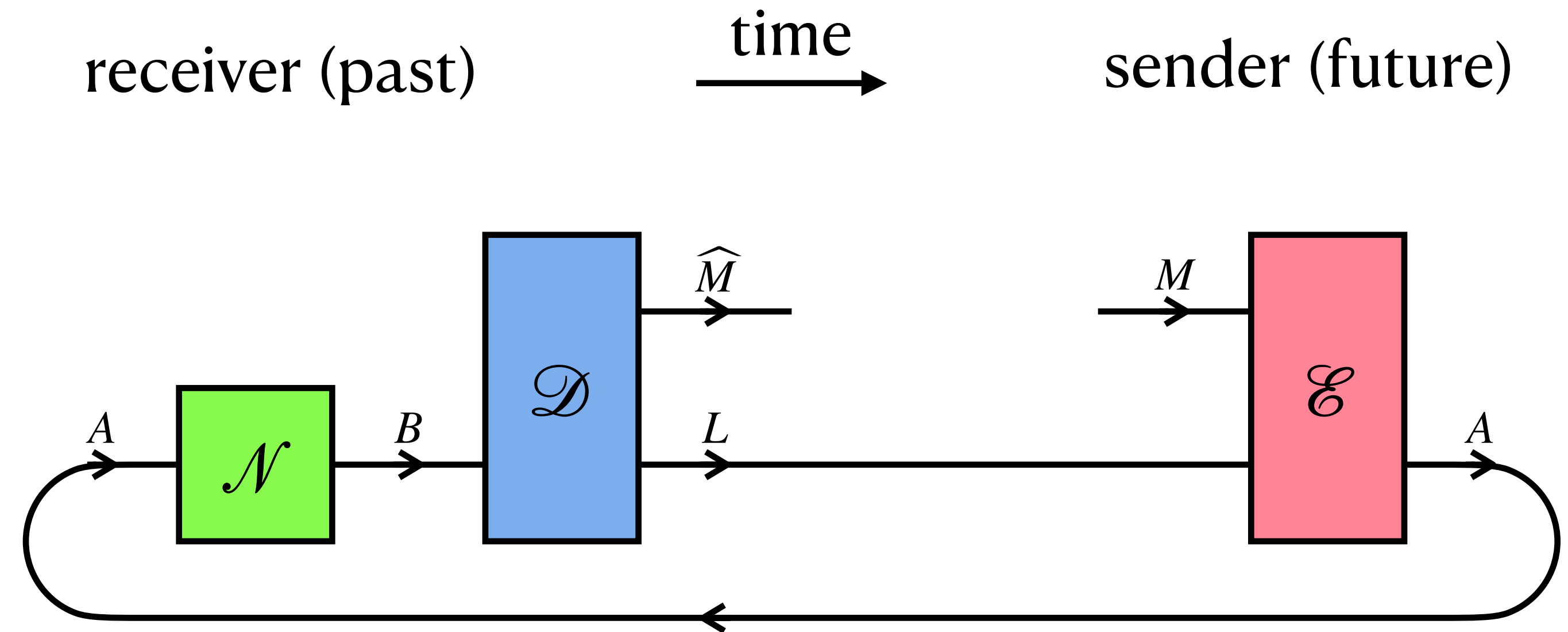


Call the entire map $\mathcal{M}_{M \rightarrow \hat{M}} \equiv \mathcal{M}_{M \rightarrow \hat{M}}^{(\mathcal{N}, \mathcal{E}, \mathcal{D})}$

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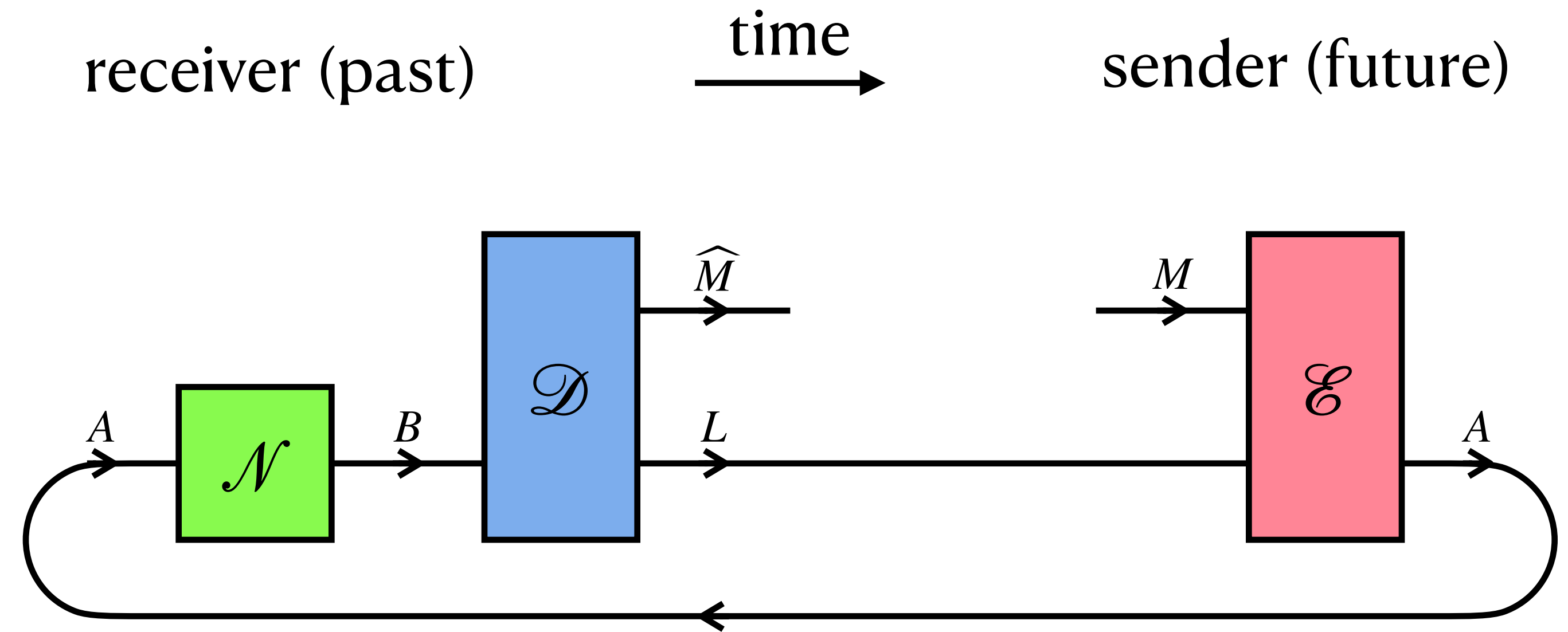


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nonlinear!

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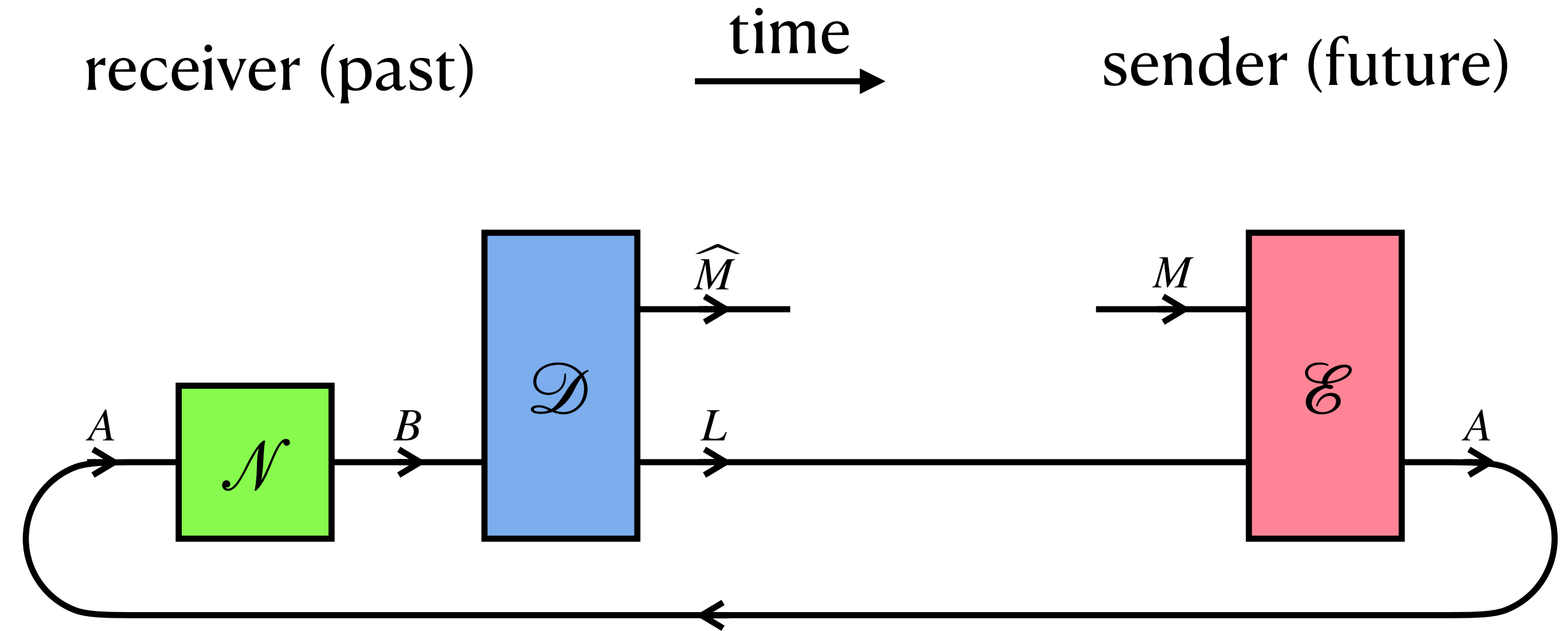
Definition of capacities



Call the entire map $\mathcal{M}_{M \rightarrow \hat{M}} \equiv \mathcal{M}_{M \rightarrow \hat{M}}^{(\mathcal{N}, \mathcal{E}, \mathcal{D})}$

nonlinear!

Definition of capacities



One-shot retrocausal classical capacity:

Call the entire map $\mathcal{M}_{M \rightarrow \hat{M}} \equiv \mathcal{M}_{M \rightarrow \hat{M}}^{(\mathcal{N}, \mathcal{E}, \mathcal{D})}$

$$C_{\text{retro}}^{\varepsilon}(\mathcal{N}) := \sup_{\mathcal{E}_{ML \rightarrow A}, \mathcal{D}_{B \rightarrow \hat{M}L}} \log_2 d_M$$

$$\text{s.t. } \min_{1 \leq m \leq d_M} \text{Tr}[|m\rangle\langle m|_{\hat{M}} \mathcal{M}_{M \rightarrow \hat{M}}[|m\rangle\langle m|_M]] \geq 1 - \varepsilon$$

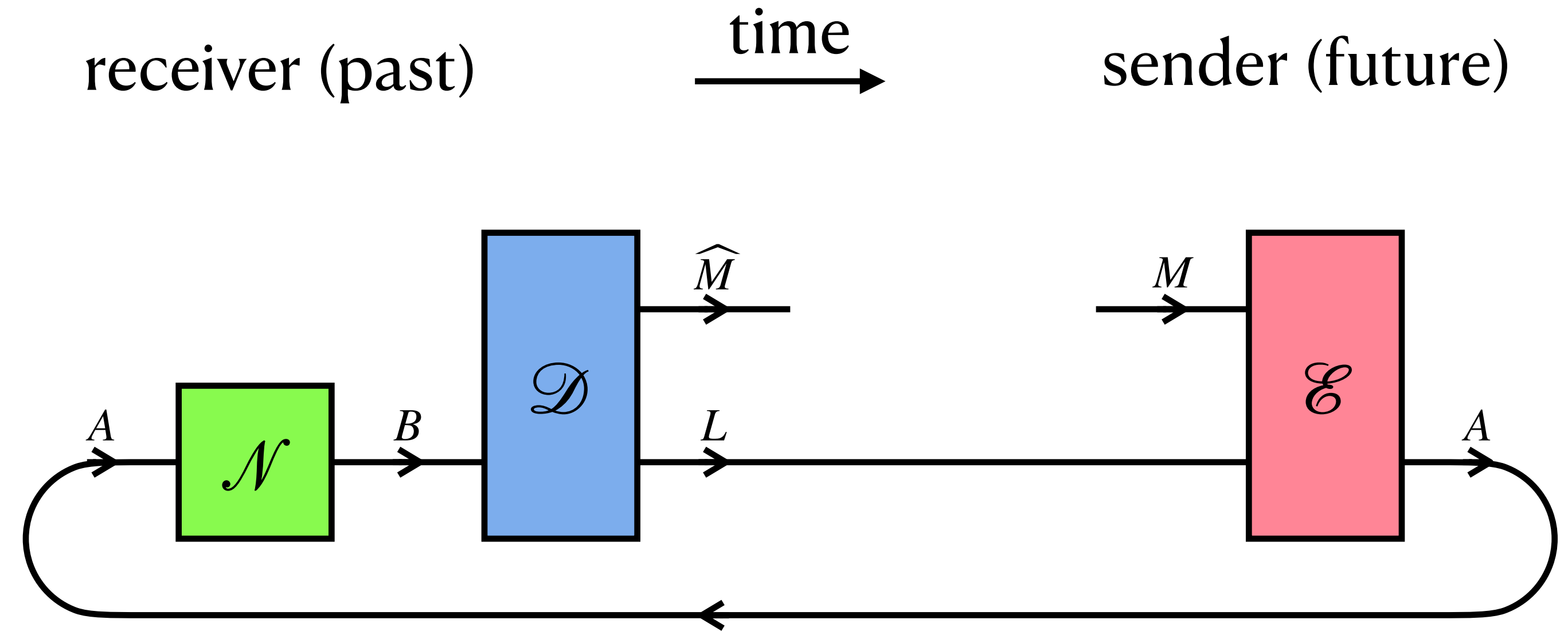
nonlinear!

Definition of capacities

One-shot retrocausal quantum capacity:

$$Q_{\text{retro}}^{\varepsilon}(\mathcal{N}) := \sup_{\mathcal{E}_{ML \rightarrow A}, \mathcal{D}_{B \rightarrow \hat{M}L}} \log_2 d_M$$

$$\text{s.t. } \inf_{\psi_{RM}} \text{Tr}[\psi_{R\hat{M}} \mathcal{M}_{M \rightarrow \hat{M}}[\psi_{RM}]] \geq 1 - \varepsilon$$



One-shot retrocausal classical capacity:

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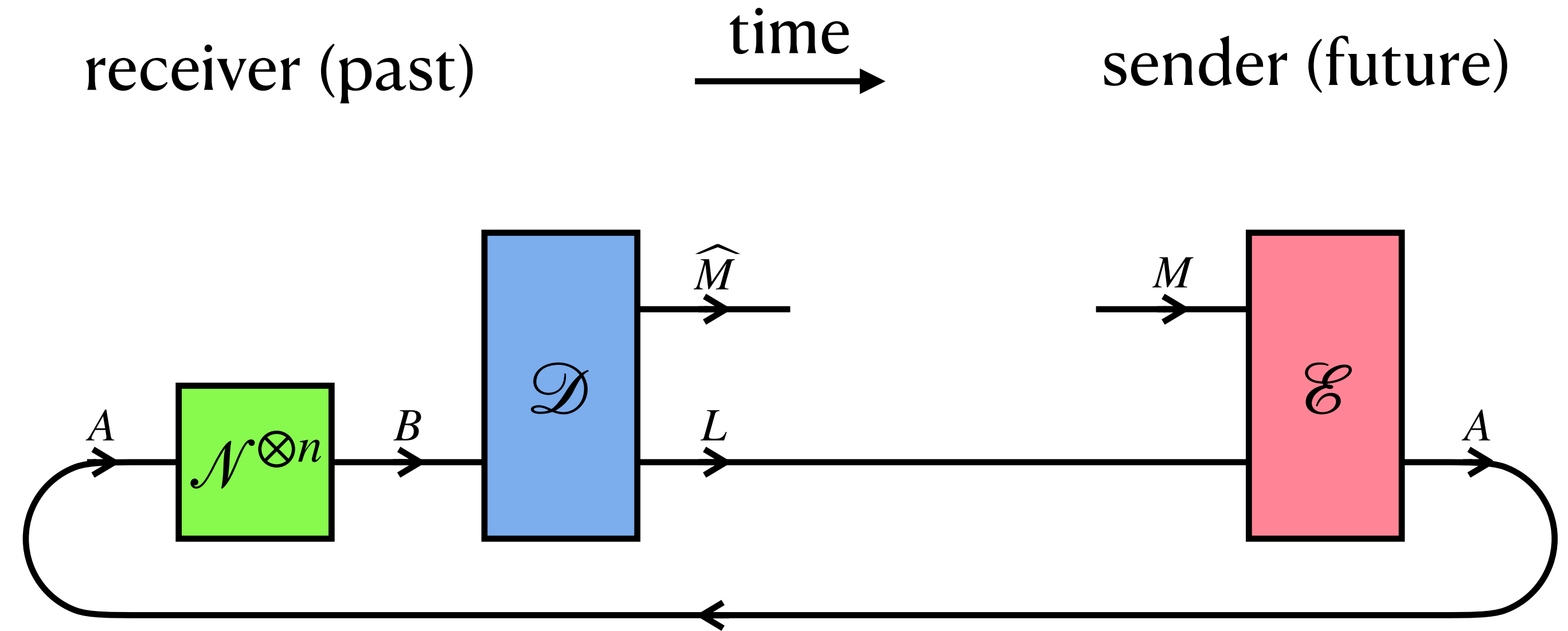
Call the entire map $\mathcal{M}_{M \rightarrow \hat{M}} \equiv \mathcal{M}_{M \rightarrow \hat{M}}^{(\mathcal{N}, \mathcal{E}, \mathcal{D})}$

nonlinear!

Definition of capacities

Asymptotic retrocausal quantum capacity:

$$Q_{\text{retro}}(\mathcal{N}) := \lim_{\varepsilon \searrow 0} \liminf_{n \rightarrow \infty} \frac{1}{n} Q_{\text{retro}}^{\varepsilon}(\mathcal{N}^{\otimes n})$$



Asymptotic retrocausal classical capacity:

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Main result:

Noisy-“time-machine” coding theorem

Main result

Main result

One-shot retrocausal quantum & classical capacity: for all $\varepsilon \in [0,1]$,

$$C_{\text{retro}}^{\varepsilon}(\mathcal{N}) = \log_2 \left\lfloor \frac{\varepsilon}{1-\varepsilon} 2^{I_{\text{max}}(\mathcal{N})+I_{\text{doe}}(\mathcal{N})} + 1 \right\rfloor$$

$$Q_{\text{retro}}^{\varepsilon}(\mathcal{N}) = \log_2 \left\lfloor \sqrt{\frac{\varepsilon}{1-\varepsilon} 2^{I_{\text{max}}(\mathcal{N})+I_{\text{doe}}(\mathcal{N})} + 1} \right\rfloor$$

+ explicit construction of
optimal comm. strategy

Main result

One-shot retrocausal quantum & classical capacity: for all $\varepsilon \in [0,1]$,

$$\begin{aligned} C_{\text{retro}}^\varepsilon(\mathcal{N}) &= \log_2 \left\lfloor \frac{\varepsilon}{1-\varepsilon} 2^{I_{\text{max}}(\mathcal{N})+I_{\text{doe}}(\mathcal{N})} + 1 \right\rfloor \\ Q_{\text{retro}}^\varepsilon(\mathcal{N}) &= \log_2 \left\lfloor \sqrt{\frac{\varepsilon}{1-\varepsilon} 2^{I_{\text{max}}(\mathcal{N})+I_{\text{doe}}(\mathcal{N})} + 1} \right\rfloor \end{aligned} \quad \begin{array}{l} \text{+ explicit construction of} \\ \text{optimal comm. strategy} \end{array}$$

Max-information: $I_{\text{max}}(\mathcal{N}) := \inf_{\lambda, \sigma_B} \left\{ \log_2 \lambda : \mathcal{N}_{A \rightarrow B}[\Phi_{A'A}] \leq \lambda \frac{I_{A'}}{d_{A'}} \otimes \sigma_B \right\}$ [König et al. 2009]

Doeblin information: $I_{\text{doe}}(\mathcal{N}) := \inf_{\lambda, \tau_B} \left\{ \log_2 \lambda : \frac{I_{A'}}{d_{A'}} \otimes \tau_B \leq \lambda \mathcal{N}_{A \rightarrow B}[\Phi_{A'A}] \right\}$ [George et al. 2025]

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$$Q_{\text{retro}}^\varepsilon(\mathcal{N}) = \log_2 \left\lfloor \sqrt{\frac{\varepsilon}{1-\varepsilon} 2^{I_{\text{max}}(\mathcal{N})+I_{\text{doe}}(\mathcal{N})} + 1} \right\rfloor$$

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unit-trace Hermitian operator!

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+ explicit construction of optimal comm. strategy

Asymptotic retrocausal quantum & classical capacity:

$$C_{\text{retro}}(\mathcal{N}) = 2Q_{\text{retro}}(\mathcal{N}) = I_{\text{max}}(\mathcal{N}) + I_{\text{doe}}^\infty(\mathcal{N})$$

Max-information: $I_{\text{max}}(\mathcal{N}) := \inf_{\lambda, \sigma_B} \left\{ \log_2 \lambda : \mathcal{N}_{A \rightarrow B}[\Phi_{A'A}] \leq \lambda \frac{I_{A'}}{d_{A'}} \otimes \sigma_B \right\}$ [König et al. 2009]

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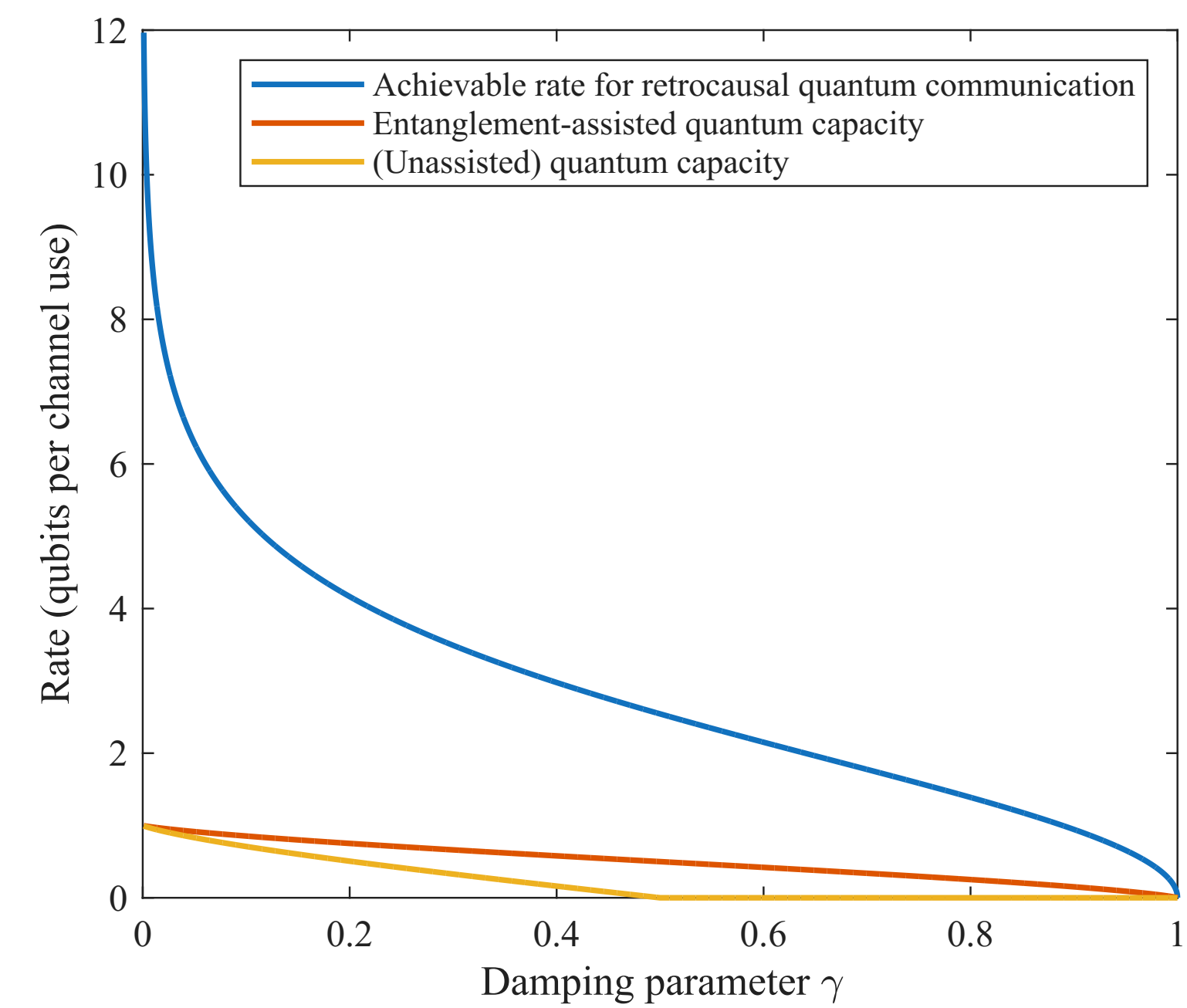
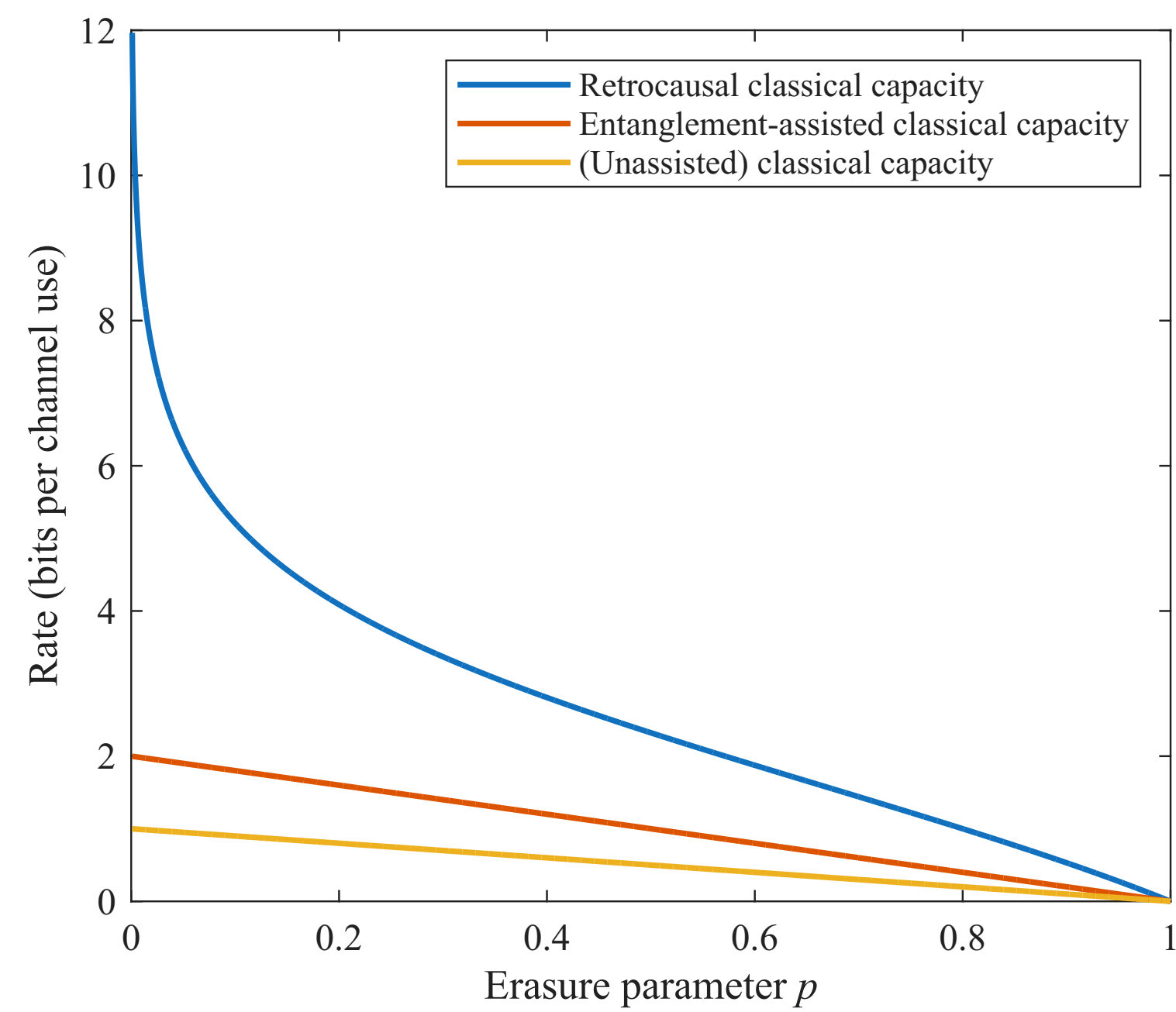
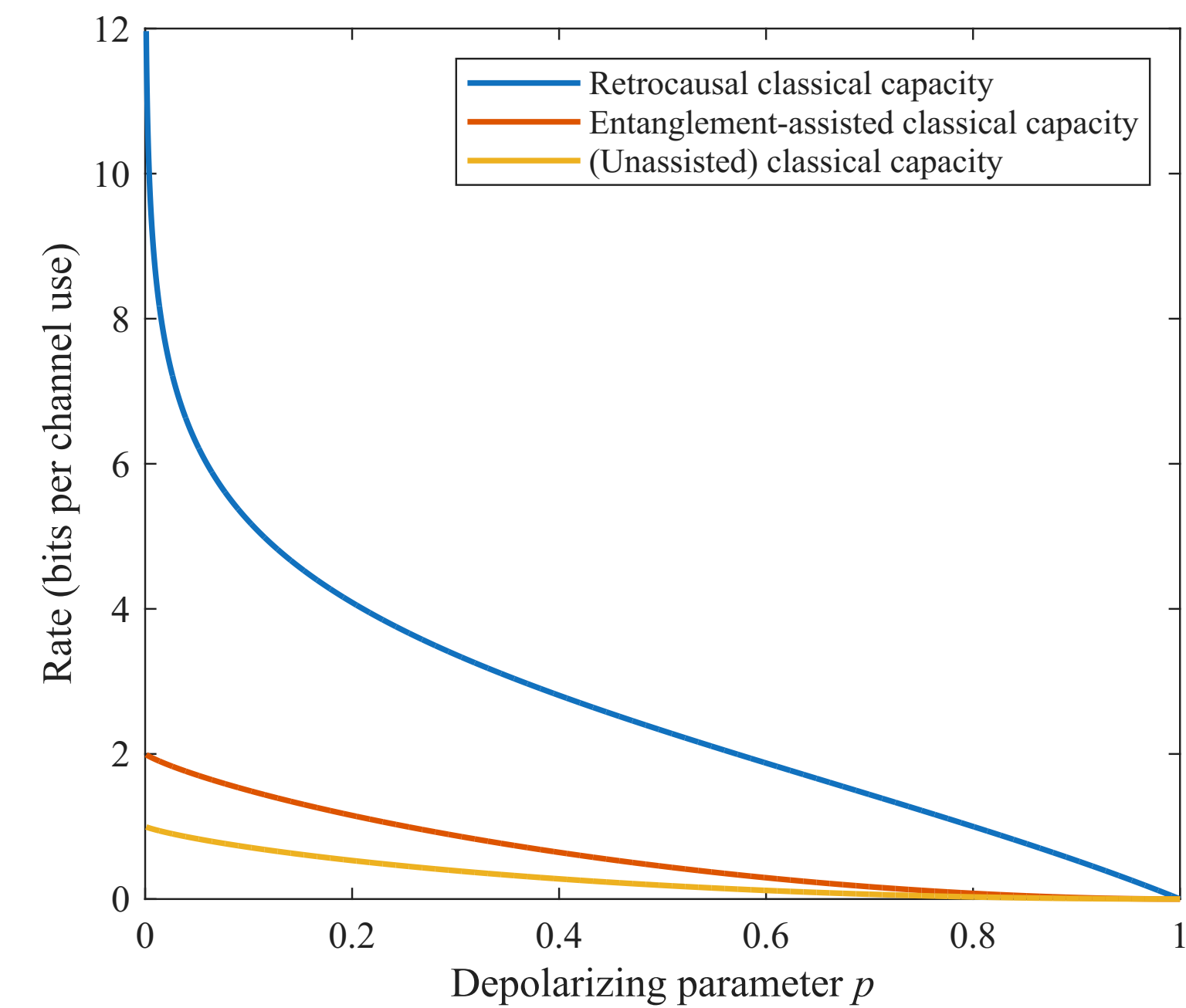
$$C_{\text{retro}}(\mathcal{N}) = 2Q_{\text{retro}}(\mathcal{N}) = I_{\text{max}}(\mathcal{N}) + I_{\text{doe}}^{\infty}(\mathcal{N}) \equiv \lim_{n \rightarrow \infty} \frac{1}{n} I_{\text{doe}}(\mathcal{N}^{\otimes n})$$

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Comparison with conventional capacities



Comparison with conventional capacities

Entanglement-assisted capacities:

$$C_{\text{EA}}(\mathcal{N}) = 2Q_{\text{EA}}(\mathcal{N}) = I(\mathcal{N})$$

Retrocausal capacities:

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Sandwiched Rényi mutual information

$$\tilde{I}_{\alpha}(\mathcal{N}) \equiv \sup_{\rho_{RA}} \tilde{I}_{\alpha}(R; B)_{\mathcal{N}_{A \rightarrow B}[\rho_{RA}]}$$

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$\alpha \rightarrow 1$

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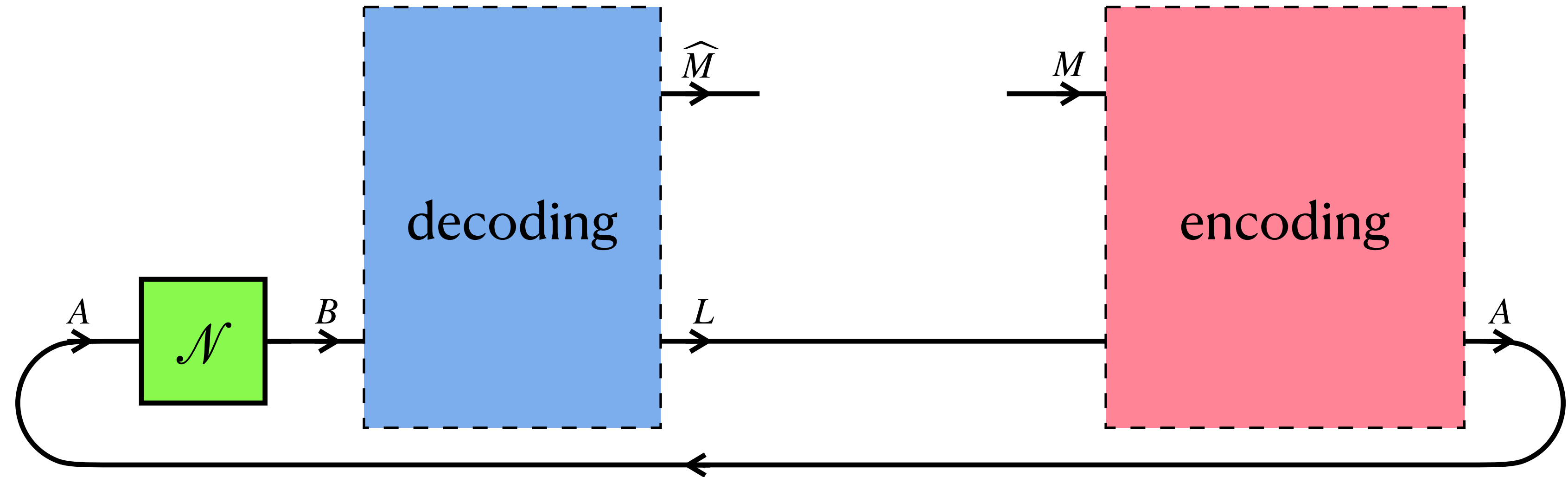
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Proof of the main result

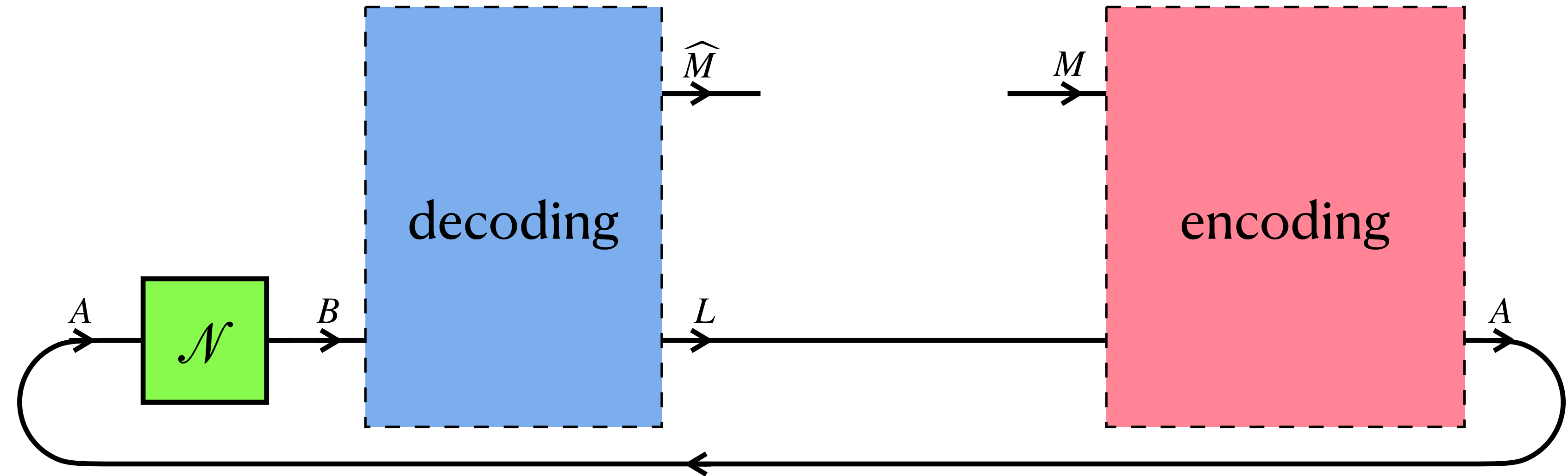
Achievability & optimality

Achievability: Amplified probabilistic teleportation



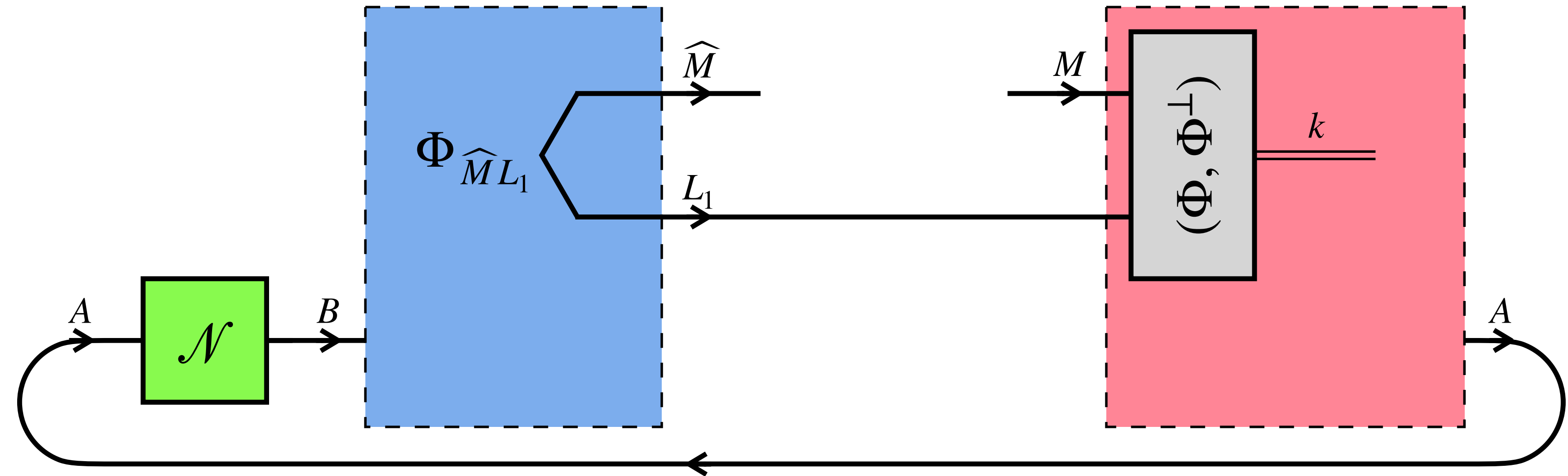
Achievability: Amplified probabilistic teleportation

Instead of encoding the message in the input of the noisy P-CTC, the sender tries to send the message via probabilistic teleportation through $M \rightarrow L_1 \rightarrow \hat{M}$



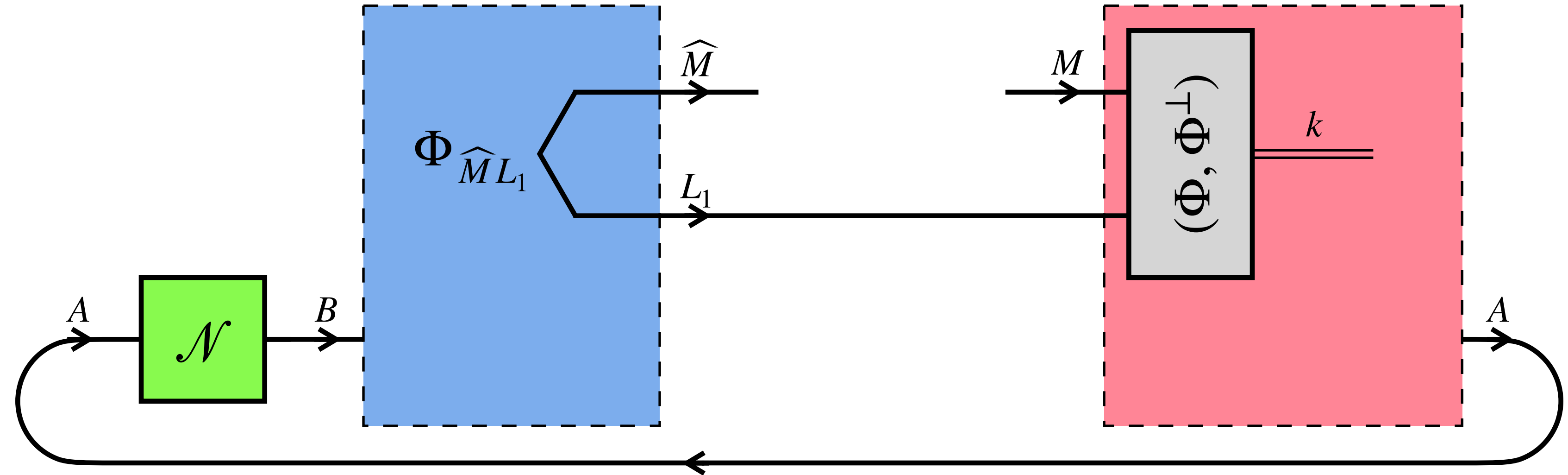
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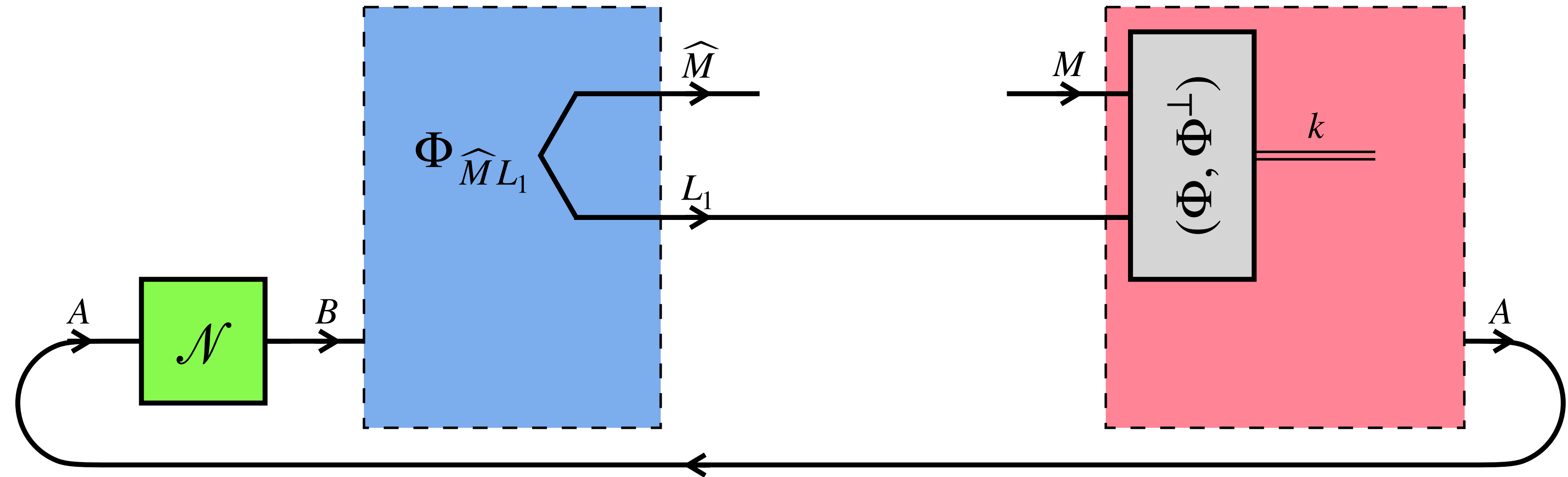
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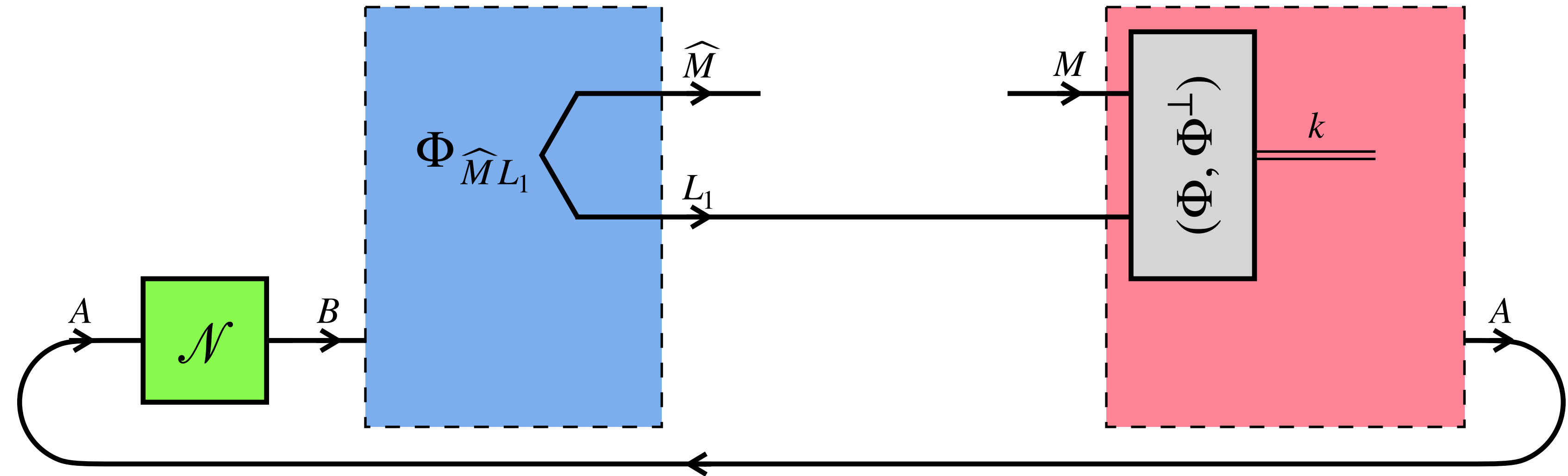
- The probability of success ($k = 0$) is of course only $1/d_M^2$

Achievability: Amplified probabilistic teleportation



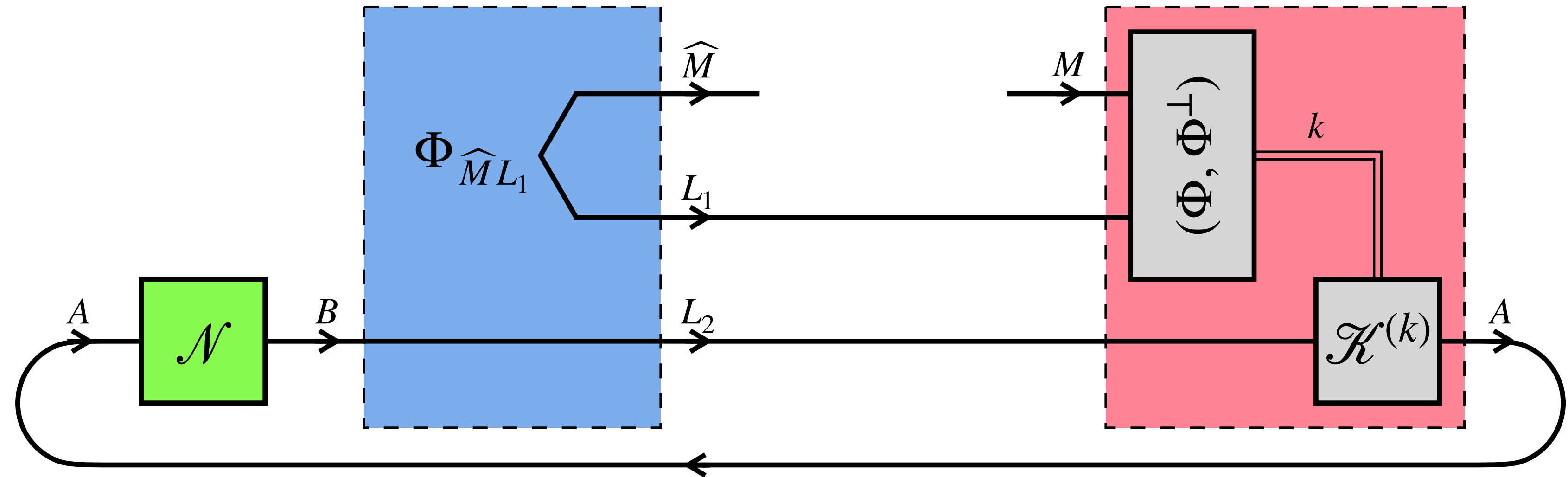
Achievability: Amplified probabilistic teleportation

Engineer a causal loop consisting of $\mathcal{N}$ and a channel $\mathcal{K}^{(k)}$, conditioned on outcome $k \in \{0,1\}$ of probabilistic teleportation



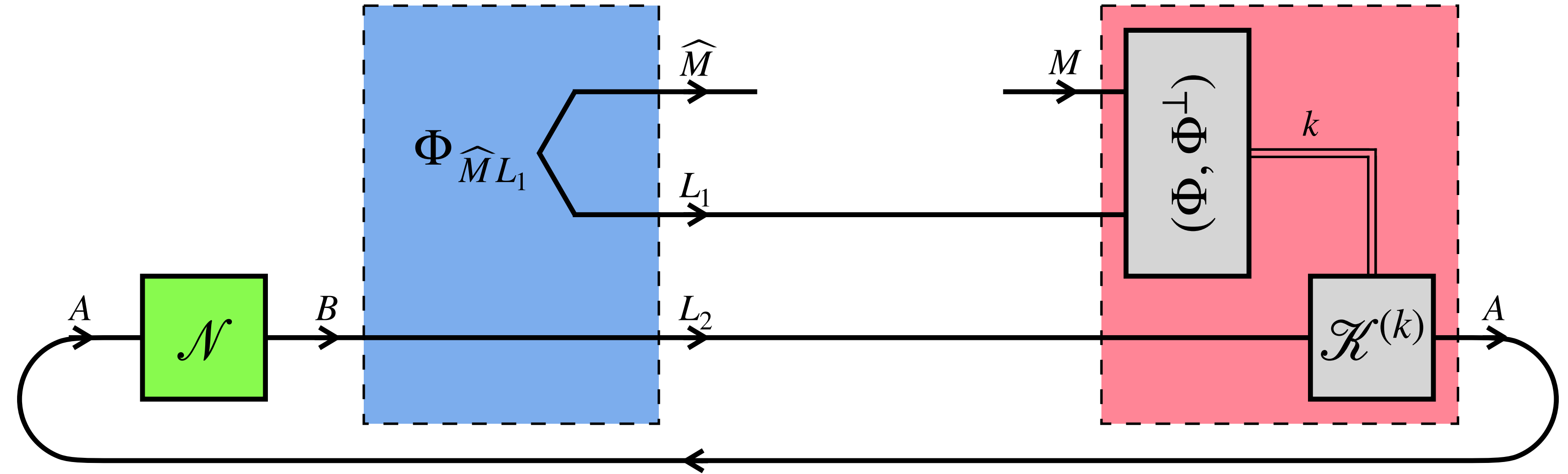
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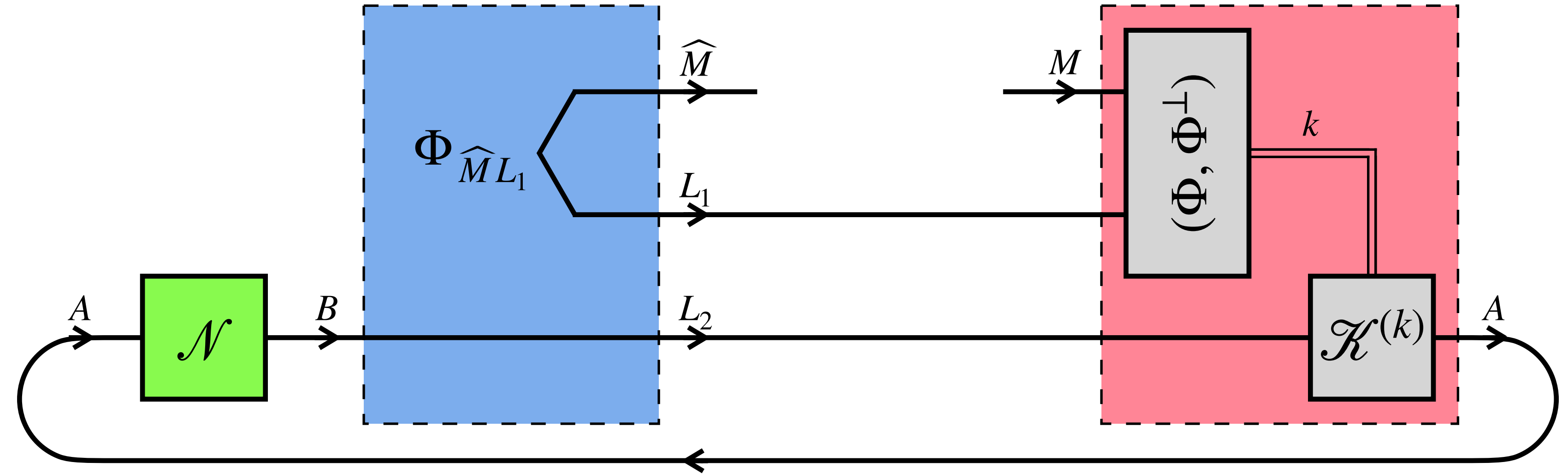
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- Due to nonlinearity, the odds of successful teleportation are amplified from $\frac{1/d_M^2}{1 - 1/d_M^2}$ to $\frac{1/d_M^2}{1 - 1/d_M^2} \cdot \frac{\text{Tr}[\Phi_{AA'}(\mathcal{K}_{B \rightarrow A}^{(0)} \circ \mathcal{N}_{A \rightarrow B})[\Phi_{AA'}]]}{\text{Tr}[\Phi_{AA'}(\mathcal{K}_{B \rightarrow A}^{(1)} \circ \mathcal{N}_{A \rightarrow B})[\Phi_{AA'}]]}$

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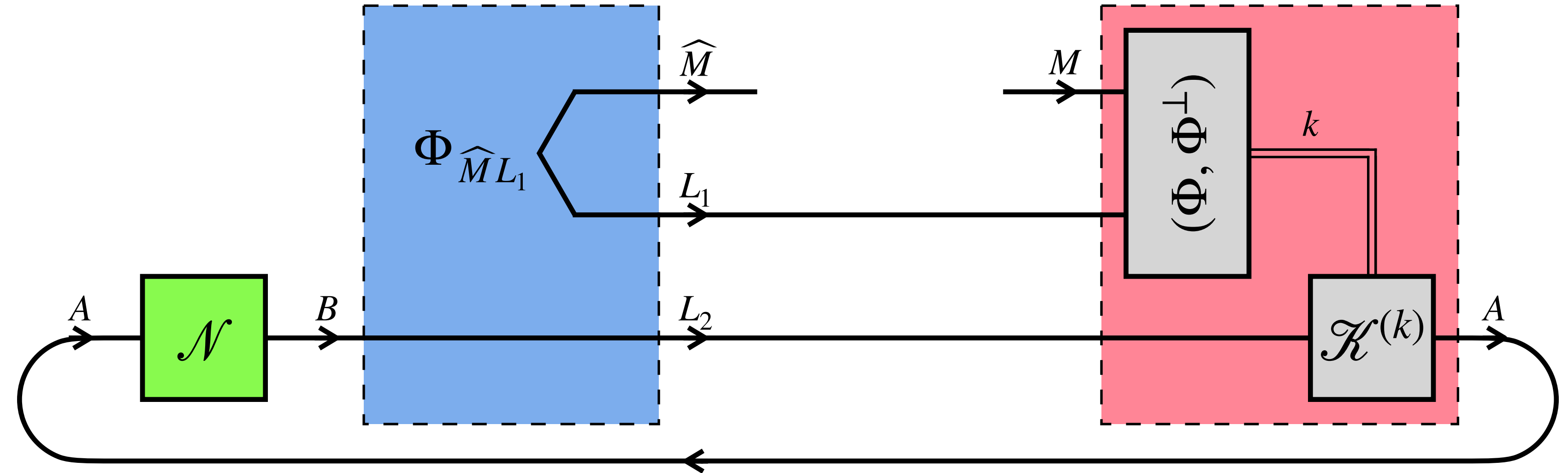
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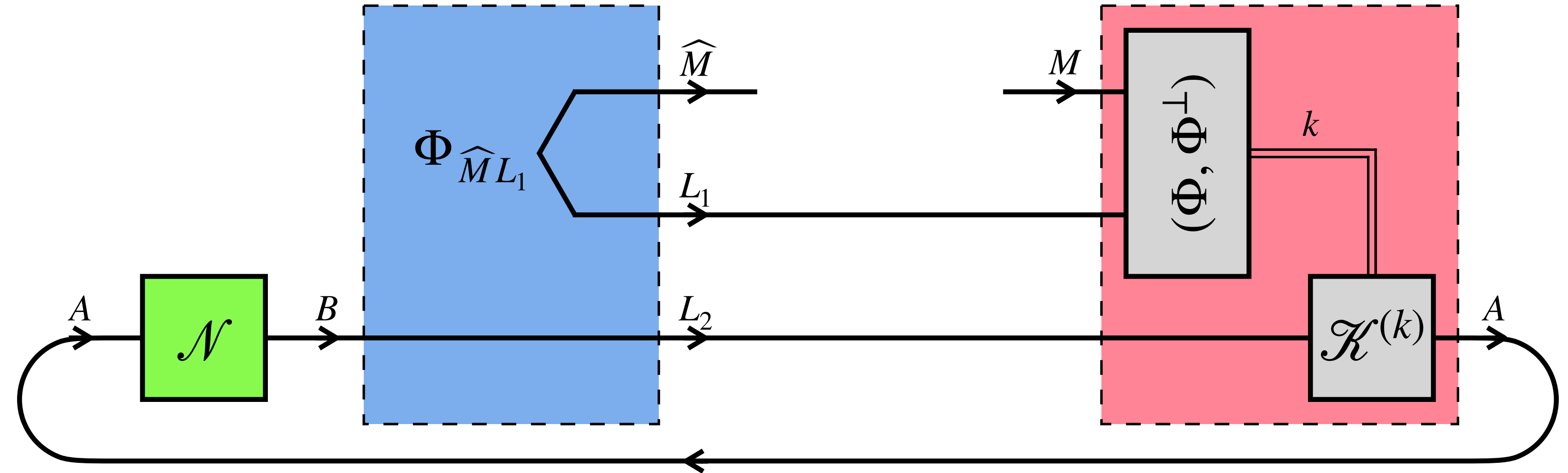
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- $\geq 2^{-I_{\text{doe}}(\mathcal{N})/d_A^2}$ [George et al. 2025]

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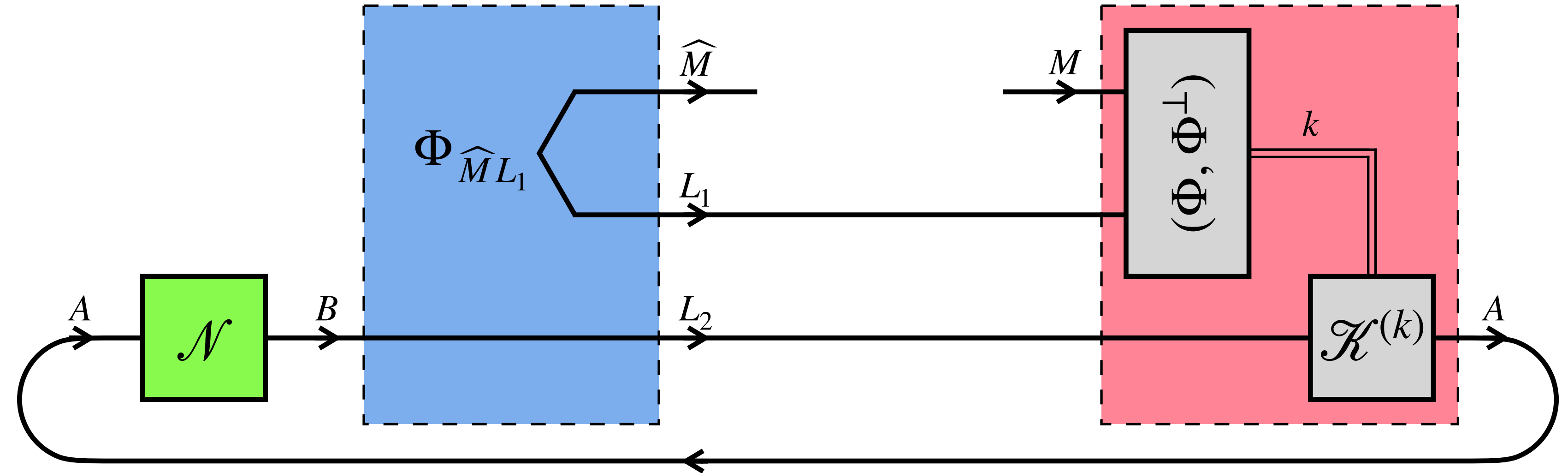
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- Choose $\mathcal{K}^{(0)}$ and $\mathcal{K}^{(1)}$ such that the above inequalities are equalized

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- Choose $\mathcal{K}^{(0)}$ and $\mathcal{K}^{(1)}$ such that the above inequalities are equalized
- Let $\log_2 d_M$ be the desired capacity formula $\Rightarrow$ the odds $\geq \frac{1-\varepsilon}{\varepsilon} \Rightarrow$ infidelity $\leq \varepsilon$

$$\log_2 \left[\sqrt{\frac{\varepsilon}{1-\varepsilon} 2^{I_{\max}(\mathcal{N})+I_{\text{doe}}(\mathcal{N})} + 1} \right]$$

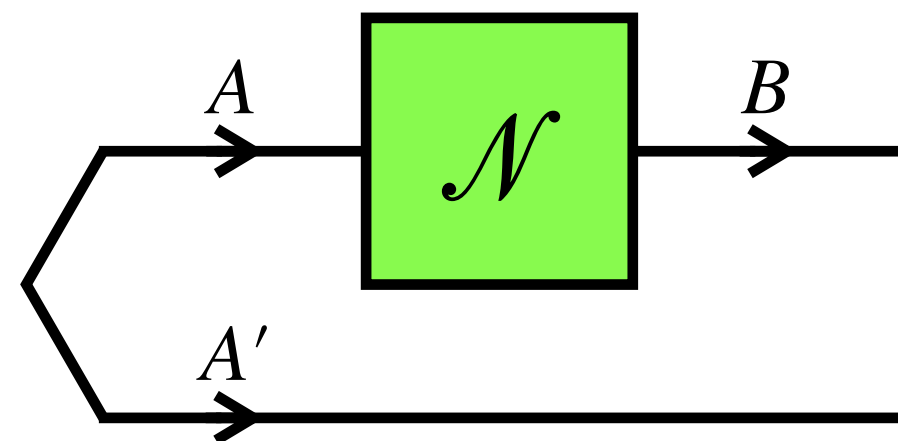
Optimality of the strategy

Prove: for every encoding & decoding with infidelity $\leq \varepsilon$,

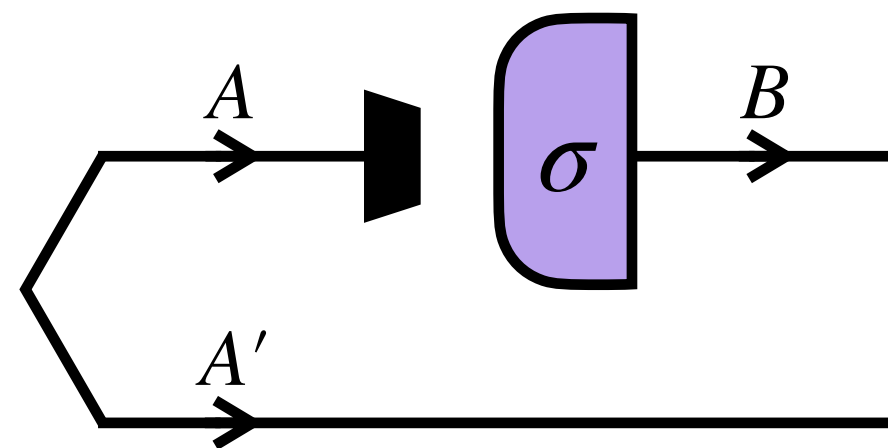
$$\text{we must have } \log_2 d_M \leq \log_2 \left[\sqrt{\frac{\varepsilon}{1-\varepsilon}} 2^{I_{\max}(\mathcal{N}) + I_{\text{doe}}(\mathcal{N})} + 1 \right]$$

Optimality of the strategy

$$I_{\max}(\mathcal{N}) := \inf_{\lambda, \sigma_B} \left\{ \log_2 \lambda : \mathcal{N}_{A \rightarrow B}[\Phi_{A'A}] \leq \lambda \frac{I_{A'}}{d_{A'}} \otimes \sigma_B \right\}$$

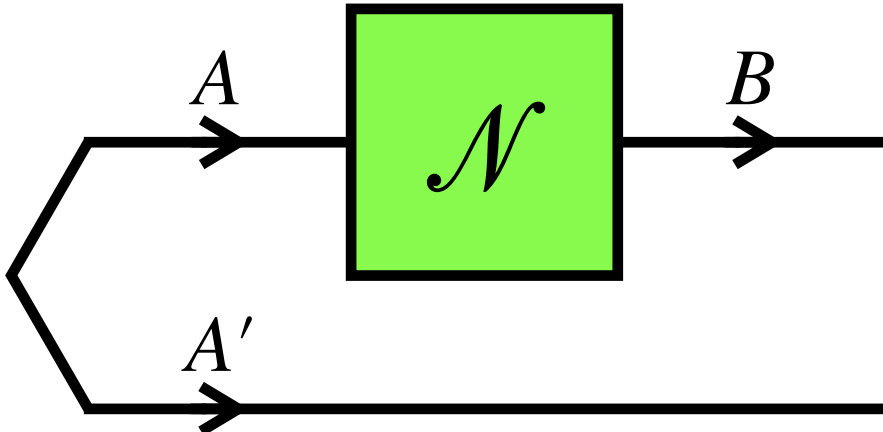


$\leq \lambda \times$

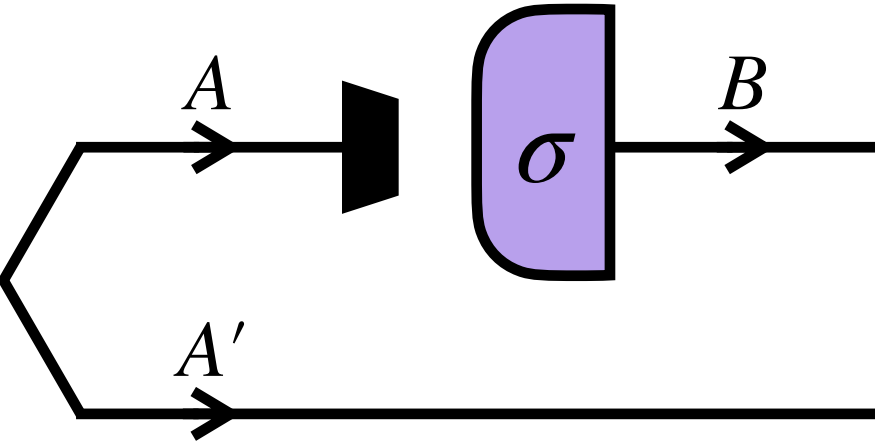


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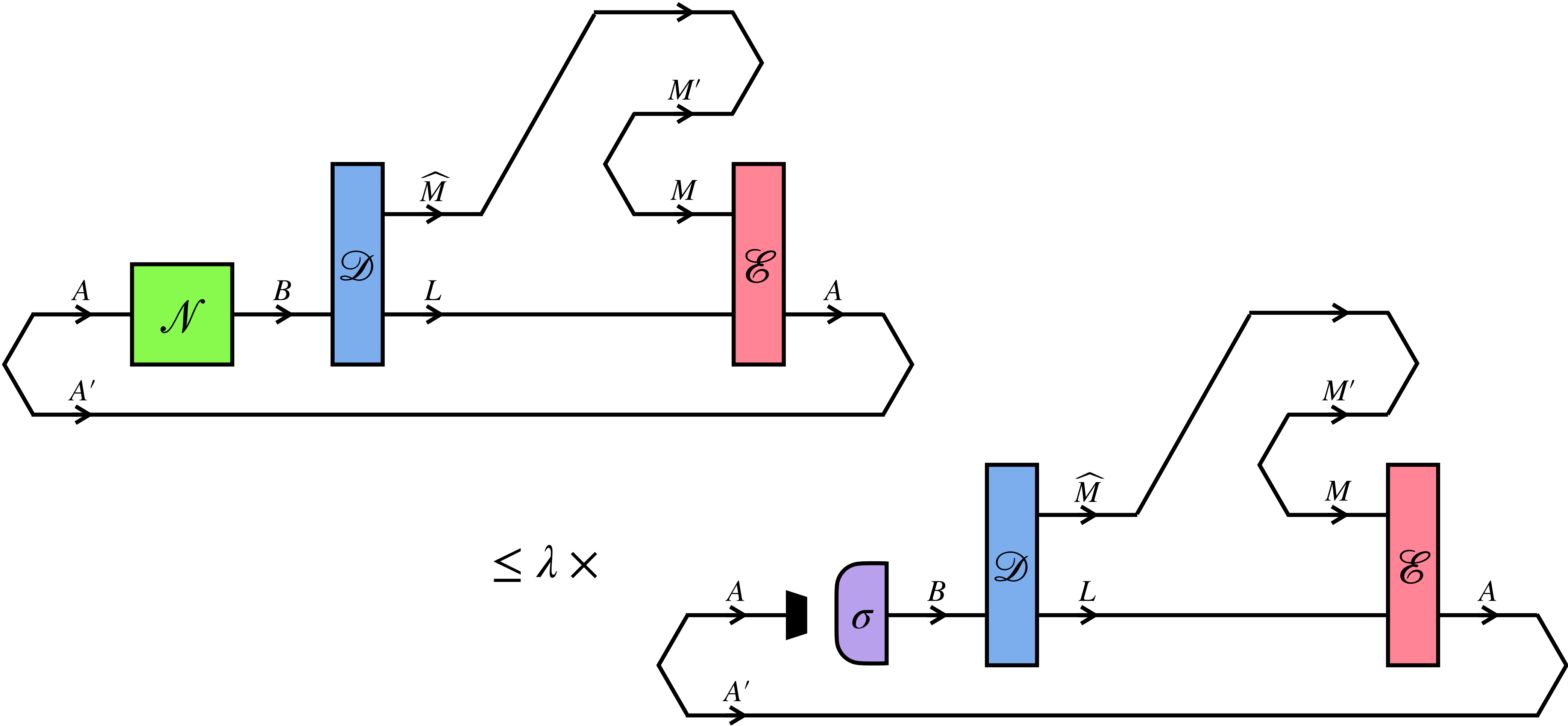


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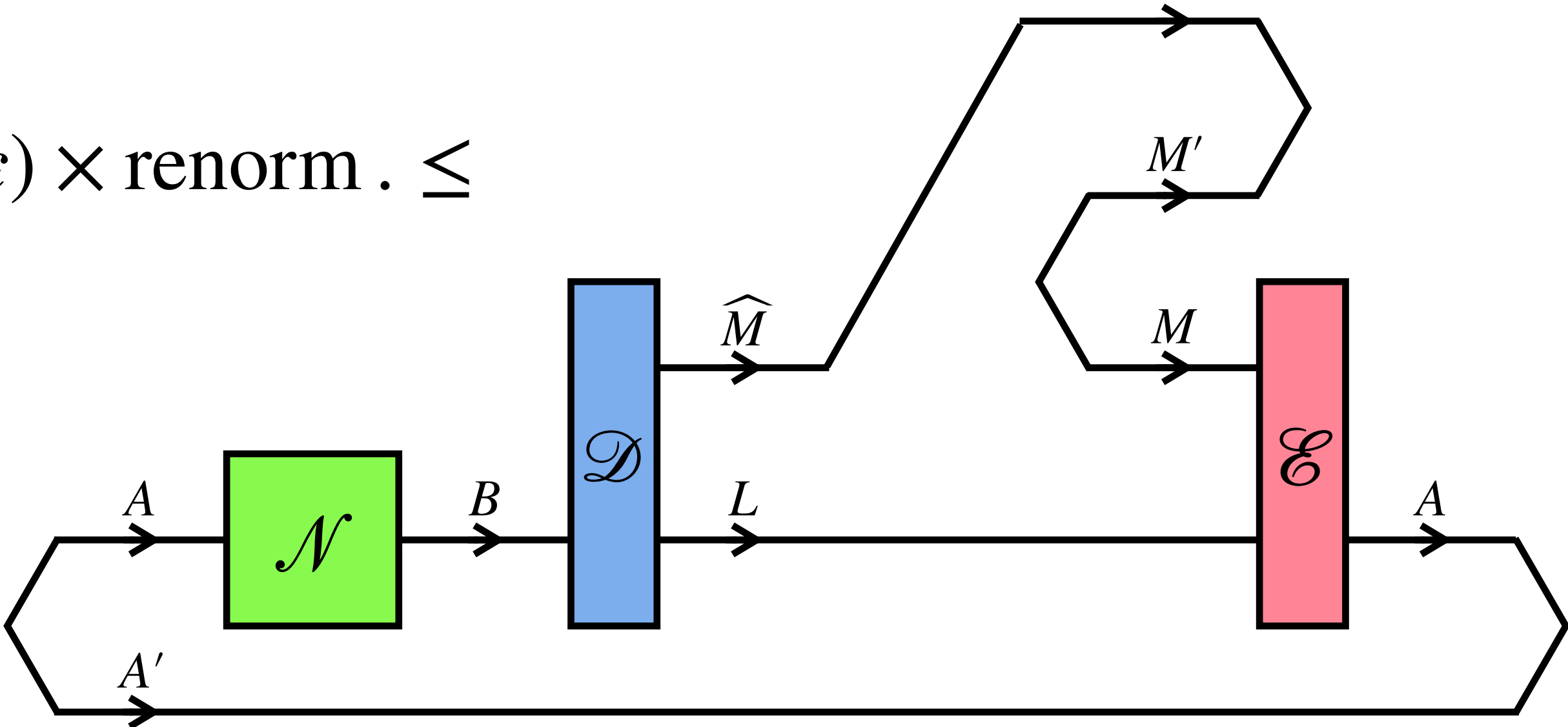
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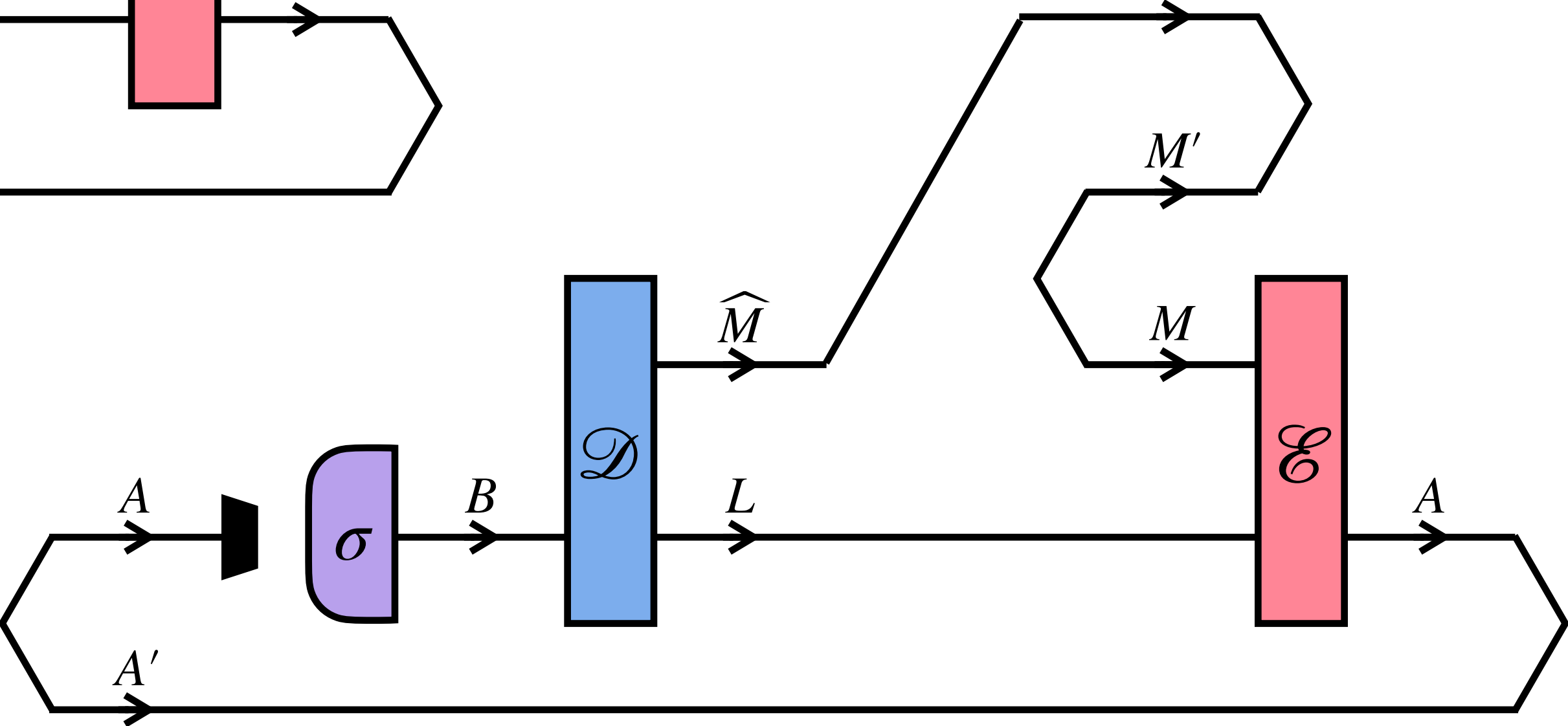
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$(1 - \varepsilon) \times \text{renorm.} \leq$



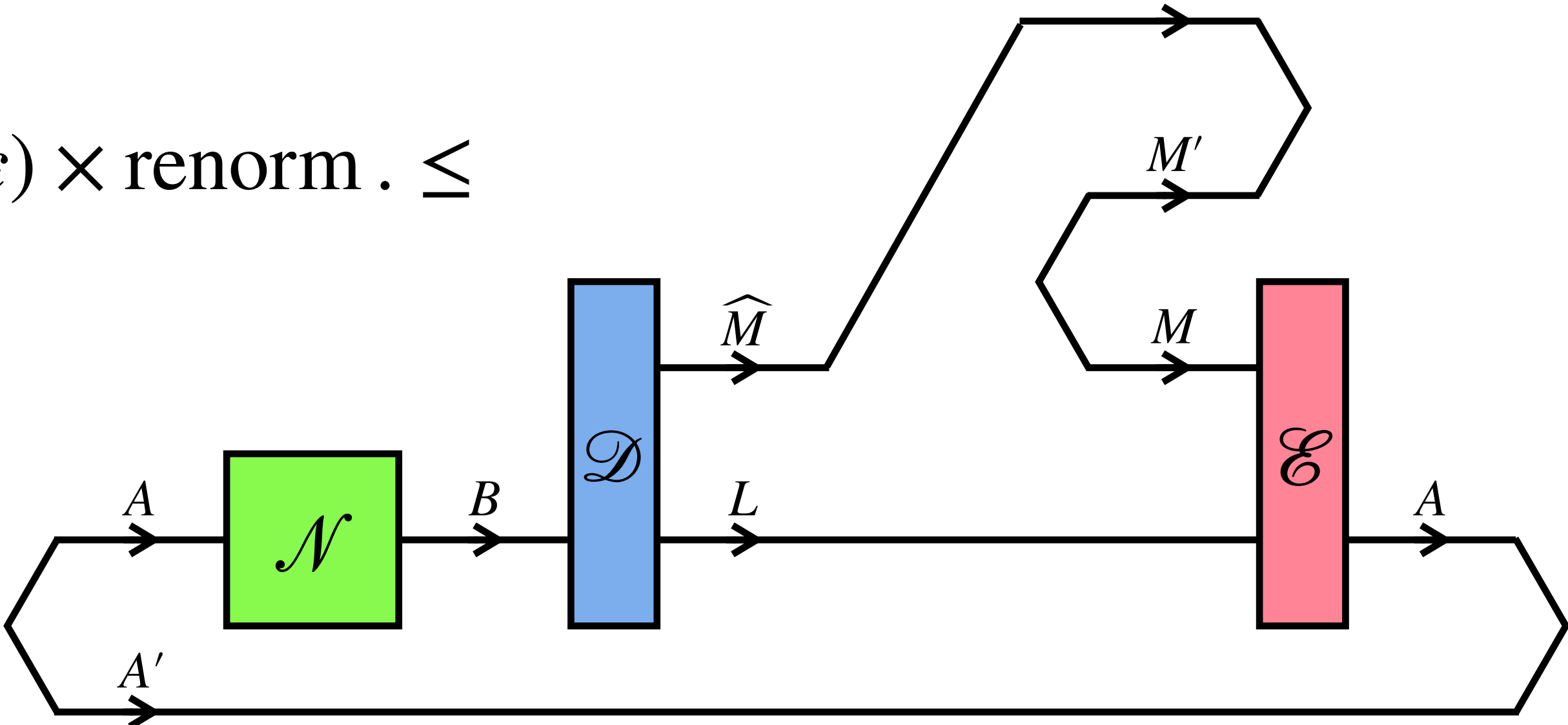
$\leq \lambda \times$



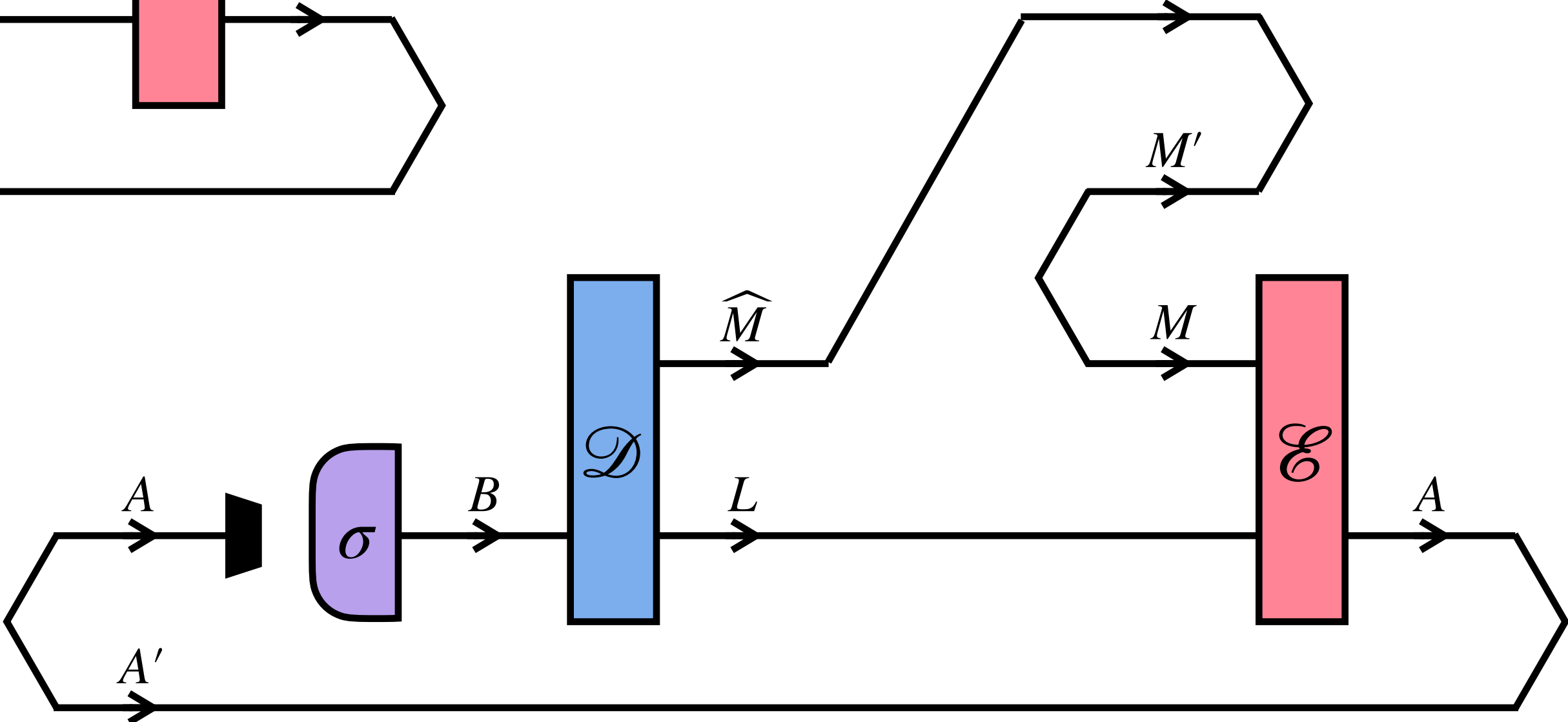
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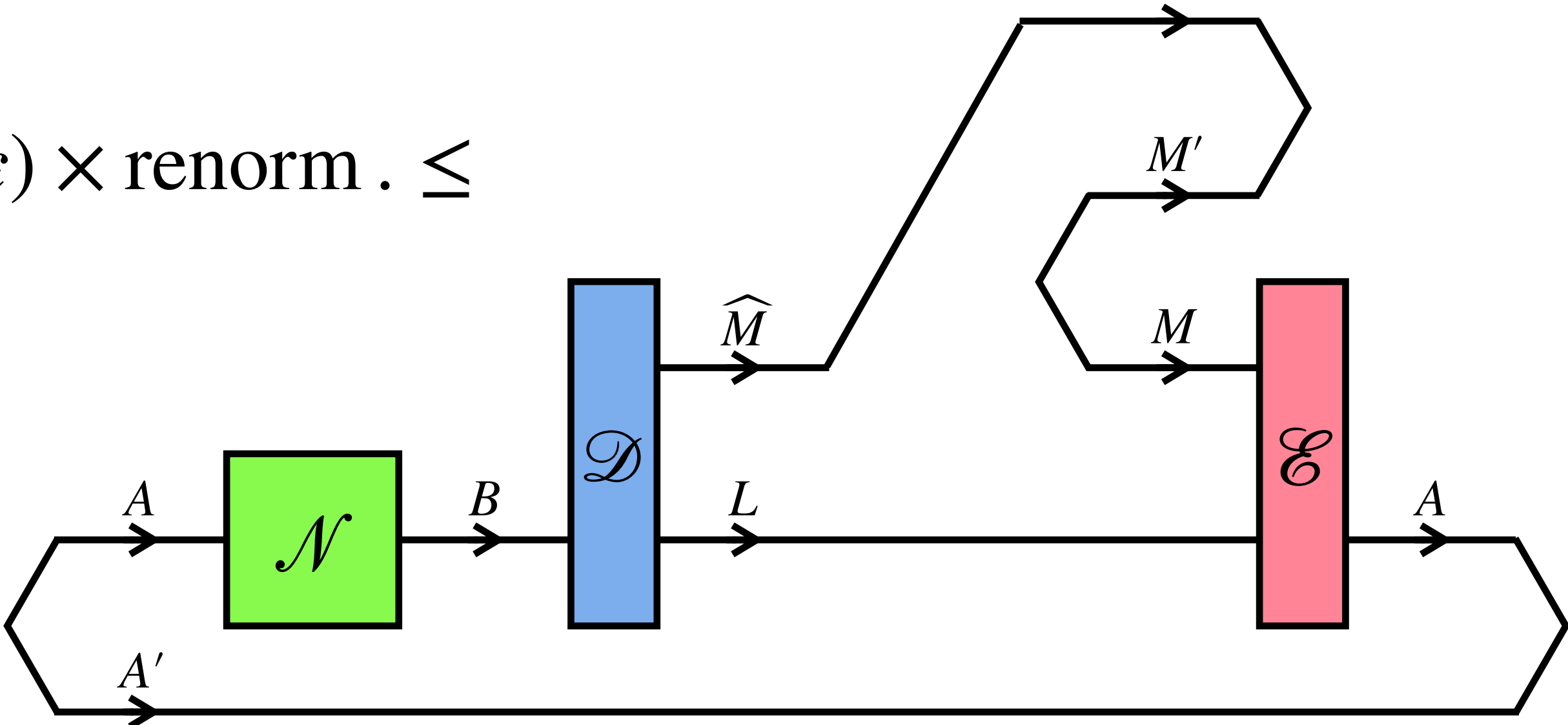


$$= \frac{\lambda}{d_M^2}$$

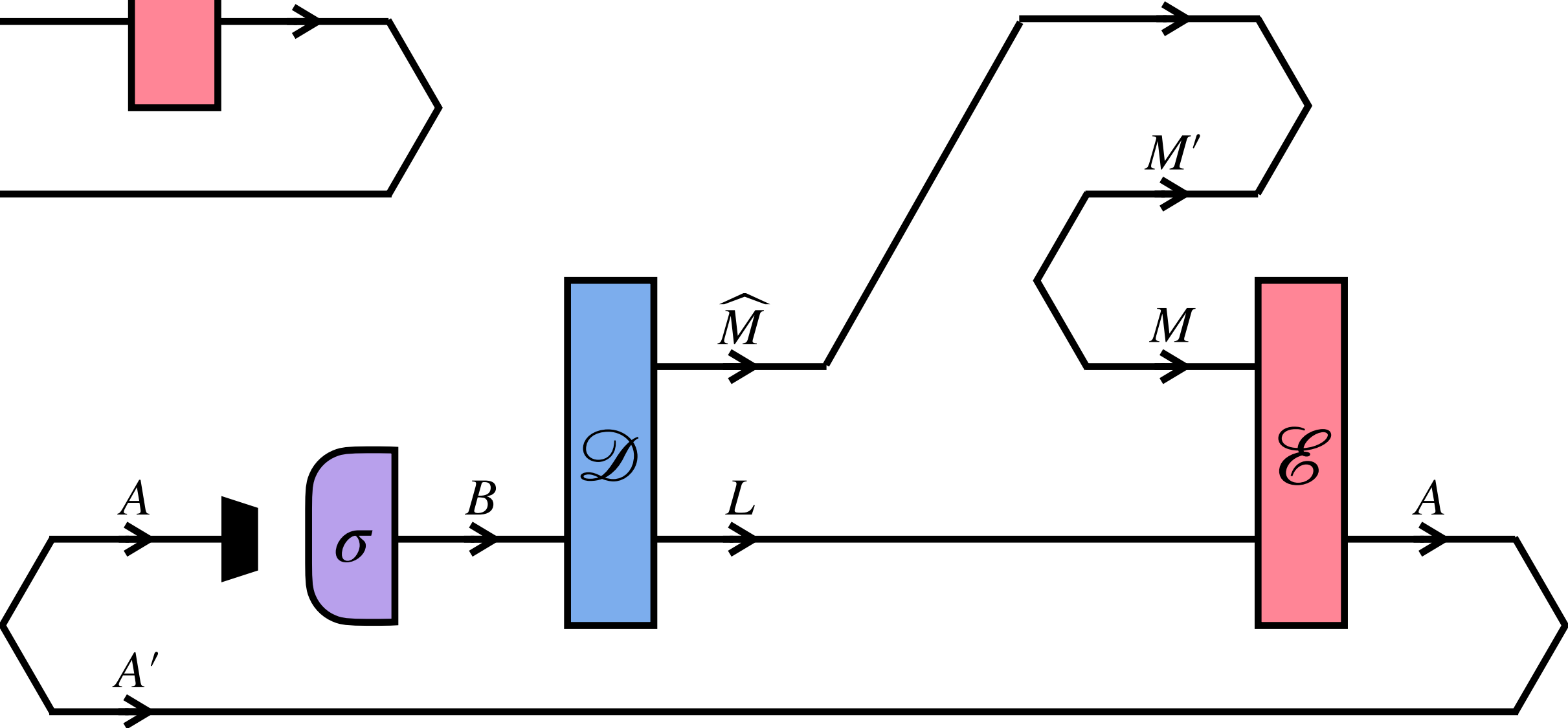
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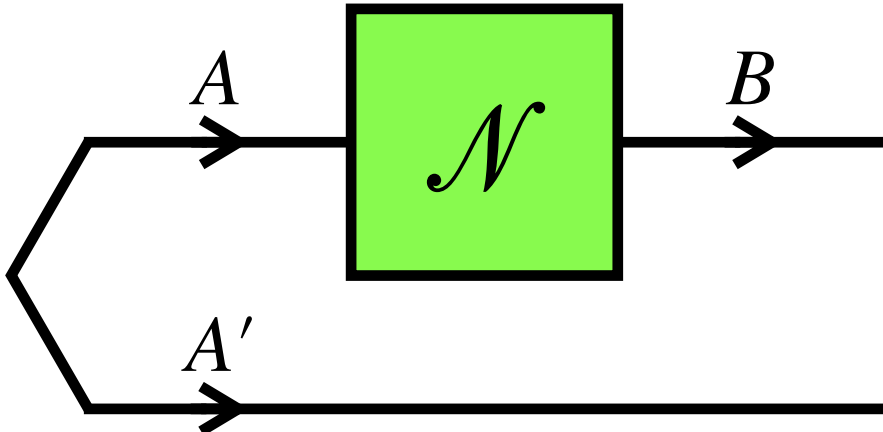


$$(1 - \varepsilon) \times \text{renorm.} \leq \frac{\lambda}{d_M^2}$$

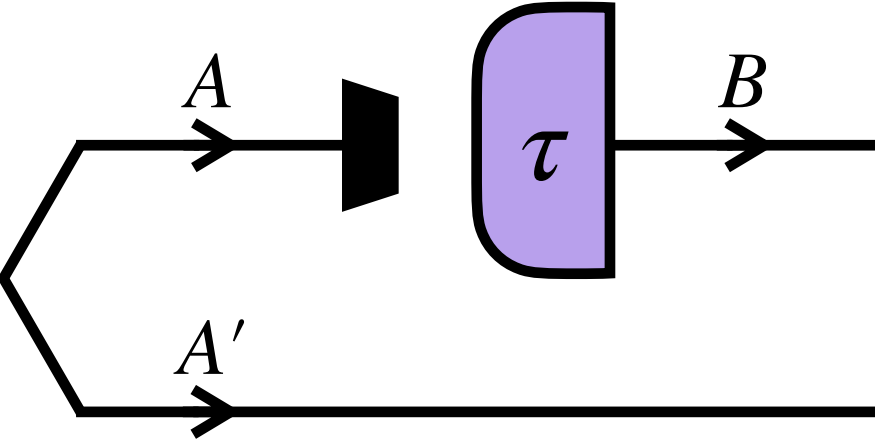
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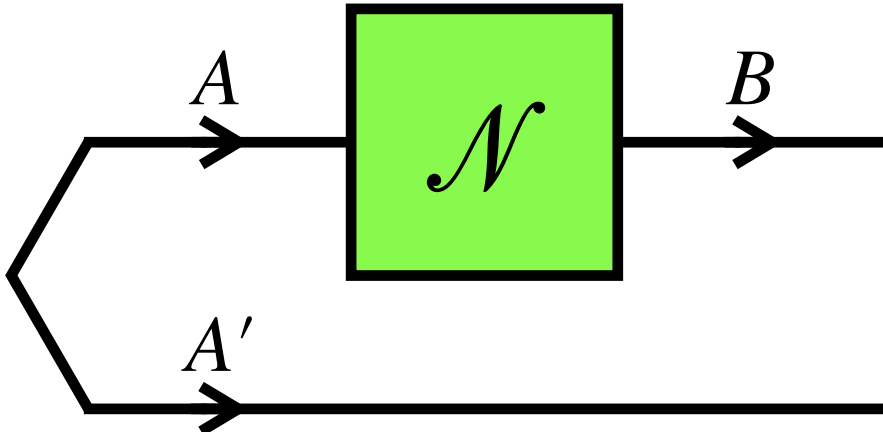
$$\geq \frac{1}{\mu} \times$$



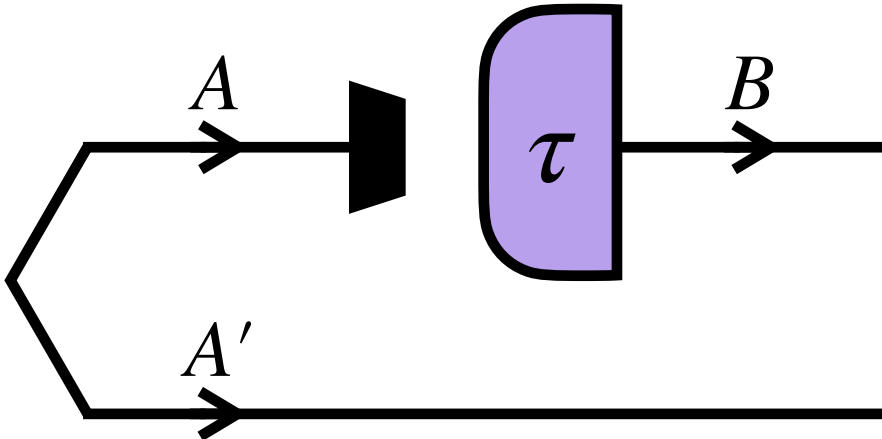
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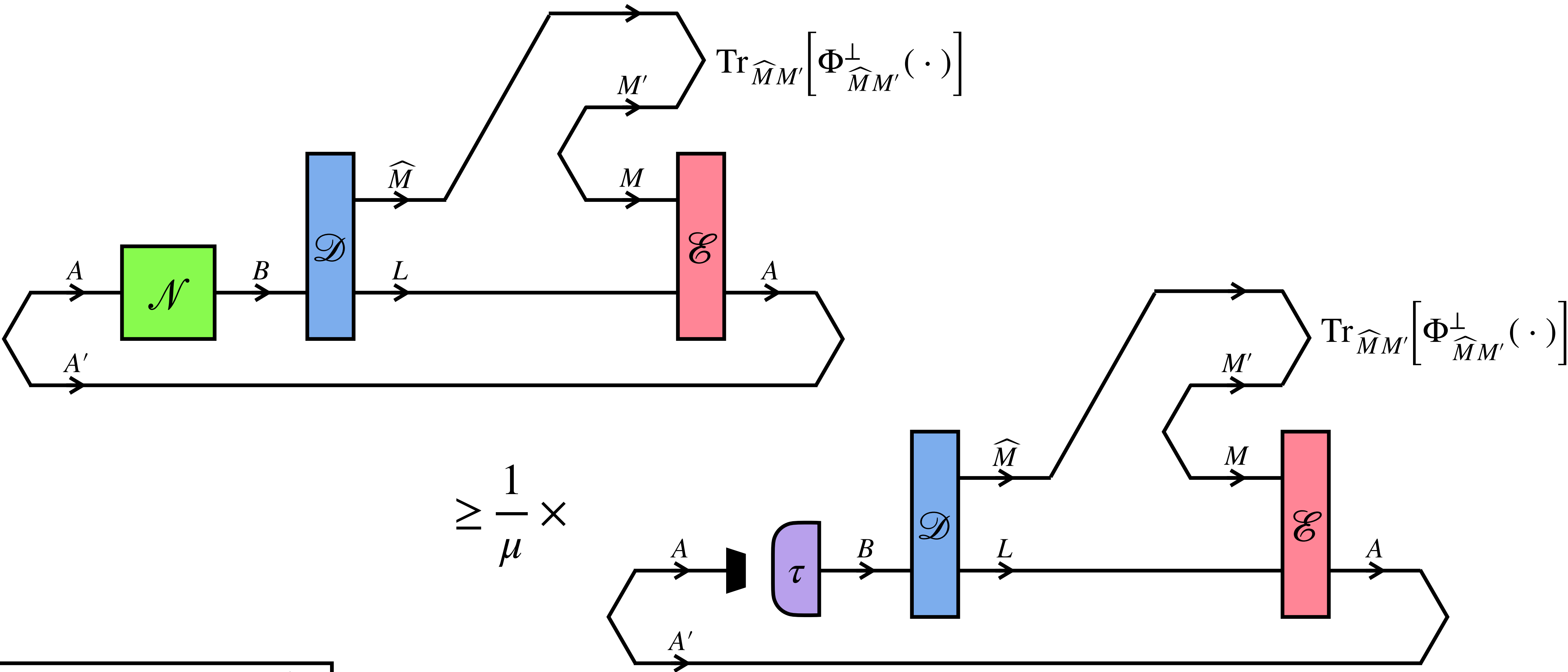
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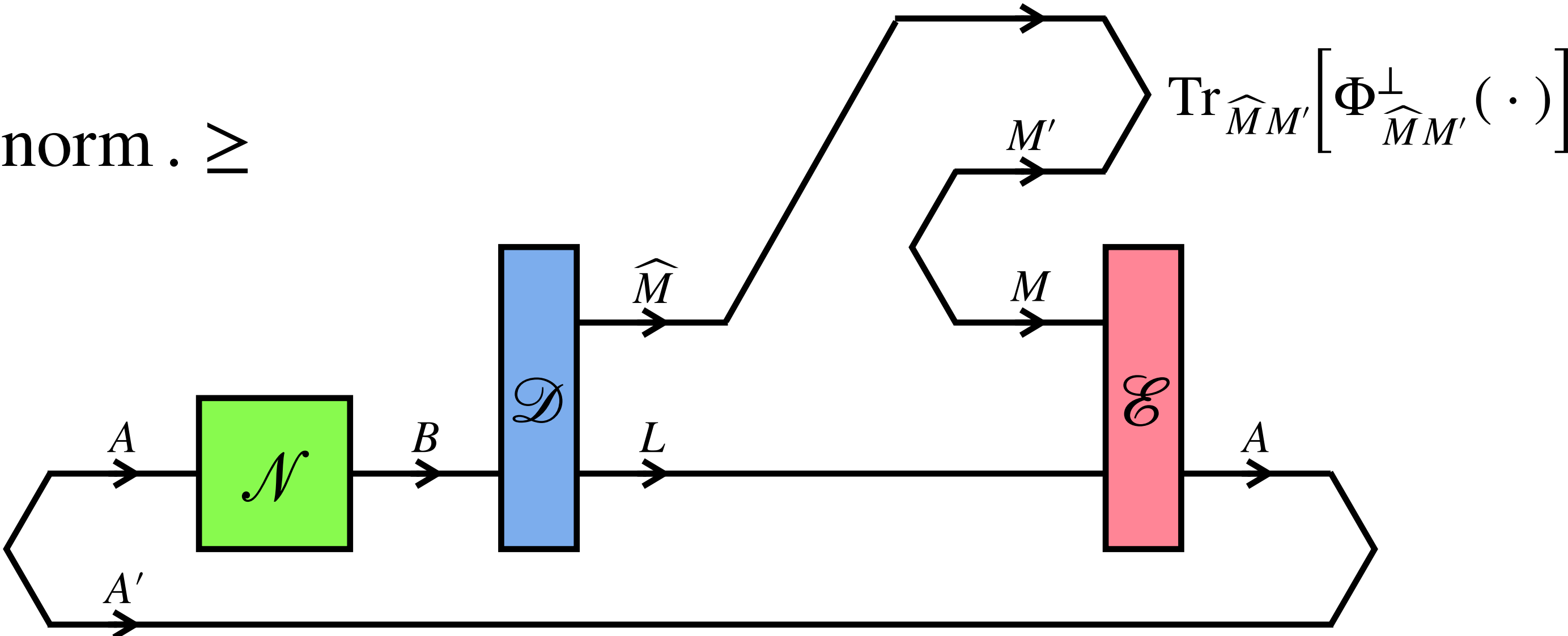


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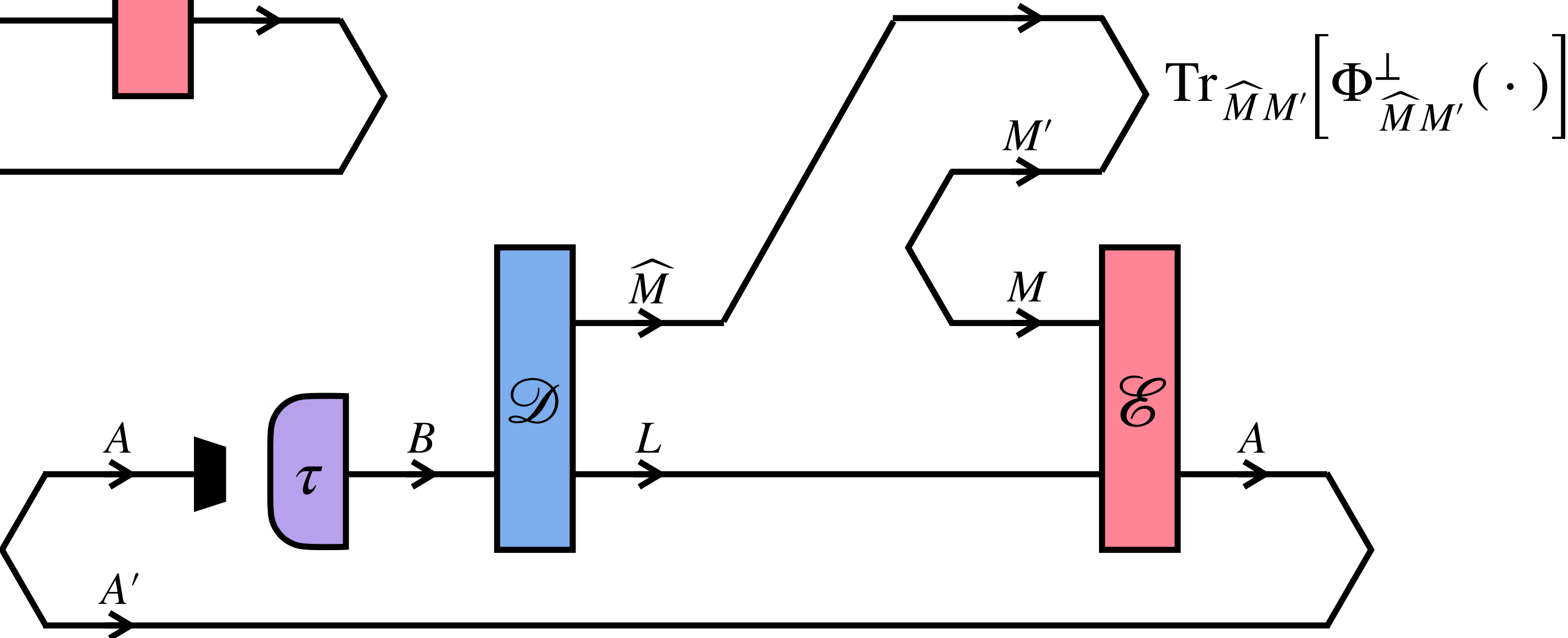
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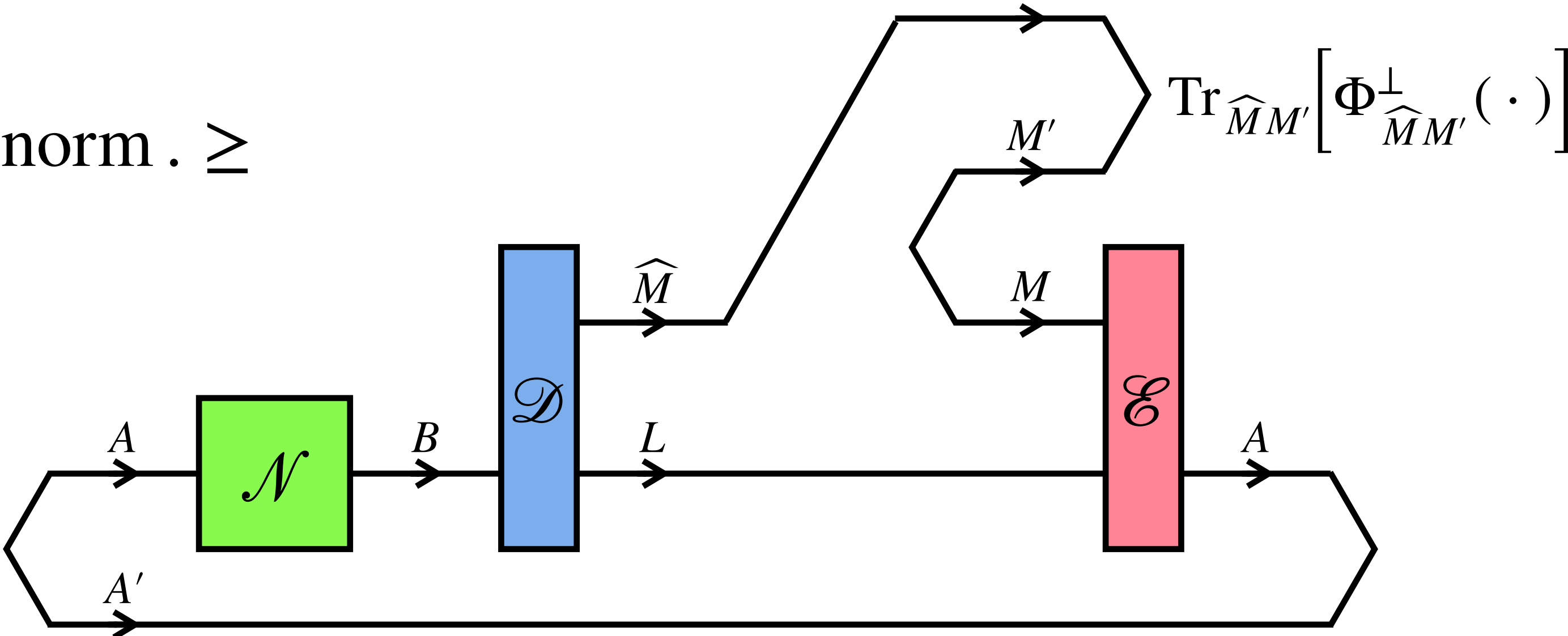


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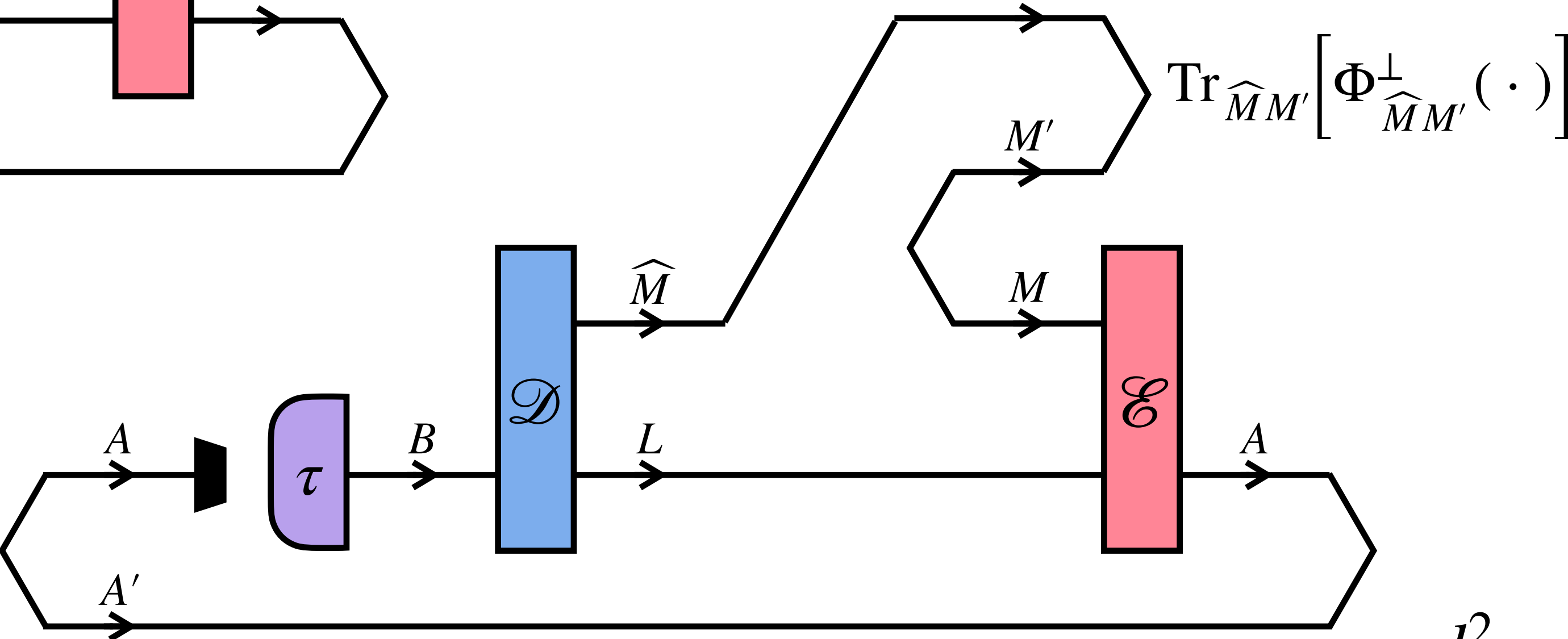
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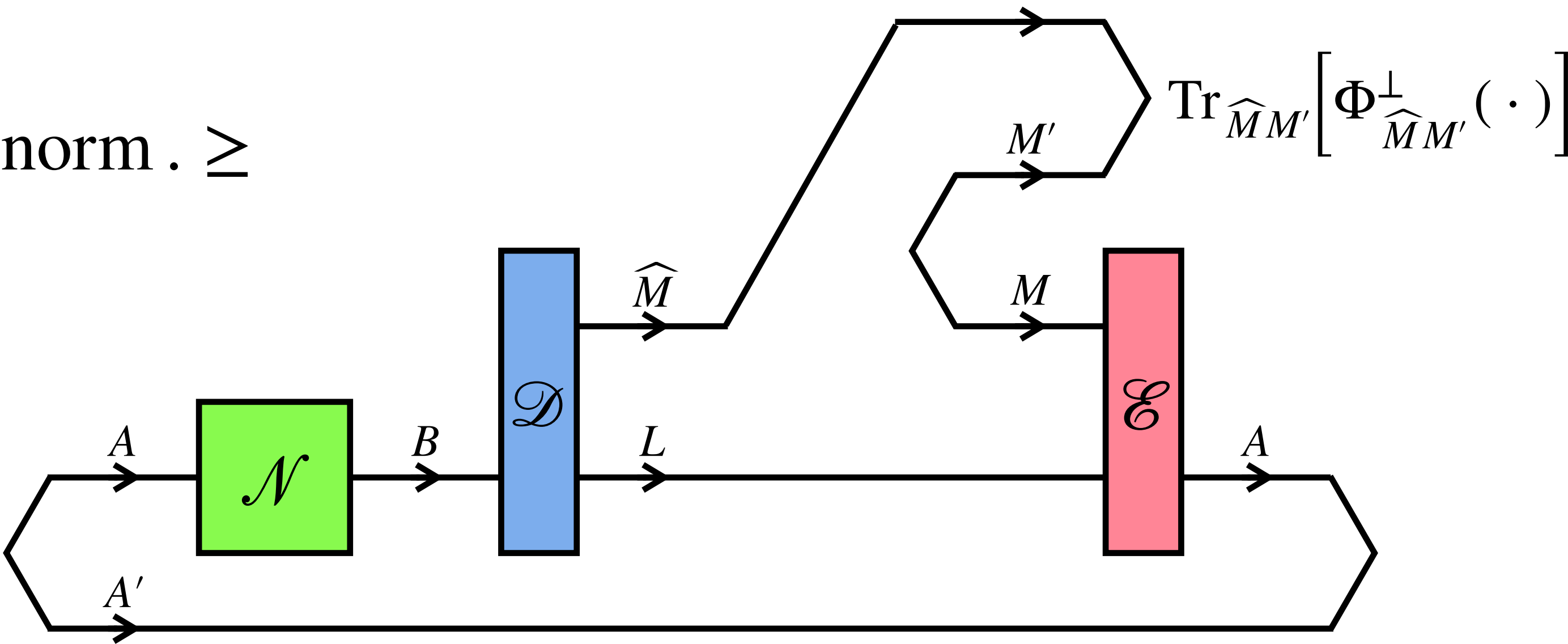
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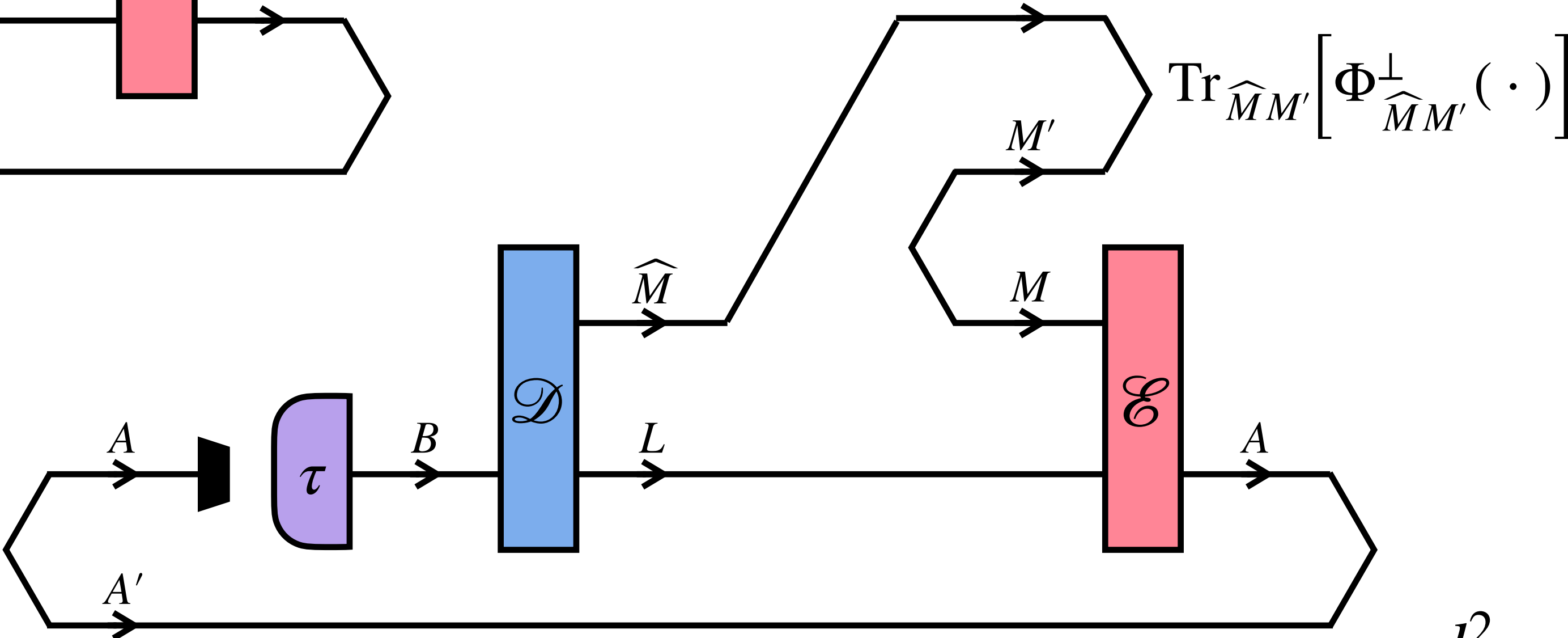
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Open questions:

- What if the memory that the daughter leaves for the father is noisy?

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[Horowitz & Maldacena 2004] [Lloyd 2006]
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Thanks!

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